

Essays on the Political Economy of Development

by

Ruimin He

Submitted to the Department of Economics
in partial fulfillment of the requirements for the degree of

Doctor of Philosophy in Economics

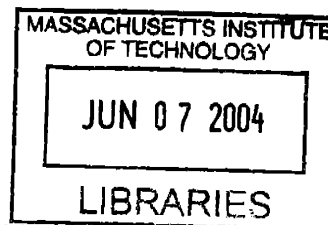
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Abstract

This dissertation presents four distinct essays. Mutually beneficial deals between government officials and merchants have negative social externalities, motivating policies to reduce this form of corruption. Chapter 1 analyzes a model with the following two components. *Guanxi*, or connections, between officials and merchants remove the risk that merchants expose corrupt officials, providing the basis for officials to treat their friends preferentially, leading to ex ante friendship formation; and supervisors demand bribes from their officials with the goal of maximizing returns from un-exposed officials. Individual merchant friendship formation imposes externalities on other merchants. I find that short-term policies can have long-term perverse effects. Furthermore, reducing bureaucratic size might non-intuitively increase corruption, and the mere perception that supervisors are corrupt reduces official-merchant corruption.

Chapter 2 proposes an explanation for the political bandwagon effect. In democratic countries where investments risk expropriation when the ruling political party changes, voters have an incentive to boost the support of the expected winning party during elections to show investors that the incoming party is more popular than it actually is, and hence more likely to stay in office after the next election. With rational expectations, investors are not fooled, but voters nonetheless over-vote in favor of the expected winner. The determinants of over-voting are explored, and an empirical exercise provides initial support for the results.

Chapter 3 is a joint work with Abhijit Banerjee. The chapter evaluates the performance of the World Bank and the Asian Development Bank, and finds no evidence of great effectiveness. The limits to effectiveness, and resistance to evidence-based policy making, are discussed. The authors argue that evidence-based aid is feasible.

Chapter 4 is co-authored with James Snyder and Stephen Ansolabehere. The chapter suggests that voters do not merely want to maximize their individual voting power, but also seek to increase the voting power of other voters who have similar spending preferences, effectively allowing other voters to virtually represent them. The authors develop a theoretical explanation, and use legislative ballot proposition data from reapportionment initiatives in California over time and at various aggregation levels for empirical support.

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I first timidly dragged my luggage through the automatic swinging doors at 77 Massachusetts Avenue on August 19, 2000 with no idea of what to expect. As a starry-eyed freshman, I never dreamt that these pages of acknowledgments would ever exist. The fact that you are reading them is a direct consequence of the love and mentor-ship and sacrifices of the individuals that these pages attempt to honor in too few words.

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Chapter 1

Policy in a world of Corruption and *Guanxi*

1.1 Introduction

Bribes in excess of a trillion dollars (US\$1,000 billion) are paid each year (World Bank Institute, 2004) around the world (Transparency International, 2003). Corruption reduces economic growth (Mauro, 1995)¹, exacerbates income inequalities (Gupta et al., 1998), and fuels political resentment², although some scholars see it in a more positive light³. The design and implementation of effective anti-corruption policies are in the populist and public interest. Too often, corruption is simply attributed to moral failings, socio-economic change, poverty (Bardhan, 1997), weakness in enforcement, and social norms (Andvig, 1991). These causes imply either despair and hopelessness, or policies such as moral indoctrination, increasing income, and strengthening policing mechanisms.

Without dismissing any of the above-mentioned causes or policy recommendations, this chapter argues that institutional and anthropological factors matter too. Specifically, I examine bureaucratic hierarchies and endogenous friendships between agents and argue that these affect both the extent and form of corruption, therefore impacting policy formulation. Policy responses must be sensitive to such factors in order to maximize efficiency and to prevent unintended consequences.

While corruption takes on many forms, this chapter focuses on corruption in the context of

¹I acknowledge Bardhan (1997) as the source of many of the literature cited in this chapter.

²Corruption is partially blamed for the Tiananmen protest (Sun, 1991), and political parties in numerous countries such as India (Seshia, 1998), Malaysia (Chin, 1996), Philippines (Case, 1999), and Egypt ('Aziz, 1998) run on anti-corruption platforms.

³Leff (1964) and Huntington (1968) argue that corruption compensates for erroneous policy and over-centralized bureaucracies, respectively, while Laffont & Tirole (1991) see corruption as a necessary response to regulatory capture.

mutually beneficial deals between government officials and private agents (merchants), which come at the expense of broader society. It is this net social loss that motivates corruption-reducing policies. In many countries, government officials enforce and administer the costly demands placed on merchants by the state. These demands take the form of monetary repatriations, either as taxes or payments for limited government licenses, or expensive actions to improve production techniques and buildings to satisfy health, safety, environmental, and fire requirements. Regardless of the exact form taken, these repatriations and actions are costly to the individual merchants concerned, while beneficial to society at large. Government officials merely facilitate this transfer – they do not privately benefit from the process.

Such a set-up endows government officials with rents (Krueger, 1974 and Klitgaard, 1988) that create the incentives for what Shleifer & Vishny (1993) term as “Corruption with Theft.” Merchants may give a side payment to the government officials if the officials allow them to forego their costly obligations to the state. For example, officials might not enforce regulation standards or they might allow taxes to be under-reported. It is hard for external agents to audit such crimes. Merchants would be willing to pay up to the private cost that they would have had to give to the government otherwise, while government officials would be willing to accept any positive side payment as they would have received nothing under the status quo. If such a side payment occurs, society at large loses out because the actual payment does not reach the government coffers, or the socially beneficial intervention is not carried out. Gong (1997), in describing Yan (1995), writes,

In 1993, the country’s (China) procuratorial organs found in their investigations that the taxation offices of at least 30 counties granted tax receipts to people who had not paid taxes at all in exchange for monetary gains, causing a major loss in government revenues.

Responsible governments therefore seek to prevent such corruption. The private nature of such transfers makes monitoring difficult for third-party anti-corruption agencies without evidence provided by the parties involved. Moreover, because both parties benefit from the relationship, there are very few incentives for either party to engage in whistle-blowing. Even so, some fraction of the population may still choose to alert and facilitate anti-corruption agencies to catch the corrupt government official. They may do so for nebulous moral reasons, for monetary rewards, or to remove and deter corruptible officials from causing losses to society at large. As in Cadot (1987), engaging in corruption is a gamble. Anti-corruption agencies might increase the probability that a

merchant “squeals” by running television campaigns or by increasing monetary rewards.

Whistle-blowers force government officials to balance the desire for wealth accumulation against the risk of getting caught. Consequently, they may limit their corruption activities and only choose to be corrupt with those that they trust the most. All else being equal, officials trust merchants that they have known for a long time more than passing acquaintances or strangers, so as to reduce uncertainty (Wellman et al., 2002) and transaction costs (Standifird & Marshall, 2000). This places an incentive for merchants to invest in connections, or *Guanxi* in the Chinese lexicon, with officials so as to signal their trustworthiness. Boycko et al. (1995) emphasize that corruption contracts in general are not-enforceable; provisions against whistle-blowing should feature prominently among examples of non-enforceability. Merchants may also hold their friends to differential moral standards. Bardhan (1997) writes,

The same people who are most vocal and genuinely worried about widespread corruption and fraud in the public arena do not hesitate at all in abusing public resources when it comes to helping out people belonging to their own kinship network.

Yang (1989) makes a similar point in the Chinese context.

Under these assumptions, the social cost of a corrupt deal is the payment to the treasury that is not made, or the socially-beneficial regulation that is not carried out. This social cost is the same regardless of whether the corruption occurred between friends or strangers. The level of familiarity between agents affects only the size of the transfer payments. Hence, net social cost is proportional to the total amount of corruption in the society – total corruption per official (both with friends and strangers) multiplied by the total number of officials.

The multi-layered nature of bureaucracies is the other ingredient of the model. Government officials report to superiors and, consequently, might have to pay bribes to them in a long corruption chain. Cadot (1987) and Andvig & Moene (1990) introduce official-to-supervisor bribes as a means of reducing an official’s probability of getting caught, leading to multiple equilibria. Hillman & Katz (1987) look at the social costs incurred in the jockeying for bureaucratic positions to receive the rents from positional jockeying further down the bureaucratic hierarchy. In this chapter, the focus is on the wealth consequences of official-to-supervisor bribes for officials, which in turn affect their risk preferences.

The model yields the following conclusions: Friendships formed by individual merchants impose both positive and negative externalities on other merchants, and the sign of the net externality is

parameter-specific. In addition, the efficacy of a campaign aimed at increasing the probability that merchant-strangers will squeal is increased if supervisors react to the campaign by re-optimizing their bribes. This occurs as supervisors have an incentive to reduce the amount of risk that their officials face, and hence the amount of corruption that their officials engage in, so as to maximize the possibility that they will be able to collect their bribes. Endogenous merchant incentives to form friendships lead to a positive feedback loop that magnifies the impact of such campaigns.

Furthermore, if long-term friendships can be cultivated, short-term credible campaigns (such as the one mentioned above or raising the wages of government officials) aimed at reducing corruption might perversely back-fire and increase corruption in the long-term instead. Also, ignoring friendship formation causes policy makers to over-estimate the impact of increasing bureaucratic size on total corruption levels, to the extent that for some parameter values, reducing bureaucratic size might non-intuitively increase corruption levels. Finally, the mere perception of corruption within a bureaucracy (i.e., the presence of corrupt supervisors who demand bribes from the officials under them) might go some way in reducing actual corruption between officials and private merchants.

This chapter contributes to the literature by examining the effects of *Guanxi* and hierarchies on the form and extent of corruption. The reasons for the existence of corruption, such as the low probabilities of getting caught, market failures, and agency problems, are black-boxed. Instead, the focus falls on the strategic interactions of agents in this environment and their effects on policies.

The rest of the chapter is structured as follows. Section 1.2 reviews the relevant literature. The model is introduced in Section 1.3, and some results are derived in Section 1.4. Section 1.5 looks at the policy implications, while Section 1.6 concludes.

1.2 Literature Review

This chapter addresses two frequently-raised issues in the corruption literature⁴. First, what determines the level of corruption in a society? Shleifer & Vishny (1993) look at the impact of government structure on the level of corruption. They find that weak decentralized governments experience more corruption and that corruption is more distortionary than taxation. Banerjee (1997) identifies corruption as a result of governments acting in cases of market failure and agency problems within governments. More market failures lead to more corruption. Tirole (1996) and Andvig (1991) note that corruption tends to persist – corruption is an unfortunate consequence of

⁴For a comprehensive overview of the existing literature on corruption, please refer to Bradhan (1997).

past corruption. Second, how should policy be sensitive to the prevailing environment? Acemoglu & Verdier (2000) theorize that if the costs of preventing all forms of corruption are too high, then it would be optimal to allow some corruption.

The Western sociological/anthropological study of *Guanxi* began with Yang's (1989) seminal ethnographic fieldwork. She defines the practice of *Guanxi* as "the exchange of gifts, favors and banquets; the cultivation of personal relationships and networks of mutual dependence; and the manufacturing of obligation and indebtedness" (Yang, 1994). Her definition portrays a network-centric climate that emphasizes symmetric reciprocity (Lin, 2002), reputations, and long-term benefits to police opportunistic behavior (Standifird & Marshall, 2000) and substitutes for the formal, legalistic completeness of Western contracts.

Sociologists have commented on the intimate link between corruption and *Guanxi*. Yang (2002) further writes,

As *Guanxi* practices shift increasingly into the business domain where business interests must engage with government officials who control the means to favorable business opportunities, the explicit material monetary calculations have the scale of monetary values transacted transforms *Guanxi* into the order of corruption... one is highly sceptical of the simple argument that *Guanxi* culture is being replaced by a new rational-legal regime.

while Ledeneva (1998) sketches the corruption-connection link in Russia. More broadly, the existing literature provides an empirical basis for the assertion that personal relationships matter. Roberts (1990) uses the death of Senator Henry Jackson as a basis for an event study that finds that a seniority/benefits relationship exists in the United States Senate. In addition, Fisman (2001) uses Suharto's declining health as an exogenous shock to put a monetary value on a businessman's connection to the former Indonesian president. Gold, Guthrie & Wank (2002) document the advantages of *Guanxi* for business groups, business-state clientelism, the job search, and job mobility.

Kipnis (2002) hints at the mercenary creation of friendships, while Chow (1997) writes,

Guanxi is a form of human capital. Having *Guanxi* is like having knowledge of which friend would be helpful when needed, having a reputation similar to that conveyed by a college degree, and having a good credit rating. Developing *Guanxi* is accumulating human capital just like getting a college degree or promoting a reputable brand name.

The cultivation of friendship takes time, and it is much too late to start building bonds when one

realizes that he needs to deal with a particular official. Hence friendships must be formed ex-ante – Lin (2002) refers to “the tactics of cultivating indebtedness.” The pervading importance of high school “old-boy networks” and college ties⁵ suggests that friendship bonds tend to form early and persist over time.

The formation of webs of friendships (*Guanxi Wang* in Mandarin) leads us to the economic literature on networks. Here Kranton & Minehart (2001) examine networks of buyers and sellers in a world where buyers and sellers need an explicit “link” to exchange goods. Buyers have uncertain valuations of the homogeneous goods that the sellers sell. Sellers use networks to pool this uncertainty, while buyers use the networks to secure better terms of trade. Buyers and sellers acting individually maximize overall welfare and efficiency, a result that does not hold in this model due to the externalities exerted by merchants on other merchants when they form friendships with officials and by officials on society at large when they engage in corruption.

1.3 The Model

1.3.1 Introducing the Model

Government transactions are organized as follows:

$$\text{Merchants} \rightarrow \text{Officials} \rightarrow \text{Supervisors}$$

Merchants deal with officials who report to supervisors, with (potential) payments flowing from left to right. Imagine a five period game. In the first period, merchants individually decide on the number of friendships that they want to form with the officials, based on expectations of the anti-corruption policy. This policy is revealed in the second period. Supervisors decide how much to demand from their officials in the third period. Officials and merchants transact in the fourth period. Officials engage in corrupt deals with some of the merchants, running the risk that some merchant-strangers might report them to the police. In the final fifth period, the officials that have been reported are caught and thrown into jail, where they enjoy a normalized zero utility. Those who are not caught pay their bribes to their supervisors and derive utility from their residual income.

Corruption here is of the “Corruption with Theft” variety described by Shleifer & Vishny (1993), where both the briber (the merchant) and the bribee (the official) benefit from the transaction⁶.

⁵George W. Bush and his Yale Skull and Bones secret society, for example.

⁶In the case of “Corruption without theft,” the briber pays the “bribee” an amount over and above what he is supposed to pay under the proper channel. The private incentive for both parties to co-operate to split the rents is

Let us describe an example. A merchant approaches an official to apply for a license. Going by the proper channels, the merchant pays a stipulated fee that goes straight into government coffers, while the official issues the license without pocketing anything. Alternatively, the merchant could pay a smaller amount of money as a side payment to the official, without anything going to the government coffers. The official issues the license and pockets the full side payment. Both the merchant and official have incentives to go through the corrupt channels, with officials charging their friends less than they charge strangers for the same service. M merchants each make T transactions from the pool of N officials ($T < N$). For simplicity, merchants have the liberty to choose the T officials that they deal with. Perhaps they just need any T signatures to get their project approved. Among the officials, the merchants choose to deal with their F friends first before going to strangers. While they would only like to deal with strangers who offer them corrupt deals, the reality is that only a fixed number of such deals are available to be split among all the merchants. Demand shortages and potential search costs (in the spirit of Diamond, 1971) may prevent officials from competing their rents down to zero. Also, if the officials charge overly exorbitant rates, the merchants may decide to take revenge by reporting the transaction to the police or can simply reject it. An alternative story is that merchants know ex-ante the full set of officials that they will have to deal with in future – but they find it cost-effective to make friends only with F of them.

1.3.2 Maximization problem for the officials

The following assumptions are implicit in this model. First, an official's friends will never turn him in, and hence corruption among friends is risk-less. Making a less extreme assumption (such as it is still risky, but less risky, to deal with friends) does not yield additional insight. Second, getting prosecuted for corruption leaves an official with a normalized zero utility. Third, the probability that a merchant stranger squeals is exogenously set as a policy variable.

Officials choose X , the number of strangers to engage in corrupt deals with, to maximize utility

$$U_O = \max_{0 < X < \frac{MT}{N} - Y} S^X G(P) \quad (1.1)$$

where

$$P = W - E \cdot Cut + AY + BX. \quad (1.2)$$

Utility G increases at a diminishing rate with total wealth P , implying $G_P > 0$ and $G_{PP} < 0$.

now lost, which leads to a very different analysis.

Total wealth P is the additive summation of its components. W is the official's gross wage received from the treasury. Cut is the bribe that each official might have to pay to his supervisor, and E is a dummy variable that takes the value of 1 if the bribe is actually paid, and 0 otherwise. Y is the number of friends that the official has. A and B denote the amount that an official earns per corrupt deal with a friend and stranger, respectively. Strangers have a probability $(1 - S)$ of reporting their corrupt dealings with officials to the police. Y is determined in the first period, while S and W are determined in the second period. The inequality $N(X + Y) < MT$ introduces a symmetry assumption that the total number of corrupt deals (with friends and strangers) made by the officials is less than the total number of transactions made by the merchants. Setting $0 < X < \frac{MT}{N} - Y$ assumes away corner solutions such as the official accepting zero, negative, or every deal from strangers.

The first order condition of Equation 1.1 yields

$$G \ln(S) + B \cdot G_P = 0. \quad (1.3)$$

1.3.3 Maximization problem for the merchants

Merchants each maximize over F , the number of friendships to form. Merchants maximize utility U_M of the form

$$U_M = \max_{0 < F < T} D(R) - C(F) \quad (1.4)$$

with

$$R = \lambda_1 F + \lambda_2 H(\tilde{W}, \tilde{S}, E \cdot \tilde{Cut}, F, F'). \quad (1.5)$$

The merchant maximizes the difference between the utility that he gets from a higher revenue R , and the cost C from a higher F . An increasing friendship cost function, $C_F > 0$ and $C_{FF} > 0$, and a diminishing (or constant) marginal utility revenue function, $D_R > 0$ and $D_{RR} \leq 0$, are assumed. Net revenue R is additively separable into the components of wealth. Constants λ_1 and λ_2 , where $\lambda_1 > \lambda_2 > 0$, are the benefits of a corrupt deal with a friend and stranger respectively, from the point of view of the merchant. When dealing with a particular official, merchants who are friends with him benefit more than merchants who are strangers to him. The benefit of dealing via the proper un-corrupt channel is normalized to zero. H is the expected number of corrupt deals to be carried out with strangers. $\tilde{W}$ is the anticipated gross wage paid to the officials, $(1 - \tilde{S})$ is the anticipated probability that a stranger will expose his dealings with particular officials, and $E \cdot \tilde{Cut}$

denotes the amount of bribe that merchants expect officials to pay to their supervisors. $\tilde{X}$ is the pre-policy announcement anticipated number of corrupt deals with strangers that each official will have in period four. F' is the number of friendships that each of the other merchants form. $F' = F$ in a symmetric Nash Equilibrium. Appendix A.3 explains why a functional form that was more general was not chosen instead.

A merchant makes F deals with his official-friends. Out of the remaining $T - F$ deals that a merchant has to make, H are corrupt. In expectation, the fraction $\frac{H}{T-F}$ must equal the fraction of corrupt deals among strangers from the point of view of the officials. It is anticipated that each of the N officials will engage in $\tilde{X}$ corrupt deals, out of a universe of $(M - 1)(T - F') + (T - F)$ potential deals. This implies

$$\frac{H}{T - F} = \frac{N\tilde{X}}{(M - 1)(T - F') + (T - F)}. \quad (1.6)$$

A large M assumption gives us

$$\frac{H}{T - F} \approx \frac{N\tilde{X}}{M(T - F')} \quad (1.7)$$

from the previous equation and allows us to assume that merchants individually are too inconsequential to affect the average number of corrupt deals that officials make with merchant-strangers, i.e.,

$$\left. \frac{d\tilde{X}}{dF} \right|_{F'} \approx 0. \quad (1.8)$$

Together, they yield

$$\frac{dH}{dF} = \frac{-N\tilde{X}}{M(T - F')}. \quad (1.9)$$

The first-order condition of Equation 1.4 yields

$$\frac{\partial D}{\partial R} \left(\frac{\partial R}{\partial F} + \frac{\partial R}{\partial H} \cdot \frac{dH}{dF} \right) = \frac{dC}{dF}. \quad (1.10)$$

Plug in Equations 1.5 and 1.9, and substitute $F = F'$ in the Nash Equilibrium to get

$$\frac{\partial D}{\partial R} \left(\lambda_1 + \lambda_2 \cdot \frac{-N\tilde{X}}{M(T - F)} \right) = \frac{dC}{dF}. \quad (1.11)$$

There is a simple monotonic relationship between the number of friends Y that the official has and the number of friends F that the merchant has, since in expectation, friends are equally distributed among all officials. Since there are M merchants and N officials, $MF = NY$ and

$$M \cdot dF = N \cdot dY.$$

1.3.4 Maximization problem for the supervisors

The supervisors that officials report to may also be corrupt and take a cut (*Cut*) of the officials' earnings. It is assumed that when officials maximize over X , they already know whether their supervisor is corrupt and if so, how much they will have to pay him. However, officials only pay this *Cut* if they (the officials) have not been caught for corruption. Enforcement mechanisms for this quid pro quo are assumed to work. They might be non-monetary; in any case they are not examined in any detail. Corrupt supervisors thus choose to maximize the *Cut* amount that they can squeeze out of each of the officials under them; however, the negative wealth effect of increasing *Cut* compels the officials to take more risk. The probability that the officials get caught then increases, preventing the supervisors from getting any *Cut* at all.

If an official under the supervisor is caught and sent to jail, it is assumed that the supervisor merely loses his *Cut* from this particular official – the rest of the supervisor's wealth is untouched. In reality, some officials might squeal on their supervisors, an issue that is examined in Sub-Section 1.5.4. For now however, assume that supervisors are too entrenched to be caught. The maximization problem then becomes

$$\max_{Cut} L(Q(Cut, T)) \quad (1.12)$$

for total monetary income,

$$Q = Mi + \sum_{Ns} Cut \cdot S^X \quad (1.13)$$

with base monetary income Mi , number of officials under a particular supervisor Ns , and diminishing marginal utility of L in Q ($L_Q > 0$ and $L_{QQ} < 0$).

Proposition 1.

$$Cut = -\frac{B}{\ln(S)} > 0 \quad (1.14)$$

Proof. See the Appendix for all proofs. □

Cut is directly proportional to B , the marginal increase to an official's wealth for every additional corrupt deal carried out with a merchant-stranger. Therefore, *Cut* must scale with B to maintain the same marginal incentives for such dealings. Furthermore, $\frac{dCut}{dS} > 0$, i.e., increasing S increases the official-to-supervisor bribe. A supervisor now dares to demand more from officials because the

supervisor knows that it is now less likely that demanding this extra dollar will cause him to lose the entire bribe.

1.3.5 Bounds to prevent corner solutions

For this section it is assumed that merchants in period one are able to correctly predict period two policy outcomes. Hence $\tilde{W} = W$ and $\tilde{S} = S$. Friendship formation decisions are always made with respect to $\tilde{X}$. In contrast, X refers to the eventual realized variable after all policy announcements are made. $\frac{\partial \tilde{X}}{\partial X} = 1$ due to rational expectations.

Neo-classical congestion theory suggests that if there are two roads, A and B, the introduction of a toll on road A causes drivers to switch to road B. If the additional congestion causes the cost of driving on road B to also increase, then the total number of cars on roads A and B decreases. This is true unless we are constrained by the corner solution: nobody is driving on Road A in the first place. Therefore, there will be no effect on Road B, even though the unconstrained first order conditions suggest otherwise. In the framework of the model, a shock that decreases the likelihood that a merchant would deal with an official-stranger should cause the merchant to substitute towards investing in friendship. At the same time, there is less corruption overall since the cost of corruption has increased. Analogously, for this to be always true, corner solutions are assumed away.

Proposition 2. *To prevent corner solutions,*

$$-1 < \frac{M}{N} \cdot \frac{\partial F}{\partial \tilde{X}}. \quad (1.15)$$

Satisfying this inequality requires sufficient “curvature” in the merchant’s cost or demand functions.

1.4 Basic results

Proposition 3. *An exogenous increase in the number of expected corrupt deals that officials engage in with merchant strangers, $\tilde{X}$, reduces the number of friendships formed by merchants, F .*

$$\frac{\partial F}{\partial \tilde{X}} = \frac{\frac{\gamma \lambda_2 N}{M} \cdot \frac{\partial^2 D}{\partial R^2} - \frac{N \lambda_2}{M(T-F)} \frac{\partial D}{\partial R}}{\frac{\partial^2 C}{\partial F^2} - \gamma \lambda_1 \frac{\partial^2 D}{\partial R^2} + \frac{N \lambda_2 \tilde{X}}{M(T-F)^2} \frac{\partial D}{\partial R}} < 0. \quad (1.16)$$

The marginal change in F is larger if the 2nd derivatives on the right-hand side are smaller in

magnitude ($\frac{\partial^2 C}{\partial F^2} > 0$ and $\frac{\partial^2 D}{\partial R^2} < 0$); the 2nd derivatives act as “brakes” on the cost and demand sides to prevent a huge merchant response to a change in the risk-preferences of the officials.

Proposition 2 was proven using the assumption that shocks that increase the cost of engagement through the deal-with-strangers channel should reduce the net usage of both channels. Usage shocks are examined next. Given our previous assumptions, if the usage Y of the deal-with-friends channel increases for exogenous reasons, then the usage X of the other channel decreases, but the total usage still increases.

Proposition 4. *An increase in Y , the number of friends that each official has, reduces X , the number of corrupt deals that the official engages with merchant-strangers.*

$$-1 < \frac{dX}{dY} = -\frac{A}{B} < 0. \quad (1.17)$$

This implies

$$0 < \frac{d(X + Y)}{dY} = 1 - \frac{A}{B} < 1. \quad (1.18)$$

As explained earlier, more friendships imply more risk-less wealth from friends, and hence officials engage in fewer risky deals with strangers. The degree of substitution between X and Y is increased if the marginal utility from either channel is more similar. To gain some intuition, assume $A = B$. Then a unit increase in Y implies a unit decrease in X , while G remains the same. As the marginal payoffs from increasing X by 1 relative to the utility loss from getting caught in this new equilibrium are the same as the old equilibrium; Y completely substitutes for X . Imperfect substitutability ($B > A$ assumed) and $\frac{dY}{dF} > 0$ imply $\frac{dX+dY}{dF} > 0$. Hence ceteris paribus, increasing friendships increases corruption.

1.4.1 Externalities

Positive and negative externalities are created when a merchant invests in an additional friendship. The net externality depends on the parameters – this sub-section describes conditions where the private incentives are greater/lesser than the social incentives, from the merchant’s point of view, for friendship creation. We refer here to the social marginal revenue with respect to the aggregate revenue of the society of merchants, and not of broader society.

A merchant’s creation of a friendship with an official imposes two types of effects on other merchants. First, this friendship translates into direct risk-less wealth for the befriended official,

causing the official to be less willing to take risks, since he stands to lose more from each risky deal that he makes. The official is then less willing to accept deals from merchant-strangers, forcing the other merchants to pay higher costs as they have to go through the proper channels instead. In economics terms, this translates to a negative externality. Second, by forming a friendship with an official, the merchant no longer has to compete with other merchants for the limited number of merchant-stranger corrupt deals available. Some other merchant then has access to the corrupt merchant-stranger deal instead of having to make the more expensive un-corrupt payment. The cost saving for the other merchant creates the positive externality.

From Equation 1.11, the private marginal revenue created per friendship is

$$\lambda_1 - \lambda_2 \frac{N\bar{X}}{M(T-F)}. \quad (1.19)$$

In the cases of deals between officials and merchant-strangers, the probability that a given deal is corrupt is $\frac{N\bar{X}}{M(T-F)}$, so the expected revenue is given by $\lambda_2 \frac{N\bar{X}}{M(T-F)}$. Any friendship that results in a corrupt deal with an official-friend which brings a guaranteed revenue of λ_1 . The difference between these two expected revenues yields the private marginal revenue of friendship creation.

Proposition 5. *The social marginal revenue created per friendship is*

$$\lambda_1 - \lambda_2 \frac{A}{B} \quad (1.20)$$

Each additional friendship provides a merchant with an additional revenue λ_1 . At the same time, it causes an official to engage in $\frac{A}{B}$ fewer corrupt deals with strangers. This implies a loss of λ_2 revenue for $\frac{A}{B}$ other merchants. The net social marginal revenue created is thus the difference between these two revenues.

When $A \approx B$, an increase in the number of friends that an official has will lead to a one-for-one decrease in the number of corrupt deals that the official will strike with a stranger. A friendship thus has a strong crowding out effect, implying a huge negative externality, causing private marginal revenues to be larger than social marginal revenues. Conversely, when $A \approx 0, B \rightarrow \infty$, a friendship has no crowding out effect. The total number of corrupt deals with stranger-officials available to merchants remains unchanged, but the number of merchants fighting for them is reduced. Each merchant now has to pay the expensive proper payment for a smaller proportion of his required transactions. This implies a positive externality to friendship creation. Private marginal revenues

are then less than social marginal revenues.

If $N\tilde{X} \approx M(T - F)$, the merchant is merely substituting a definite corrupt-stranger deal with a definite corrupt-friend deal. His private benefit from a friendship is thus less than would be the case if he had a low probability of engaging in a corrupt deal with a stranger ($N\tilde{X} \ll M(T - F)$); as friendships are more valuable when there is more uncertainty over whether a merchant can engage in a corrupt deal when he interacts with a stranger-official.

1.5 Policy

An anti-corruption campaign takes the form of reducing S – making it more likely that a merchant stranger will turn the official in, or by increasing W – causing the official to have more to lose if he is caught. An anticipated and executed anti-corruption campaign reduces the number of corrupt deals that officials are willing to have with merchant-strangers, which in turn reduces the probability that the deal that an merchant engages in with the official-stranger is a corrupt one. This reduction in turn decreases the merchant's expected benefit from dealing with a official-stranger. As a result, merchants choose to practice towards corruption with friends, but the overall level of corruption decreases. The next two propositions satisfy this intuition.

1.5.1 Stick approach (Reducing S)

Proposition 6. *A decrease in S causes a decrease in X and, a correspondingly decrease in total corruption.*

$$\frac{dX}{dS} = -\frac{G}{S(B \cdot G_{PP} + \ln(S)G_P)(B + A\frac{M}{N} \cdot \frac{\partial F}{\partial X})} + \frac{E}{S(\ln(S))^2(1 + \frac{MA}{NB} \cdot \frac{\partial F}{\partial X})} > 0 \quad (1.21)$$

This proposition confirms the intuition that more corrupt deals are taken when it is safer to do so. The marginal effect is larger with bigger $\frac{G_P}{G}$. Each successful additional corrupt deal has a large positive utility effect, and hence it is a risk that officials are more willing to take. A more negative $\frac{G_{PP}}{G}$ implies that the next risk is relatively less worth taking than the previous one. In other words, the marginal utility gained by an official from an additional corrupt deal with a stranger will be decreased, in turn reducing X .

More interestingly, the effect of an increased squealing rate (lower S) on lowering corruption (lower X) is greater when supervisors are corrupt (bigger E). Larger Cut paid to supervisors leads

the sub-ordinate official to take on more corrupt deals with merchant-strangers. With a smaller S and a given Cut , the probability that the official is caught is larger. Therefore the supervisor is less likely to receive the Cut . Hence supervisors will want to encourage their officials to take less risk, which they do by reducing Cut . Reducing Cut in turn increases the guaranteed take-home pay of the officials, so they have a smaller desire for additional pay from corrupt dealings, thereby further reducing both X and total corruption.

1.5.2 Carrot approach (Increasing W)

Proposition 7. *An increase in gross wealth, W causes a decrease in X , and correspondingly a decrease in total corruption.*

$$\frac{dX}{dW} = -\frac{1}{B + A \cdot \frac{M}{N} \cdot \frac{\partial F}{\partial \tilde{X}}} < 0. \quad (1.22)$$

Fewer corrupt deals are undertaken when the potential loss to wealth is greater. This result parallels the Shapiro-Stiglitz (1984) efficiency wage hypothesis. In contrast to the stick approach, a change in official wages W does not affect the supervisor bribe, Cut . This result comes from the additive separability functional form assumption used to compute net wealth.

The denominators of Equations 1.21 and 1.22 hint at a positive feedback effect. An increase in wages (higher W) or increase in the probability that a merchant-stranger squeals on an official (lower S) decreases X immediately (even when the friendship formation term remains fixed). From the merchant's perspective, the opportunities for corruption with stranger-officials decreases, thereby decreasing the expected proportion of deals that will be corrupt out of all the transactions between the merchant and officials who are strangers. Hence, the incentive for merchants to form friendships with officials. This creates not only a higher F , but also a higher Y . The increased friendship formation further increases the wealth of officials (increase P) and makes them even less willing to take risk. Officials are now even less willing to accept corrupt deals from merchant-strangers (a decrease in X leads to a decrease in $\tilde{X}$ due to rational expectations), leading to more friendships formed by merchants (higher F)... This positive feedback loop, stemming from the additional *Guanxi* channel, magnifies the initial effect.

What determines the size of the positive feedback in this proportional control system, given $-1 < \frac{M}{N} \cdot \frac{\partial F}{\partial \tilde{X}} < 0$ and $B > A > 0$? Following Lundberg (2001), the feedback is stronger if $\frac{M}{N} \cdot \frac{\partial F}{\partial \tilde{X}}$ is more negative or $\frac{A}{B}$ is more positive. These make intuitive sense. When an increase in merchant-friendships occurs, officials will decrease the number of corrupt contracts with merchant-

strangers by a bigger degree, if $\frac{A}{B}$ is more positive. This result is caused by a stronger wealth effect, further increasing the incentives of befriending an official. From Proposition 2, a more negative $\frac{M}{N} \cdot \frac{\partial F}{\partial X}$ requires more curvature in the demand/cost functions. These “brakes” to the incentives for friendship formation force the additional friendships formed in each iteration to converge to zero at a faster rate. With $A = B$ and zero second derivatives in the cost/demand functions, it is possible that instability results. A marginal increase in W or decrease in S leads to an infinite decrease in F . At that point the corner solutions are reached.

1.5.3 The perverse effect of an anticipated but un-executed anti-corruption drive

Tirole (1996) uses a dynamic model to argue that a one-shot reduction in corruption has negligible impact, in the long run, because bad collective reputations take a long time to disappear. In this model, a short-term attempt at reducing corruption might not only be ineffective, it might be counter-productive instead.

This sub-section looks at the effects of an anti-corruption campaign that was widely anticipated but did not actually happen. This inconsistency might occur because political parties may campaign to eradicate corruption, but lose the political motivation or capital to do so after attaining office. In commenting on the transience of China’s anti-corruption campaigns in the 1980s, Chen (1994) writes that “disciplinary organs are inclined to wink at incidents that would have brought harsh punishment in the past” (translated by O’Brien & Li (1999)). Alternatively, the original agents for change are bought out or side-lined. More generally, they might be perceived to be less effective than they were expected to be. Even under a rational expectations framework, if the population assigns a 50% probability that there was going to be a tough anti-corruption campaign ($S = 0.2$ for example), and a 50% probability that the status quo were to remain ($S = 0.8$), friendships will be formed based on an anticipated $\tilde{S} = E(S) = 0.5$. Even if the status quo prevailed eventually, this difference ($E(S) = 0.5$ versus actual $S = 0.8$) still represents a mismatch between expectations and reality.

Simplistic intuition would suggest that a short-term anti-corruption campaign might reduce corruption levels in the short run, with no long-run effects. The model suggests an alternative hypothesis.

Proposition 8. *A government promise to crack down on corruption that was believed but not carried out results in more corruption compared to the case in which no promise was made at all.*

If merchants believe that officials will be less forthcoming in engaging strangers with corrupt deals, merchants will invest more heavily in costly long-term friendship formation than they would have otherwise. When the short-term campaign ends, two effects occur. First, officials regain their original confidence to engage with strangers. Second, they continue to benefit from their engagement with their (increased number of) friends. Even though the resultant wealth effect causes them to reduce their engagement with strangers, the total corruption level increases.

In sum, credible short-term policies aimed at reducing corruption have perverse long-term effects. This effect is absent without friendship formation. This aspect of policy in general seems to be under-emphasized in the literature, which dictates that policies must be credible to be effective. Instead, this chapter highlights the peril of credible short-term policies: a society's reaction to a policy change might yield the intended objectives in the short term, but this same reaction might have long-lasting effects such that if the policy is removed after some time, the original reaction might worsen the problem that the policy was trying to solve in the first place⁷.

1.5.4 Effect of bureaucratic size on corruption levels

What happens when the bureaucracy's size increases? In this section, a different argument for expanding bureaucracies independent of the probability of getting caught is examined. In a world without friendships, each official is only willing to have corrupt deals with X merchants to maximize his expected utility. X is only governed by an official's fixed wealth and the fixed probability of being exposed for each corrupt act. Thus, X does not vary with bureaucratic size, i.e., N only enters Equation 1.1 through its effect on Y , and correspondingly F . Setting $F = Y = 0$ implies that $\frac{\partial(X+Y)}{\partial N}|_{Y=0} = 0$. Total corruption per official $X + Y = X$ is independent of the total number of transactions that the official has to process. With more officials, total corruption NX increases proportionately to bureaucratic size, N (for NX less than the total number of transactions that need to be made, MT).

⁷As a further illustration, in Singapore, the SAT was introduced as an additional entry criteria for universities two years ago as a means of reducing the stress over the A level exams. In March 2004, the SAT was dropped. As a result, students who had substituted away from studying for the A levels now found that they were even more unprepared to take an exam that mattered more than they had expected, thereby increasing stress levels instead. Similarly, a new subway station at a housing estate was widely anticipated to reduce traveling times for residents, so residents substituted consumption away from car purchases. The sudden decision not to open the station due to insufficient capacity therefore caused a net increase in traveling time compared to a scenario in which the station was not built at all. However such disastrous short-term policies might be the optimal response of bureaucrats whose performances are measured by the number of new schemes implemented during a short tenure (fast-climbing civil servants given a sampling of different positions) at a particular position. The unceremonious termination of policies implemented by one's predecessor might be an institutionally promoted response in a bureaucracy with limited funds and other resources.

How does the introduction of friendships change the effect of a larger bureaucracy? Start with a certain stock of friendships generated by the merchants. When N increases, this stock of friendships is split up among more officials. The number of friends, and consequently the income (apart from income stemming from corrupt deals with strangers) of each official is reduced, making each official more willing to take risks. Each official is now willing to deal with more strangers-merchants than before. Together with an increase in the total number of officials, the probability that merchants will be successful in engaging in corrupt deals with each stranger-official increases, reducing the incentives to form friendships. The same positive feedback highlighted in Section 1.5.2 also applies here, magnifying the original effect of the increase in N . Increasing bureaucracy size thus leads to a less than proportional increase in the amount of corruption in the society. The next few paragraphs formalize this intuition.

Proposition 9. *The decrease in F caused by an increase in N is given by*

$$\frac{dF}{dN} = \frac{\frac{\dot{X}}{N} \frac{\partial F}{\partial \dot{X}}}{1 + \frac{M}{N} \cdot \frac{\partial F}{\partial \dot{X}} \cdot \frac{A}{B}} < 0, \quad (1.23)$$

Again, the denominator shows the same positive feedback as before. The effects of N on corruption levels per official is given by

$$\frac{d(X + Y)}{dN} = \frac{\partial(X + Y)}{\partial Y} \cdot \frac{dY}{dN} = \frac{\partial(X + Y)}{\partial Y} \cdot \left(\frac{M}{N} \cdot \frac{dF}{dN} - \frac{MF}{N^2} \right) < 0. \quad (1.24)$$

The product rule nature of $\frac{dY}{dN}$ yields further insight. Increasing N affects Y via two channels: Incentives to form friendships are curtailed (the $\frac{dF}{dN}$ term), and holding F fixed, the amount of Y per official decreases because MF is now spread out among more N .

Hence, the total amount of corruption rises less than proportionally to the increase in the number of corrupt officials. The number of friendships formed drops, reducing the *uncatchable* type of corruption, Y . This reduction is equivalent to a decrease in the guaranteed wealth of an official, leading him to be more risk-taking, increasing the amount of *catchable* corruption, X . However this increase in corruption does not completely offset the decrease in the amount of *uncatchable* corruption. For illustrative purposes, the graphs of X, Y and $(X + Y)$ against N (for a particular functional form and parameters satisfying all the above conditions⁸) are shown in Figure 1-1. Lines without/with '+' indicate corruption levels assuming an honest/corrupt supervisor

⁸ $G(P) = P^\alpha, C(F) = F^2, W = 5, A = 1, B = 10, \alpha = .8, S = .8, M = 100, T = 3, \lambda_1 = 5$ and $\lambda_2 = 2$

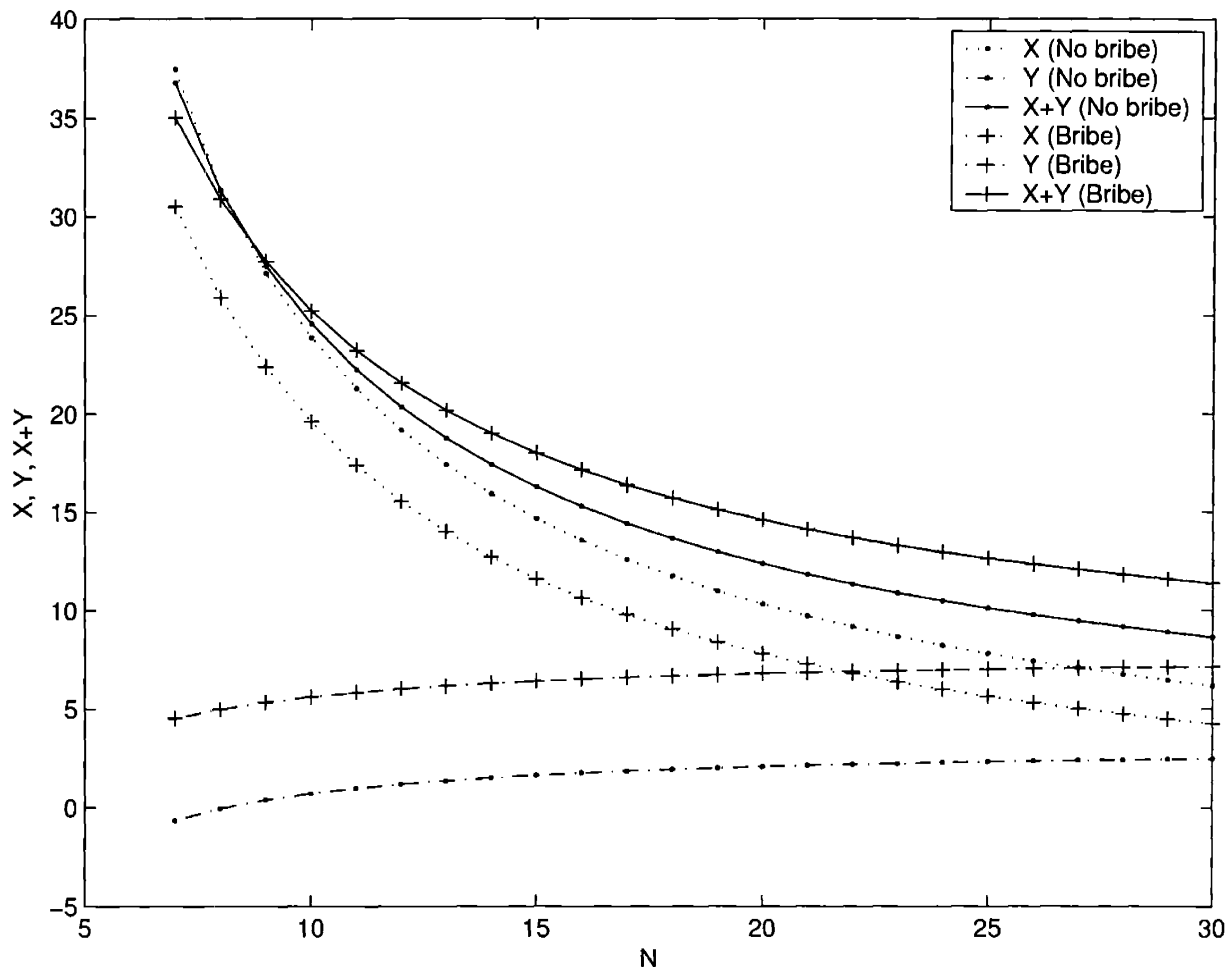


Figure 1-1: Graph of X , Y and $(X + Y)$ against N

($E = 0, 1$ respectively). Regardless of the absence or presence of supervisor corruption, the graphs confirm that an increase in N leads to an increase in X and a decrease in both Y and $(X + Y)$.

Therefore, with friendship formation, an increase in N leads to a less than directly proportional increase in $N(X + Y)$, the total amount of corruption. The policy implication is that the trade-offs involved in bureaucracy design must take friendship formation into account. Otherwise the cost-benefit tradeoffs involved in bureaucratic size design problems are distorted, and bureaucracies will be too small. In fact, surprisingly, in some circumstances increasing N might actually decrease total corruption $N(X + Y)$. Figure 1-2 plots the graph of $N(X + Y)$ against N for a given functional form assumption and parameter values⁹. With an honest bureaucracy, the value of N that minimizes corruption is larger than zero.

⁹Same parameters and functions described in the previous footnote.

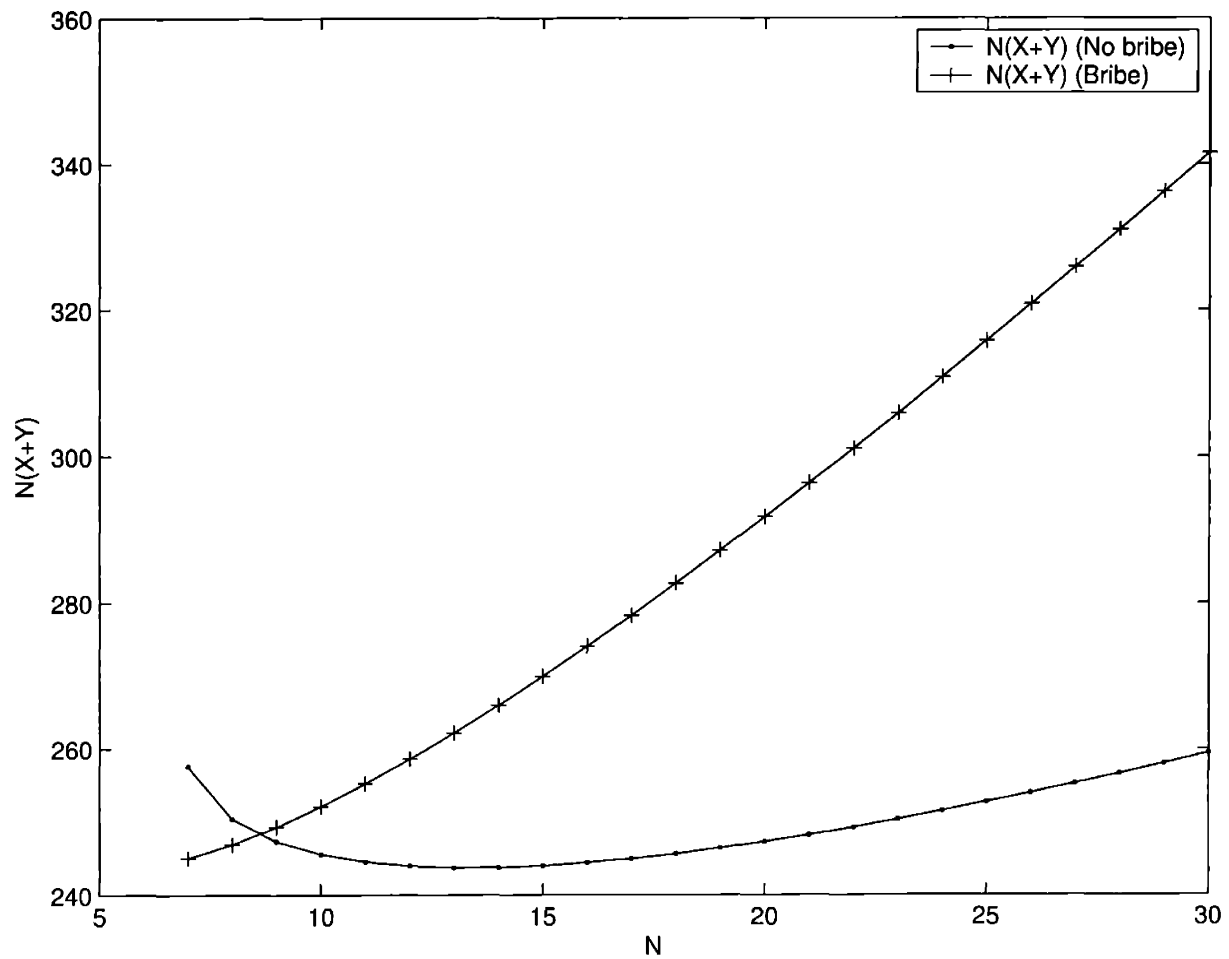


Figure 1-2: Graph of $N(X + Y)$ against N

Furthermore, this result suggests that proportionately more corruption is carried out among friends than strangers in small countries as compared to large countries (Venezuela and Argentina compared to the United States); and in local governments than national governments (in America, progressive machine politics in Chicago compared to the Federal government; in China, local communist bureaus compared to the central bureau in Beijing). While this implication can be trivially explained by the fact that the cost of befriending an official is lesser in a small community (perhaps the matching technology is better – potential businessmen are more likely to meet potential government officials if they all went to the same school, and the same government officials are more likely stay in the same locality and hence be useful to businessmen if the country is smaller). However, the model offers a deeper explanation for this phenomena.

1.5.5 Optimal bureaucracy size

To a certain extent, the size of a bureaucracy is an endogenous choice variable of policy makers. Different motivations affect preferences controlling the number of officials, in view of the above-mentioned results.

First, assume that N is set by agents who are completely dedicated to reducing corruption and have no access to bribes. In this case, optimal N is not necessarily 0. This result might be true regardless of whether supervisors are corrupt or not.

Second, N could be set by a corrupt supervisor who wants to maximize his net Cut from the officials under his charge. As seen in Proposition 1, Cut does not vary with N . Hence, if the supervisor is beyond prosecution (he is too high in the political hierarchy to be caught, but is unwilling/unable to protect the officials under him), then increasing N linearly increases $N \cdot Cut$. Optimal N approaches infinity. The supervisor wants to get the same expected Cut from as many officials under him as possible. Eventually, however, with a huge N , each official only handles such a small number of transactions that he engages in a corrupt deal with every single merchant-stranger. Increasing N just further reduces the official's wealth, until it constrains him from paying his Cut , thereby placing an upper bound on N .

Alternatively, the corrupt supervisor can be thrown into jail too together with the corrupt officials if he is caught. Assume that there is a finite non-zero probability that officials report their supervisor to the anti-corruption agency. In this case, the choice number of officials that a corrupt supervisor wants to demand bribes from is an isomorphic problem to the choice number of merchants that a corrupt official wants to demand bribes from. Instead of a marginal increase of B

for each successful bribe from a merchant stranger, the expected marginal wealth increase, *Profit*, for each successful bribe from an official is

$$Profit = Cut \cdot S^X \quad (1.25)$$

as S^X is the probability that the official is not caught and is thus actually going to pay the bribe to his supervisor. In this case, N is affected by the same comparative statics discussed earlier for X .

Third, N could be set by a political party. Imagine the following scenario: Each politician has jurisdiction over a certain region/department of officials. The politician is part of a larger political party. Voters vote the entire political party into office, so collectively politicians value a smaller total amount of corruption across all regions, as higher voter dissatisfaction increases the likelihood that the party will be overthrown. A politician has a utility function that increases in his total bribes and decreases in the probability of getting kicked out of office. As seen earlier in this sub-section, a larger N increases the total amount of bribes that an individual supervisor can receive. In large political parties, the social cost of a large N (through higher total corruption) is not completely internalized by individual politicians. Thus if politicians were left on their own, N would be too large relative to the social optimum for the party. Perhaps this externality partially explains why politicians have severe restrictions over the size of their staff. The political-party optimal N is still necessarily larger than the minimum point on the $N(X + Y)$ graph because the collective political party also places some value on more bribes for each of its members. This large-political-party assumption admits the simplifying assumption that *Cut* is set solely with the objective of maximizing bribes. The individual politician's utility loss from increasing the probability that the party is removed via his action of increasing X through *Cut* is miniscule and ignored by politicians when they individually optimize over un-monitorable *Cut*.

1.5.6 Effect of corruption among supervisors

The presence of a corrupt supervisor taking a bribe from his official is just like a reduction in the wage W of the official. If supervisors take the bribe and the public realizes it, then the presence of the bribe increases total corruption. However, if an honest supervisor ($E = 0$) is able to manipulate

beliefs such that anticipated ($\tilde{E} = 1$), then

$$\frac{d(X + Y)}{d\tilde{E}} = \frac{\partial(X + Y)}{\partial Y} \cdot \frac{\partial Y}{\partial F} \cdot \frac{\partial F}{\partial \tilde{X}} \cdot \frac{d\tilde{X}}{d\tilde{E}} < 0, \quad (1.26)$$

holding actual E constant (and unobserved), i.e., total corruption can be reduced if governments can cause merchants to perceive that $\tilde{E}$ is big.

With friendship formation, governments that want to minimize net corruption will want to pretend that they are actually more corrupt than they actually are. Private merchants are thus led to believe that officials have to pay more bribes to their supervisors (larger perceived Cut), reducing the net wealth of officials. As a result, officials will be perceived to be more willing to take on corrupt deals with strangers (increased $\tilde{X}$), hence reducing the incentives for friendship formation (lower F). If supervisors are actually *not* corrupt, a situation of large actual net wealth and small numbers of friendships is obtained. Both these factors reduce merchant-to-official corruption. Conversely, the very act of projecting a corruption-free image (for vote-getting political economy reasons) creates more corruption. Differences between perception and reality are clearly possible and manipulable. Oldenburg (1987) finds that perceptions of corruption differed from actual official corruption during the land consolidation program in India; the middlemen pocketed the money that was supposedly going to the government – they had the incentive and ability to credibly distort perceptions.

While it is relatively obvious that governments will want to exaggerate the number of corrupt officials to reduce corruption, it is less obvious that governments will also want to exaggerate the number of corrupt supervisors in an attempt to reduce merchant-to-official corruption. Conversely, a society with high-ranking supervisors that are perceived to be clean but are actually corrupt, leads to more corruption than a society where corrupt supervisors are perceived to be corrupt, at least in the short run before perceptions catch up with reality.

1.6 Conclusion

This chapter emphasizes the need to be sensitive to institutional setups and anthropology in formulating policy in general. Institutional factors may increase the efficacy of policies through positive feedback loops, and short-term policies may be perverted in the face of long-term reactions. Decreasing the number of agents causing a problem (the officials) may increase the total magnitude of the problem if there are sufficiently strong substitution dynamics, and unfair perceptions of poor performance might actually improve performance. The endogenous incentives of bureaucratic

growth are also discussed.

Returning to the narrower question of the effects of *Guanxi* on corruption, one wonders, with the modernization and formalization of the Chinese and Russian (and many other countries') legal institutions, will *Guanxi* face a "declining significance," as Guthrie (1998) claims? This author subscribes to the opposite view; that *Guanxi* will find a way to remain significant in the future. Yang (2002) writes,

Looking back on the years since I studied *Guanxi*, there have been many new uses: obtaining passports and exit permits to leave the country, finding job opportunities with the decline of state job assignments and unemployment, linking up with relatives overseas for business and emigration, locating sources for loans to finance a new economic venture or purchase a home, and attracting overseas Chinese investors, to name just a few in urban contexts.

A growing economic pie also implies greater opportunities for corruption, as the "corruption of today often results from direct economic activities of corrupt government offices" (Gong, 1997). Potter (2002) suggests that *Guanxi* is expected to play an increasing role as China moves towards a legal system that is based on common law instead of codified civil law. Furthermore, Wong (2004) concludes that the minimal consideration given to *Guanxi* in the drafting of the PRC Contract Law suggests a distinct possibility that *Guanxi* will continue to co-exist alongside more formal laws. A culture of informal contacts between businessmen will naturally spill over to businessmen-official contacts.

Furthermore, modernization magnifies the potential damage that such connections might have. In commenting on Ledeneva (2000), Yang (2002) describes the devastating effect of *blat*, the Russian equivalent of *Guanxi*:

Where once *blat* was functional as a way to make the austere state command economy more reasonable for ordinary people, where it was based on personal ethics, and where *blat's* damage to social equity was limited by its modest goals of personal consumption, today, the profit motive and monetary calculations in *blat*-corruption practices, and its linking of the business and official worlds and the criminal underworld magnifies the scale of its destruction to Russian society as a whole.

Understanding these connections and how their effects are shaped by institutional factors like bureaucratic hierarchies may play an even more crucial role in informing future policy debates.

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Chapter 2

Jumping on the Political Bandwagon: A policy consistency explanation

2.1 Introduction

“Jumping on the political bandwagon” – voting for a particular candidate because he is expected to win – is a well-documented and important¹ phenomenon in the political science literature. This effect has been found both experimentally² and empirically³. Many reasons, including psychology, information, and pork-barrel politics, have been proposed to explain the bandwagon effect. However, these “theories are imprecise and underdeveloped” (Kenny & Rice, 1994).

This chapter proposes an alternative political-economy model. Voters care about a candidate’s ability and ideological alignment. If voters only care about these two factors (their “true preferences”), then each election yields a unique equilibrium. In addition, however, voters might want to attract foreign direct investments because of the resultant multiplier job creation and capital injection. Investors consider expropriation risks, including those due to changes in government policies, when they make investment decisions. After all, governments face dynamic inconsistency problems. Ex post nationalization and heavy taxation of foreign-owned sunk assets (such as factories and railways) bring immediate wealth. Ex post re-negotiation of country-customized specific investments made by firms in areas like defense might give the host country more bargaining power, and hence

¹Simon (1954) hints at the need to correct for bandwagon and underdog effects when making predictions.

²See, for example, Navazio (1977), Fleitas (1971), Marsh (1985).

³Nadeau, Niemi and Amato (1994), and Allister and Studlar (1991) find it in the British General Elections. Skalaban (1988) finds it in the United States 1980 Presidential Elections, while Robinson (1932) highlights the folklore of winning the Maine Presidential primaries for similar reasons.

in the process achieving a better deal. Ex ante, however, such policies deter foreign investment. Often, a politician's honor is the only guarantee against ex post policy changes, especially in countries with weaker avenues for independent legal recourse. Ceteris paribus, compared to incumbents, newly-formed governments are less bound by the policies and promises made by prior governments. They are less credibility-constrained.

Given the above environment, investors are more likely to invest in countries where politicians enjoy broad-based support, because the politician is more likely to win subsequent elections, increasing the probability of policy consistency. If voters value such investments, they will then not simply vote for the politician that they like best. Instead, the effects of their votes on investment biases them to vote for the party/politician that they believe will eventually win, so as to promote an aura of future stability. Over-voting is defined here as the additional support that the winning party gets due to this bias.

With rational expectations, investors discount such behavior. However, voters are still biased to vote for the expected winner in spite of the lowered payoffs. In this paper, we investigate the resultant equilibria, and examine the determinants of over-voting. Over-voting creates equilibria in which less able candidates win for a bounded range of relative abilities between the two candidates. The existence of multiple equilibria is a direct consequence of voting for the expected winning candidate. The model suggests that more over-voting occurs in countries with lower levels of economic freedom, resulting in a wider range of relative candidate abilities that admit multiple equilibria. Hence countries with fewer independent safeguards against expropriation might have political competition processes that are inherently less efficient in selecting the most able candidates for the job.

This result is surprising because one might expect the comparative statics to go in the other direction. Politicians matter more in countries with centralized governance structures due to the concentration of power in their hands. Voters thus have greater incentives to select the best candidate (participate in discussions, pro-actively search for information, etc.), making it more likely that the political process causes the better candidate to win. The mechanism suggested in this chapter suggests that this conventional intuition is not necessarily correct.

We then report a simple regression on post Communist transition economies suggesting that economic freedoms are negatively correlated with winning margins during parliamentary elections. This provides initial support for the comparative static that more over-voting occurs when there are less economic freedoms.

This chapter contributes to the broader literature on political competition. Periodic elections introduce the possibility of turnover. On the one hand, this competition allows societies to correct mistakes when they choose their politicians, select more able candidates when they appear, and reduce the politician's moral hazard to under-perform and steal. On the other hand, the inability of current politicians to write contracts governing the conduct of future politicians introduces inefficiencies. If the preferences of future policy makers are unknown to or different from current politicians, deficits are too large (Persson & Svensson, 1989 and Alesina & Tabellini, 1990), the degrees of legislative control over regulatory agencies are skewed (McNollgast, 1999), and public investments are not made (Besley & Coate, 1998). Leblanc et al. (2000) offer an alternative explanation for the under-investment in public goods, attributing it to the nature of majority-rule distributive politics. In this chapter, we assume that too little private investment occurs because current administrations are unable to commit against future expropriation. Equivalently, current policy makers have to give costly near-term sweeteners to induce private investment. In addition, while the literature cited in this paragraph looks at the responses of policy makers vis-à-vis each other in the face of commitment problems, this chapter focuses on the reactions of private agents (voters and investors) within a model that explicitly incorporates investment and elections.

The rest of the chapter proceeds as follows. Section 2.2 reviews the literature, while Section 2.3 sets up the model. Regressions in Section 2.4 and anecdotes in Section 2.5 provide empirical support. Section 2.6 concludes.

2.2 Literature review

Various explanations have been used to account for the observed Bandwagon phenomenon. However, these come mainly from the American context. Kenney & Rice (1994) look at the psychological reasons behind the Bandwagon effect in the US 1988 Presidential primaries, such as getting “caught up in the excitement of the moment” and the favorable “feeling of supporting the favorite,” or voting for “the one with the best chance” to win the general elections. In addition to backing the front-runner, the media might generate momentum and excitement that aid the psychological processes (Bartels, 1998). The front-runner might be given (or have more donations to buy) more air time, helping him communicate his positions and personality to the electorate. Alvarez (1999) writes that people are generally less likely to vote for someone about whom they are unsure. More media exposure bridges the information asymmetry gap. Further emphasizing this imperfect infor-

mation point of view, McKelvey & Ordeshook develop models where polls and endorsement data reveal information (1985a), and examine sequential selection with limited information (1985b). Fey (1997) looks at pre-election polls as information and coordination devices.

In a different light, Straffin (1977), and Brams & Riker (1972) develop theoretical models of Bandwagon effects among voting blocs competing for uncommitted voters to increase their strategic power. However, this primarily applies to legislators, and scarcely applies to the multitude of voters in a general election.

2.3 The model

2.3.1 Setting up

The structure of the repeated election game (one election in each time period) is as follows. The exogenous qualities of the two candidates contesting election t are revealed. Election t then occurs with all voters simultaneously casting their votes. The winner and vote shares of election t are announced; investors from period $t - 1$ discover if their $t - 1$ period investments are expropriated, while period t investors decide investment levels based on the election results and immediately invest. The cycle repeats itself, with $t + 1$ replacing t .

The following assumptions are made. After every election, investors evaluate the results of the previous election and the incentives that the new politician offers to entice investments. The investments made now only yield returns after the next election, for exactly one period. Including more periods complicates the math but does not provide more insight. A re-elected politician (for election $t + 1$) is committed to maintain the contracts (tax reliefs, etc) made with investors in period t for credibility reasons. Anyone else elected in $t + 1$ might renege on the promises that the previous politician made; the extent that he is free to do so exogenously depends on the formal institutions of the country. Stronger institutions better guarantee investor rights across administrations and thus increases the probability, D , that investments are secure. Only two candidates compete in each election, the incumbent and a challenger. The politician that gets more than half of the votes wins.

Let $\beta_{i,t}$ and $\beta_{c,t}$ denote the incumbent's and challenger's perceived ability (by the population) respectively in period t . Denote the person who wins/loses the election t as the winner/loser for election t . The election t winner becomes the incumbent for election $t + 1$; his new opponent in election $t + 1$ is then the election $t + 1$ challenger. Subscripts i, c, w and l refer to the incumbent,

challenger, winner and loser respectively.

Since the election $t + 1$ incumbent won election t ,

$$\beta_{i,t+1} = \beta_{w,t}. \quad (2.1)$$

The election $t + 1$ challenger does not have to be the election t loser, but is the best competitor available from the non-incumbent's party. Assume that the expected ability of the best competitor in the next period is the ability of the loser this period, and the two abilities are related by

$$\beta_{c,t+1} = \beta_{l,t} + \epsilon. \quad (2.2)$$

For analytical tractability, let ϵ be uniformly distributed between $[-c/2, c/2]$. Alternatively, the election t winner's perceived ability ($\beta_{w,t}$), could differ from his perceived ability ($\beta_{i,t+1}$) at election $t + 1$ if the population receives new information about him during his term in office. Such shocks can be folded into the ϵ term.

Proposition 10. *Given $\beta_{w,t}$ and $\beta_{l,t}$, the probability of the incumbent having a higher perceived ability than the competitor for election $t + 1$ is*

$$\Pr(\beta_{i,t+1} > \beta_{c,t+1}) = \frac{\beta_{w,t} - \beta_{l,t}}{c} + \frac{1}{2} \quad (2.3)$$

if $c > 2|\beta_{w,t} - \beta_{l,t}|$.

Higher relative ability today ($\beta_{w,t} - \beta_{l,t}$) predicts higher relative ability tomorrow, with predictive power decreasing in the uncertainty c of the competitor's ability.

2.3.2 Maximization problem for investors

Let $\alpha_{p,t}$ denote the vote share of Candidate p for election t . Immediately after election t , investors maximize expected returns given election t 's results (specifically based on $\alpha_{w,t}$, the winning percentage of the election t winner)

$$\max_I E\left\{\Pr(\alpha_{i,t+1} > \frac{1}{2}) + (1 - \Pr(\alpha_{i,t+1} > \frac{1}{2}))D|\alpha_{w,t}\right\}kI - \frac{1}{2}I^2 \quad (2.4)$$

where k is the earnings on an un-expropriated investment, and D is the probability that investments are not expropriated if a new government is formed after the next election. $\Pr(\alpha_{i,t+1} > \frac{1}{2})$ refers to

the probability that the winner of election t (the incumbent going into election $t + 1$) wins election $t + 1$. $\frac{1}{2}I^2$ is the cost of investment. Solving the first order condition yields

$$I(\alpha_{w,t}) = kE\{\Pr(\alpha_{i,t+1} > \frac{1}{2}) + (1 - \Pr(\alpha_{i,t+1} > \frac{1}{2}))D|\alpha_{w,t}\}. \quad (2.5)$$

Investments increase with higher returns k , higher expected probability that the incumbent wins the 2nd election ($\Pr(\alpha_{i,t+1}) > \frac{1}{2}$), and higher investor guarantees D .

2.3.3 Maximization problem for voters

A stylized fact of elections in general is that even if A is more able than B, A does not win 100% of the vote. Clearly voters are heterogeneous. Voter heterogeneity is modeled the same way that Hotelling competition is modeled in the Industrial Organization literature. Assume n voters with personalities distributed uniformly on $\theta = [0, 1]$, with $\theta_1 = 0, \theta_n = 1$. Without loss of generality, a smaller θ corresponds to voters who are more similar to the incumbent, and a larger θ corresponds to those that are more similar to the competitor. Think of this as the traditional Republican versus Democrat heterogeneity.

Voters additively value a more able and ideologically similar politician, and also value more investments. In election t , if voter j does not value investment, the incumbent is valued by an amount $\beta_{i,t} - G\theta_j$, while the competitor is valued by an amount $\beta_{c,t} - G(1 - \theta_j)$, for exogenous $G > 0$. G controls the importance of similarity relative to competence (the β s) in deciding vote proportions. Ceteris paribus, a smaller G implies that ability has a relatively larger effect on vote share. The amount of investment $I(\alpha_{w,t})$ is a function of the vote share of the winner of the t election. Voters attach utility γI to investment I . Ex ante either the election t 's incumbent or the competitor could win election t . Voters vote for the politician that gives them higher utility. Denoting the utility from voting for either politician as $\{Utility_i, Utility_c\}$, voters choose between

$$\{\beta_{i,t} - G\theta_j + \gamma I(\alpha_{i,t}), \beta_{c,t} - G(1 - \theta_j) + \gamma I(\alpha_{i,t} - \frac{1}{n})\} \quad (2.6)$$

if they believe that the incumbent will win ($\alpha_{w,t} = \alpha_{i,t}$), or choose between

$$\{\beta_{i,t} - G\theta_j + \gamma I(\alpha_{c,t} - \frac{1}{n}), \beta_{c,t} - G(1 - \theta_j) + \gamma I(\alpha_{c,t})\} \quad (2.7)$$

if they believe that the competitor will win ($\alpha_{w,t} = \alpha_{c,t}$).

Given that voting considerations are dependent on voter expectations, we define a voter-level equilibrium as one where the beliefs of voters (regarding the winner of the election) lead them to vote in such a way that results in an outcome that is consistent with their beliefs. A game-level equilibrium is one that incorporates investors who fully anticipate over-voting and correct for it, with voters who realize that investors are correcting for over-voting, and with voter-level equilibrium existing as defined above. Pure strategy occurs when voters believe with certainty that a particular politician will win and hence vote in a way that reflects these beliefs; a mixed strategy occurs when voters believe that a particular politician will win with a probability that is strictly between 0 and 1.

2.3.4 Equilibria when the Investors do not expect over-voting

This sub-section solves the model assuming that investors do not think that voters value investments ($\gamma = 0$) when they make their voting decisions. In other words, voter-level, but not game-level, equilibria exist. Immediately after the election t , the investor does not know the results of election $t + 1$, but can only predict it from the results of election t . As an outsider, he is unable to observe any of the β s directly.

Proposition 11. *If $\gamma = 0$, or investors believe that $\gamma = 0$, investment*

$$I(\alpha_{w,t}) = k \left(\frac{G(2\alpha_{w,t} - 1)}{c} \right) (1 - D) + I', \quad (2.8)$$

where $I' = \frac{k(1+D)}{2}$ corresponds to the amount of investment when the winning vote-share is 50% and assuming that the β s are not too different, i.e., $|\beta_{i,t} - \beta_{c,t}| < G$.

Higher k implies that the rate of return for an un-expropriated investment is higher, and hence both base investment I' , and the effect of a change in perception about government stability on investment, are higher. A higher D implies that investments are secure independent of the change of government, making information about the winning candidate for future elections less important for the investor. However, a higher D increases the base level of investment I' because holding vote share constant, a country with higher investor protection attracts more investment. Higher uncertainty (larger c) implies that the current election yields less information about future elections, so investors place a smaller weight on the information revealed in this period. Larger G implies that ideological preferences are more important than relative abilities. Therefore a given shock

to relative ability in future elections matters less, implying that information from this particular election is less revealing, and hence makes a smaller difference to I .

If, in reality, voters do not over-vote, then $\alpha_{w,t}$ is an accurate representation of ability and voter heterogeneity, and is uniquely determined for each $\beta_{i,t} - \beta_{c,t}$ – i.e., only one unique equilibrium exists. In addition, the I given above is the rational response of investors. If, instead, voters over-vote (because a higher I gives utility to the voter, $\gamma > 0$ in reality) without investors recognizing and correcting for it, then investors will maintain the same rule but voters will end up over-voting instead.

Proposition 12. *The amount of over-voting that occurs, regardless of whether the more able candidate wins or not, in a voter pure strategy equilibrium when investors are oblivious to over-voting, is*

$$OV = \frac{\gamma k(1-D)}{nc}, \quad (2.9)$$

for $-Fl \leq \beta_{w,t} - \beta_{l,t} \leq Rf$, where

$$Rf = G - \frac{2G\gamma k(1-D)}{nc} \quad (2.10)$$

and

$$Fl = \frac{2G\gamma k(1-D)}{nc}. \quad (2.11)$$

If $\gamma = 0$, voters do not get any utility from investment, and hence there is no vote-inflation, i.e., $OV = 0$. A larger γ corresponds to voters caring about investments more, and hence placing a greater value on over-voting. Larger c implies greater uncertainty over the ability of candidates in the following period. Investors thus use the information from previous elections less when making investment decisions. Therefore, voters over-vote less. When D is large, investments are secured regardless of the winner of the next election. Voters see less utility in over-voting, and thus do less of it.

Furthermore, changing perceptions of government expropriation lead to more investments when k , the un-expropriated return to investments, is higher. Voters thus have a greater incentive to change this perception, and hence over-vote more. k might be higher during recessions, implying more over-voting during recessionary times. With less people (n) voting, each voter's marginal effect on investments is greater, so they have a greater incentive to over-vote⁴. This predicts that

⁴This model does not deal with the critique facing most voting models – that n is approximately infinite for most

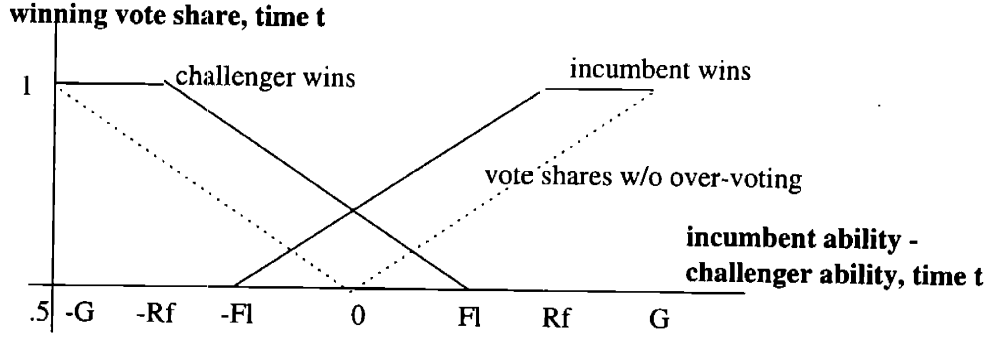


Figure 2-1: Plot of relative vote shares for various relative β s

over-voting is a small-country and local-election phenomenon. Of course, γ and k are all dependent on country size too, and hence the overall effects of size on OV are ambiguous. Finally, the amount of over-voting is independent of how strongly people feel about party affiliations (G) as opposed to candidate ability, as voters are trying to affect beliefs over β and not θ . For $G > |\beta_{i,t} - \beta_{c,t}| > Rf$, over-voting is capped when the winning vote share reaches 100%. The additional support from over-voting is still insufficient for the weaker candidate to attain a 50% vote share if $Rf > |\beta_{i,t} - \beta_{c,t}| > Fl$. Multiple equilibria in which either party can win occurs when $Fl > |\beta_{i,t} - \beta_{c,t}| > 0$. Note that multiple equilibria is absent without over-voting.

Proposition 13. *In a voter mixed strategy equilibria, when investors are oblivious to over-voting, both candidates get 50% of the vote. A mixing equilibria exists only if*

$$|\beta_i - \beta_c| < \frac{2\gamma k G(1 - D)}{nc} = Fl. \quad (2.12)$$

Mixed strategy equilibria can only occur in the range that has multiple pure strategy equilibria. A mixed strategy equilibrium is very fragile and will not be emphasized in the rest of the paper. Figure 2-1 consolidates the voter-level sub-game equilibria discussed above. The dotted line shows what investors expect the mapping between winning vote-share and relative abilities to be, if they do not factor in over-voting. The solid lines denote the two pure strategy equilibria with over-voting.

elections. Hence, each individual vote does not affect either the choice of the winner, or investment levels in the context of this model, making it irrational to vote at all.

2.3.5 Equilibria when Investors expect over-voting

When investors expect voters to over-vote, they understand that there exists equilibria where the less able politicians win. This alters the dependence of vote-share on their investments, and changes the incentives for voters to over-vote, consequently affecting the presence of multiple equilibria in the first place. This sub-section shows that for a broad class of selection rules governing the choice of a particular equilibrium out of multiple equilibria, the signs of the comparative statics of the amount of over-voting as a function of various exogenous parameters remain the same in a game-level equilibrium (as previously defined) as in the case of a voter-level equilibrium.

Figure 2-1 shows that for $\beta_{i,t} - \beta_{c,t}$ between $[-Fl, Fl]$, multiple equilibria exist. Among pure strategies, voters can coordinate and vote with the self-fulfilling expectation that the incumbent wins, or the challenger wins. Alternatively the median voter can employ a mixed strategy with non-zero probabilities that either candidate (barely) wins. The amount of over-voting that occurs depends on alternative rules of equilibrium choice among multiple equilibria.

Assume that for each $\beta_{i,t} - \beta_{c,t}$, there is a certain probability that a particular equilibrium is chosen. Let $\Phi(\beta_{i,t} - \beta_{c,t})$ be the probability that the incumbent wins, $(1 - \Phi(\beta_{i,t} - \beta_{c,t}))$ be the probability that challenger wins, and mixed strategies occur with probability 0. When only one equilibrium exists, $\Phi(\beta_{i,t} - \beta_{c,t}) = 1$ for $\beta_{i,t} > \beta_{c,t}$, and 0 otherwise.

Proposition 14. *The amount of over-voting under a probabilistic selection rule, when investors expect over-voting (game level equilibrium), is*

$$OV = \frac{\gamma k(1-D)}{nc} E(\Phi(\beta_{i,t} - \beta_{c,t} + \frac{c}{2}) - \Phi(\beta_{i,t} - \beta_{c,t} - \frac{c}{2})). \quad (2.13)$$

Note that under this more general selection rule, the comparative statics described after Proposition 12 still hold, as the previous intuition still carries through. However, the amount of over-voting is smaller when $(\Phi(\beta_{i,t} - \beta_{c,t} + \frac{c}{2}) - \Phi(\beta_{i,t} - \beta_{c,t} - \frac{c}{2})) < 1$. Voting shares are less consequential to investment decisions when candidates with lower abilities can win.

As a special case, if voters over-vote for the incumbent if and only if $\beta_{i,t} > \beta_{c,t}$ (i.e., $\Phi(\beta_{i,t} - \beta_{c,t}) = 1$ for $\beta_{i,t} > \beta_{c,t}$, and $\Phi(\beta_{i,t} - \beta_{c,t}) = 0$ otherwise), the amount of over-voting is

$$OV = \frac{\gamma k(1-D)}{nc}. \quad (2.14)$$

The amount of over-voting is the same regardless of whether investors expect the over-voting or not.

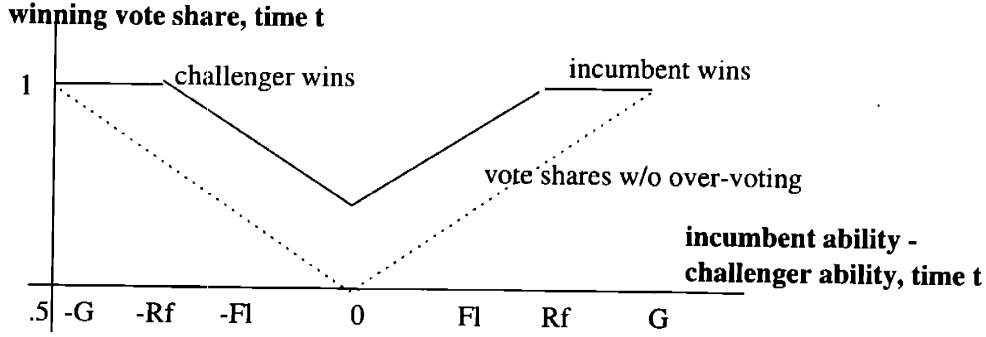


Figure 2-2: Plot of relative vote shares for various relative β s

Figure 2-2 thus shows the vote share of the winner as a function of relative β s for this special case. In between boundary conditions $-Rf < \beta_i - \beta_c < Rf$ (i.e., the difference in candidate ability isn't too great), investors who expect over-voting discount the vote share and invest $I(\alpha_{w,t} - OV)$ instead of $I(\alpha_{w,t})$. Furthermore, since voters only care about their marginal voting effect on investments (which remains the same), over-voting tendencies do not change in response to whether investors expect over-voting. The amount of over-voting for sufficiently different candidate abilities ($Rf < |\beta_{i,t} - \beta_{c,t}| < G$) is discussed later in the chapter.

Returning to the general case, investor expectations affect the payoffs to over-voting for voters, resulting in a feedback loop that affects the amount of over-voting and the range of relative abilities that admit multiple equilibria. Selection rules take many forms: for example they could be biased towards equilibria where the incumbent wins, or they could be neutral to either candidate.

Consider a neutral selection rule where in the presence of multiple equilibria, the higher ability candidate only wins with a probability r that is strictly less than 1, regardless of whether the higher ability candidate is the incumbent or not. Investments are then less conditional on the vote-share for the prior election. Incentives for voters to over-vote in the prior election are mitigated. Given that every election is a prior election for another election, the possibility that the incumbent can win when he is less able is reduced, decreasing the range of $\beta_{i,t} - \beta_{c,t}$ admitting multiple equilibria. Evidently, the results depend on how an equilibrium is chosen out of multiple equilibria.

Let us formalize this intuition. For illustration consider the case where candidates win with probability $0 < r < 1$ whenever they are better than their opponent and multiple equilibria exist. Then by symmetry, let multiple equilibria exist between $[-W, W]$. Then $\Phi(\beta_{i,t} - \beta_{c,t}) = 1$ for $\beta_{i,t} - \beta_{c,t} > W$, $\Phi(\beta_{i,t} - \beta_{c,t}) = r$ for $0 < \beta_{i,t} - \beta_{c,t} < W$, $\Phi(\beta_{i,t} - \beta_{c,t}) = 1 - r$ for $0 > \beta_{i,t} - \beta_{c,t} > -W$, and $\Phi(\beta_{i,t} - \beta_{c,t}) = 0$ for $\beta_{i,t} - \beta_{c,t} < -W$. $-W$ is the minimum $\beta_{i,t} - \beta_{c,t}$ where the incumbent

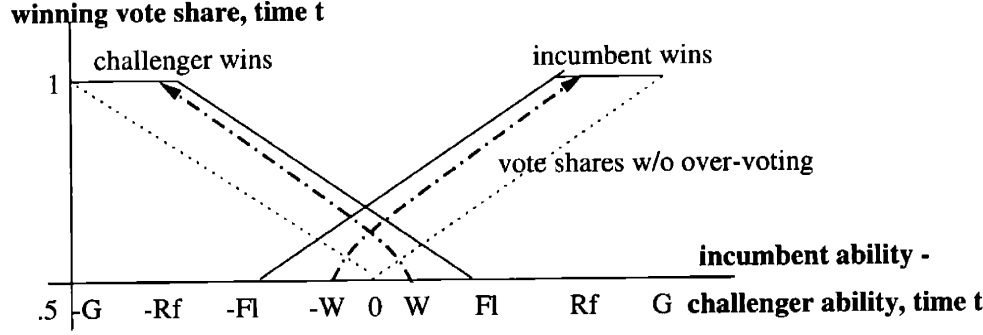


Figure 2-3: Relative vote shares in a rational expectations framework

might possibly win. Any $\beta_{i,t} - \beta_{c,t} < -W$ only has a single equilibrium where the challenger wins.

Proposition 15.

$$W = \frac{2G\gamma k(1-D)}{nc} \Phi\left(-W + \frac{c}{2}\right), \quad (2.15)$$

where $\Phi(-W + \frac{c}{2}) = r$ or $\Phi(-W + \frac{c}{2}) = 1$.

The same factors that cause an increase in over-voting cause an increase in the range in which multiple equilibria exist. This makes perfect sense as it is the over-voting that is responsible for multiple equilibria in the first place. In particular, a smaller D (less economic freedoms) widens the range of relative abilities that make it possible for the weaker candidate to win. As emphasized in the introduction, this counter-intuitively reduces the efficiency of the political competition process in selecting the most able candidate for the job.

Furthermore, higher weights on ideological similarity (larger G) imply that the impact of a larger difference in relative β s does little for vote share, hence admitting a larger range of $\beta_{i,t} - \beta_{c,t}$ with multiple equilibria. More generally, the higher the probability of the natural voter-level sub-game equilibria (the probability that the more liked candidate wins), the larger the range of possible multiple equilibria. Figure 2.3.5 denotes the voting function for general symmetric Φ . The amount of over-voting is less than in the special case, and over-voting slowly increases with higher $\beta_{w,t} - \beta_{l,t}$ due to the monotonically non-decreasing Φ assumption. The range of multiple equilibria is smaller than in the special case, where multiple equilibria in which the incumbent could win even though he is of lower ability than the challenger, exist between $[-Fl, 0]$. However, these equilibria occur with probability 0.

Finally, Φ functions do not have to be symmetric. Instead, they can be biased towards the incumbent (or the challenger). For illustration, consider the case where $\Phi(\beta_{i,t} - \beta_{c,t}) = 1$ if $\beta_{i,t} -$

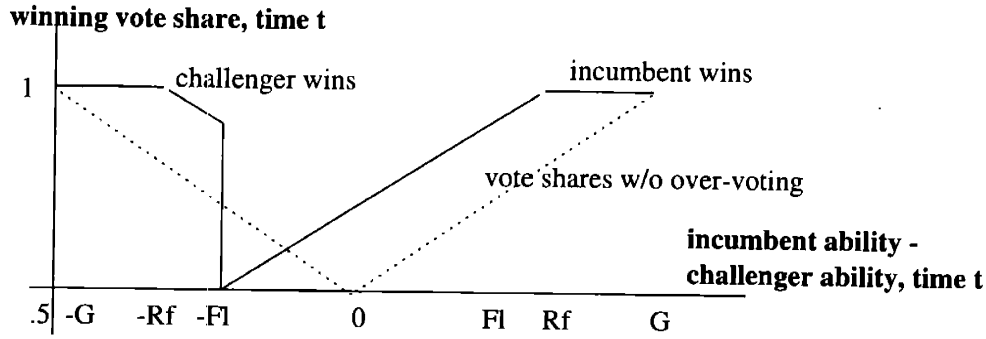


Figure 2-4: Relative vote shares with incumbent bias

$\beta_{c,t} > -W$, and 0 otherwise, i.e., the incumbent wins whenever there are multiple equilibria even if the incumbent is of lower ability. Then from Propositions 14 and 15,

$$-W = -F_l = -\frac{2G\gamma k(1-D)}{nc} \quad (2.16)$$

and

$$OV = \frac{\gamma k(1-D)}{nc}. \quad (2.17)$$

Figure 2.3.5 plots winning share against the incumbent's ability relative to the challenger for this selection rule. It shows that, ceteris paribus, the smaller the value of D , the more the incumbent is likely to win, as he is able to beat a wider range of stronger challengers. Hence, if a country has a selection rule where the incumbent is always favored over the challenger for relative β s that admit multiple equilibria (for example, because of a biased media), then it might be in the interest of the incumbent to take steps to reduce D . A smaller D implies that investors value governmental stability more, leading the population to over-vote for the incumbent more, providing insurance for the incumbent in the face of a stronger candidate. For countries with more neutral media, weak institutions do not provide the incumbent with an advantage. This chapter thus provides a new explanation for endogenously weak institutions.

In the Fundenberg and Tirole (1986) signal-jamming article about firm behavior, entrant firms who are signal-jammed are not fooled because of rational expectations. However, signal jamming has the effect of reducing 1st period profitability, hence deterring entry ex ante. A parallel exists under this over-voting framework.

Many elections have filing fees or signature collections as eligibility requirements. Ansolabehere & Gerber (1994) find that more stringent requirements increase the frequency of un-competed

elections. Over-voting makes it harder for challengers to find members of the public willing sign petitions, deterring entry. If filing fees are high but are conditional on some minimum election vote share, then over-voting might exacerbate the frequency of un-competed elections. In Singapore, a candidate is required to deposit a few thousand dollars ($\approx$ US\$8,000 for a constituency of 20,000 people) when he registers. This deposit is forfeited if he fails to garner at least 15% of the votes during the elections. With over-voting, some candidates might be deterred from competing in the elections because of the likelihood of losing the deposit. For example, if a person would have received 20% of the votes without the over-voting effect, but only 10% of the votes with over-voting, then the person might decide not to compete in the elections at all, giving the incumbent a walkover. Walkovers happen a lot in Singapore. In the last two elections there were contests in less than half of the seats up for election. Without actual competition (the parallel to entry), the investor is unable to accurately deduce the winner's actual support levels. He only knows that the winner has a level of support that results in a winning vote share that is more than 85%. This potentially causes over/under-investment relative to their optimal levels.

2.4 An empirical test

This section tests the model's prediction that there should be greater over-voting in countries with less economic freedom (i.e., smaller D). In practice, presidential elections are very idiosyncratic, so parliamentary elections are examined instead. Winning percentages are used as proxies for the amount of over-voting. Analysis is carried out at the country level as town/country level elections are invariably affected by pork-barrel politics. Furthermore, within-country variations in the risk of expropriation are probably minuscule.

Countries are at different stages of their economic development and have different political systems, introducing many unobserved variables that affect parliamentary vote share. Hence we look at a more homogeneous subset of countries. Rich countries all tend to have established independent legal institutions and have no real desire for foreign investment. Democracies in poor countries are often dysfunctional with little chance of election-induced changes in policy-makers – regime changes in those countries often occur as military coups. The correlation between the likelihood of a military coup and the vote share of the incumbent policy-maker is less obvious. These issues leave us with middle-income countries. Among them, Latin American countries tend to have strong presidents and weak legislatures, nullifying any incentive to over-vote for parliamentary

elections. Presidential terms there are often limited, and presidents have had dubious success with designating successors, preventing policy continuity. We thus choose to focus on the post-Communist transition economies in Europe⁵.

For the dependent variable, data from <http://www.electionworld.org/> gives the percentage of votes (*pvote*) won by the first place winner in the most recent election (all occur between 1999 and 2003). In two-house systems, the lower house is used for analysis.

A couple of variables are used as the independent variables in the regression. The Heritage Foundation's 1998 Index of Economic Freedom⁶ is used as a measure of Economic Freedom, with an overall score between 1 and 5. We invert their counter-intuitive measure by subtracting their index score from 5 to yield our (*ecofree*) measure. With our measure, Estonia has the most freedom with a score of 2.70, while Azerbaijan occupies the other end of the spectrum with a score of 0.7. A larger *ecofree* value now indicates more economic freedom, proxying for a larger D . Larger D implies less over-voting, and thus a smaller vote share. The chapter thus predicts a negative correlation between *pvote* and *ecofree*.

A dummy variable, *same*, is used to correct for the incumbent effect. This variable takes on a value of 1 if the winning party was the incumbent party. The correlation between Economic Freedom and Political and Civil freedoms is partialled out through the use of the *polfree* and *civilfree* indices respectively. These 1998 indices come from the Freedom House⁷. We again invert the numbers such that a higher index value signifies a higher level of political/civil liberties. Summary statistics are given in Table 2.1.

Table 2.2 shows a negative coefficient on the *ecofree* variable, as predicted by the theory, even after correcting for incumbency effects, and political and civil liberties. A decrease in economic freedom by 1 index point leads to an increase in vote share by 11 to 17 percentage points for the winning party, depending on the controls used.

More rigorous empirical tests of the data involve searching for an exogenous shift in D , and the counter-factual vote shares without over-voting. Exogenous shocks to voting rules might change the incentives to over-vote, potentially yielding information on the counter-factual. Alternatively, the utility of the marginal investment, γ , might differ across different voting groups. Capitalists

⁵The countries are Albania, Armenia, Azerbaijan, Bulgaria, Croatia, Czech Republic, Estonia, Georgia, Hungary, Kazakhstan, Kyrgyzstan, Latvia, Lithuania, Moldova, Poland, Romania, Slovakia, Slovenia, Tajikistan and Ukraine. Bosnia and Yugoslavia were dropped because of recent wars, Economic freedom index was unavailable for Macedonia, and voting numbers were unavailable for Belarus, Turkmenistan and Uzbekistan.

⁶<http://cf.heritage.org/index/pastScores.cfm>

⁷www.freedomhouse.org/ratings/

Table 2.1: Summary statistics of the Parliamentary vote share regression

Variable	Obs	Mean	Std. Dev.	Min	Max
<i>pvote</i>	21	36.05	12.25	19.5	64.5
<i>ecofree</i>	21	1.62	.54	.7	2.7
<i>same</i>	21	0.57	.51	0	1
<i>polfree</i>	21	4.14	1.80	1	6
<i>civilfree</i>	21	3.67	1.20	1	5

Table 2.2: OLS Regressions of Parliamentary vote share on Economic Freedom

	(1)	(2)	(3)
<i>ecofree</i>	-11.38* [4.8]	-11.51* [5.2]	-17.72* [7.8]
<i>same</i>	-	-.4060 [4.8]	2.889 [5.1]
<i>polfree</i>	-	-	-.5946 [3.7]
<i>civilfree</i>	-	-	4.963 [6.0]
<i>Constant</i>	54.52* [9.2]	54.97* [10.5]	47.43* [16.1]
<i>n</i>	21	21	21
<i>R</i> ²	.25	.26	.30

Notes — Robust standard errors are in parentheses. * Variables significant at the 5% level.

and workers who have skills that complement capital will value investments more than farmers and workers whose jobs that are substitutes for capital. Voter groups with higher γ will tend to over-vote more. Unfortunately voter groups differ from each other along many other different dimensions concurrently, making it hard to separate the effects of different propensities to over-vote from different policy preferences.

2.5 Some more evidence of the comparative statics

This section is very speculative, as it tries to present (even less rigorous) evidence in support of the model, which predicts that the voting shares of incumbents should increase when faced with a crisis, providing a reason for the rallying-round-the-flag phenomenon. The incumbent politician is not the incumbent due to a random assignment process, but because he is already the current winning politician. Any crisis during his term will initially tend to increase his vote share. Of course, the vote share might decrease with time as the conflict prolongs, as voters might feel that the politician is incompetent (revealing information about his β) and therefore demand change. Evidence of rallying-round-the-flag during various forms of crises (natural disasters, war, and economic crises) are highlighted, together with examples of political fragmentation when the crisis disappears. While the alternative explanations presented in Section 2.2 explain the Bandwagon effect, they do not explain the rallying-round-the-flag phenomenon. In contrast, the proposed explanation in this chapter does both.

In terms of natural disasters, Abney & Hill (1966) look at Hurricane Betsy's effect on the mayoral campaign in New Orleans, and find that "the political repercussions of a natural disaster are not automatically detrimental to the government in power." Of course, they give other reasons such as,

the political reaction may depend upon the government's capacity to neutralize the resulting stresses on the political system and upon certain relevant aspects of the political culture. Mayor Schiro's capable administration of the recovery operation, his skillful handling of his public image, and his hard work enabled to retain his adherents and perhaps win a few Fitzmorris supporters.

After a huge natural disaster, public infrastructure is destroyed, the population is poor and hence values investment more, therefore increasing support levels for the incumbent.

The rallying-round-the-flag effect has been well documented in the war literature. Mueller (1971) finds evidence of this effect in the Korea and Vietnam wars, while Oneal & Bryan (1995) find it in the Persian Gulf War. A country's position is stronger at the negotiating table and on the battlefield if the opponent knows that the politician carries full weight of home public support. This brings utility to the generic voter, who may end up supporting the politician more.

For economic crises, Gasiorowski (1995) looks at regime change in the 97 largest Third World countries, and finds that "Inflationary crises inhibited democratization from the 1950s through the early 1970s but may have facilitated it in the late 1980s, and that recessionary crises facilitated democratic breakdown but had no effect on democratic transition throughout this period." The move away from democracy and towards autocracy during recessions supports the notion that people are more accepting of strong unified rule during periods of economic difficulties. *Ceteris paribus*, a politician who receives 80% of the popular vote will find it easier to implement draconian autocratic rules compared to a politician that receives 51% of the vote. Here, voters are trying to signal to potential investors to invest during a recession. There is no clear-cut value in signaling to investors during inflations.

The vote share of the ruling People's Action Party (PAP) in Singapore has declined over the years. However, the PAP won 75.3% of the popular vote in the 2002 elections, a 10% increase over the previous general election in 1997. Did the party suddenly grow stronger, and the opposition simply collapse during this 5-year period? Or was it simply a case of more people throwing their weight behind the PAP so as to boost investor confidence in the face of a recession? During the campaign, the ruling party tried to persuade the population to "give the government a strong mandate so that we can reassure investors and concentrate on dealing with the recession" (or something to that effect).

In addition, Seshia (1998) writes that India's BJP grew because "the social and economic disruption produced by economic modernization, particularly in the seven years since India's liberalization, helped lend resonance to the BJP's appeals for Gandhian socialism and a certain amount of economic nationalism." This is the converse story. When a country modernizes, investments take a back-seat and the imperative to rally round the flag decreases, leading to decreased vote shares for the incumbent, starting off the transition towards functional democracies.

In sum, political scientists claim that cross-cutting cleavages are essential to a functional, multi-party democracy (Neto & Cox (1997)). It is hard to believe that the cleavages that any society faces realign drastically in the face of a short term hurricane, international war, or even a recession.

For example, a recession might accentuate the differences between the rich and the poor. However, if the rich stem from various racial and religious groups, the cross-cutting nature of the divide remains. Hence, the move towards one-party rule in the face of the crisis cannot be due to the triumph of one party over another in realigning itself to the new majority class. However, economic imperatives could cause people to place a greater weight on the “investment” (or stability, or war triumph) variable in their utility functions. Over-voting results.

2.6 Conclusion

This chapter suggests an alternative explanation for the Bandwagon effect, and examines the relevant comparative statics.

The presence of the bandwagon effect suggests that polls are very dangerous as they can seriously affect public opinion. Indeed, politicians have tried to buy polls to mislead voters (Huang, 2000), leading others to lobby against these polls (Pierce, 1940).

The model developed in this chapter predicts that if investments face a significant risk of expropriation, then these investments are affected by the level of support for a politician. If investments are regarded in the conventional sense, this phenomenon should not be evident in the US as the danger of expropriation from electoral term to electoral term is very low. Instead, if elected parties in the United States have a limited amount of resources to distribute among states, then the parties can take the role of investors. Their investment now comes in the form of the money, time and effort that they can potentially distribute across states. These investments bring direct utility to the voters, but remain valuable to the party only if the party continues to do well in that particular state (otherwise credit for a newly-built highway, for example, could be claimed by the new politician).

Voters then have an incentive to signal that they like the winning politician more than they actually do, so that the winning politician will invest more in their state. For reputational reasons, each party distributes money to the areas that gave it strongest support in the previous election, rather than areas that are evenly split between opposing parties. The empirical predictions from this hypothesis then offer an alternative explanation to the pure pork-barrel politics stories used as explanations for the empirical results by Dasgupta et al. (2001), and Ansolabehere & Snyder (2002), who find evidence in India and America respectively that political parties give more governmental transfers to areas where they have stronger electoral support as opposed to places with closer

races. Alternatively the “investors” can be the party faithful whose effort levels at the local party offices depend on whether they believe that their congressman will win the next election. Voters in the street will benefit from the favors that the party faithful provide in solving their day to day problems.

A final application of the model could be the choice of leaders within political parties. These leaders have agenda-setting power and influence party direction. Strong leaders signal to voters that the party is in order and has a clear vision, increasing the likelihood that they will vote for the individual politicians representing their constituency. Individual politicians thus have an incentive to over-vote, especially since the number of voting politicians in a political party (n) is small.

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Chapter 3

What Works: Making the Most of Foreign Aid

(Joint with Abhijit Banerjee)

One would think that giving away money ought to be easy. After all, there are so many who need it so badly and we have a pretty good idea of where they are. Yet, rather remarkably, we seem to have arrived at a point where more or less everyone agrees that aid-giving is not working the way it ought to.

A part of the problem is that most people are not actually giving away money—or rather, while they often give away money, the ultimate recipients do not get it as a straight gift of money. Somewhere along the chain an NGO or a government is responsible for turning this money into schools, or hospitals or roads, or whatever the people are supposed to get.

3.1 What makes giving away money hard?

The bias against just giving people money stems, in part, from the feeling that the best use of the money may not be to spend it on consumption. This is plausible, and indeed likely: There is some relatively convincing evidence that many people do not invest as much in their businesses and their children's human capita as the rewards to such investment seem to warrant. What is less clear is why people could not be relied on to make the right investments on their own—in which case it would be enough to hand them the money. One possible reason may have to do with the lack of

self-control: It may be too tempting, especially for poor people, to spend the money on something they need right away. The incompleteness of the intra-family “contract” is another reason not to trust the family with the money: Parents may put too little weight on improving their children’s earning capacity because they do not expect to share in their children’s prosperity. And of course, people may not know what is good for them.

There was a time when many of these kinds of arguments could not be made among respectable economists—it was seen as a transgression against the freedom of the individual, and bad economics to boot. Now the pendulum has swung to the other end—it is now an item of faith in the development community that no one should be giving away money. It is not clear, what, if any, evidence lies behind this shared conviction. Certainly no one has done the experiment of showering large gifts of money on poor people in poor countries and then following them to see what they do with the money and what happens afterwards.

A very different kind of argument for giving people goods rather than money comes from the fear that if you are seen handing out money, even those who have enough of their own may want to pretend to be needy. The advantage of providing public services rather than money is that the non-poor may not want them enough to make it worth their while to simulate poverty. The very rich in the U.S., after all, choose to pay for their children to go to private schools, even though their children are entitled to go to public schools at no charge, simply because they feel that the public education system is not quite up to the standards to which they aspire. This saves the government the cost of finding teachers for these extra children. On the other hand, these parents do not hesitate to claim any tax deductions that they may be entitled to, which suggests that if the government was giving away money instead of schooling, the rich would be in the queue with everyone else.

It is, however, not at all obvious that one cannot give away money without opening the flood-gates. After all, the rich value their time: Making it necessary to queue up in order to collect the money should discourage those who really do not need it.

The broader point here is not to deny that giving away money has significant disadvantages, but to emphasize that we know very little about how serious these disadvantages might be. In particular, are these costs necessarily large enough to outweigh the significant costs of trying to give away anything other than money?

3.2 Delivering goods and services to the poor

If you do want to give away money but still think it is worth trying to help the poor, you would have to give them things, i.e., roads, schools, banks, hospitals, fertilizer. Giving away things is more work than giving away money for the simple reason that someone needs to produce them: roads have to be built, teachers hired and trained, fertilizer produced and hospitals kept in good repair. It would be simple if it were just a matter of paying for the roads, fertilizer, etc., but that is just the first step. Then we would have to make sure that the roads are built to the required standards, that the teachers teach, the fertilizer gets to the right people, that the process of delivery is not subverted by corruption and/or bureaucratic incompetence

What is perhaps even more difficult is to be reasonably confident that the money is being spent on things that are really worth getting, in the usual sense of being at least as good as any other way of spending the same amount of money. After all, to take one popular example, there are many ways one could spend money on promoting education: One could build more buildings, hire more teachers, provide free textbooks, distribute free uniforms, put flip-charts in classrooms, set up computer labs in every school, provide a bonus for teachers who teach well, serve hot meals in school and much more. Indeed every one of the many interventions listed here (and many more) has been tried somewhere in the world in just the last few years. This diversity reflects, in part, differing needs, but often it is just ignorance: Lacking a clear sense of what works, well-meaning donors will choose what their intuition suggests, even though it may be very different from what the donor next door believes. Both seem to believe that they are right.

There is also no guarantee that it makes sense to spend the money on directly promoting education. The fact that people are not getting an education on their own might reflect the lack of jobs for those who have an education. Pushing out more graduates who will not find jobs either may actually be counter-productive, because it might reinforce the lack of faith in the value of education. The best way to promote education may be to create jobs.

Or we may even want to look beyond education. Perhaps one should invest in health and leave education to private initiative. Or forget about both and go for fertilizers. How should we decide where to go?

It is therefore no surprise that the process of helping the poor by giving them access to goods and services is fraught with difficulty. There needs to be a system for picking the right project and a system for making sure that the project is carried out as it should be; and for figuring out how

much people are getting out of it, and whether it continues to be what they need or want.

For this, donors will need to get involved in the process of decision-making and delivery at the ground level, though the exact nature of the involvement can vary substantially: The actual production of the good is usually contracted out, though even this is not always the case. Both the process of delivery and its impact has to be assessed, though once again there is a choice between doing it yourself and contracting it out. And the broader strategy needs to be worked out, based, one presumes, on knowledge of the situation on the ground. This might involve consulting local experts. Even carrying out some new research as a prelude to the intervention is not out of the question.

In all of this, the donor will typically work with one or more local organizations, be it government departments or NGOs. Some of these local partners may have their own sources of funding, which will allow the donor to leverage its resources.

Whatever the exact strategy, whether the donor does the monitoring or preliminary research, or whether these are contracted out, these involve very substantial expenditures over and above the direct cost of delivering the good (or service) to the ultimate beneficiaries. To get a sense of how large these expenses might be, we note that between the years 1996 and 2001, the World Bank administrative budget averaged US \$1,401 million per year (World Bank, 2001 Appendix 5, and World Bank, 2003b Appendix 1), while the total World Bank IBRD and IDA loan per year (World Bank, 2000b pg. 10; and World Bank, 2002b pg. 26) averaged US \$15,615 million and US \$6,154 million respectively. The Operations Evaluation Department of the Bank calculates ex post Economic Rate of Returns for the projects that it evaluates (World Bank, 2002a Statistical Appendix Table 12), but only for certain sectors¹, and finds that the median revised² economic rate of return for both the IBRD and IDA/blend lending operations for Fiscal Years 1996-2001 exits³ was about 20%. The aggregate economic return on the Bank's portfolio was 20% of \$15.615bn + \$6.154bn, i.e., about \$4.400bn. It follows that the benefits from World Bank lending are lower by almost a third because of administrative costs.

Of course, as we will discuss later, we should not expect to entirely avoid these costs by switching to donating money rather than goods. For now, however, the relevant point is that these costs are large.

¹ Agriculture, Electric Power & Other Energy, Environment, Mining, Oil & Gas, PSD/Industry, Telecommunications & Informatics, Transportation, Urban Development, and Water Supply & Sanitation.

² Ex post.

³ "Date that the project leaves the World Bank's active portfolio."

3.3 Evaluating donor effectiveness

Given everything that they spend on the design and management of aid projects, are donors getting what they hoped for? The short answer is, unfortunately, we do not know. A part of the problem is data: As Appendix C.1 shows, most of the larger “public” donor organizations (as opposed to private foundations), do evaluate their projects, but they usually stop short of a summary quantitative assessment of the social impact of the project, such as a rate of return. Of the donors listed in this Appendix, only the World Bank reports rates of returns, and then only for certain sectors; education, health and nutrition, for example, are left out. This reflects, in part, doubts about whether it makes sense to try to reduce the many dimensions of a project outcome to a single rate of return. In part it also reflects the inherent difficulties of coming up with a rate of return: How do you come up with the right counterfactual which tells you what would have happened in the absence of the project?

What some of these donors do instead is to assess overall project performance combining both process evaluation (“the right number of schools were built”) and impact evaluation (“the children’s test scores improved”) on a set scale⁴ (Appendix C.1).⁵ However, this evaluation is often carried out by those who are also involved in the implementation of the program, making it somewhat hard to know what to make of the results. Finally, many of these organizations do not allow the public access to their assessments. Of the eight organizations listed in Appendix C.1, only the World Bank and the Asian Development Bank (ADB) have their projects assessed by a formally independent organization, report project level assessments on a set scale and allow the public access to the assessment results. These are the only two, therefore, that offer the possibility of delving deeper.⁶

This is very far from ideal. The World Bank is not just any donor organization: It is probably the most visible organization of this class, with all the constraints that come from being in the public eye. It is formally responsible to those who provide its financing, which are the governments of a handful of rich countries. Moreover, it only gives loans (albeit on very attractive terms). It is also an organization that attracts and employs many of the best minds thinking about development today. Perhaps most importantly, it sees itself as a leader in the efforts to promote development in

⁴Such as satisfactory/unsatisfactory or successful/unsuccessful.

⁵Others, like USAID, just report their assessment of how well the country is doing.

⁶EBRD published their results online for the first time in late 2003. Even though they provide item-by-item ratings for all projects evaluated since 1993, they do not provide the information needed to match these projects to funding information and therefore the data is not yet usable.

the world: This probably means that its projects need to be evaluated not just in terms of what they directly achieve but also in terms of how they shape efforts outside the Bank to promote development. In a previous paper, Banerjee and He (2003), we try to evaluate the Bank's achievement as a leader and conclude that there is no evidence that others are following its lead. But we also argue that the Bank is ideally placed to take this leadership role and that it is important that it does so, which obviously implies that we must take it seriously in evaluating the Bank's performance.

The ADB is also quite special, being the one multilateral funding organization that has close ties with the Japanese government. We may expect it to have been influenced by the Japanese government's rather distinct view of economic policy.

Given that the World Bank and ADB evaluations are on different scales and the evaluators have potentially different standards, there is no point in trying to compare these two organizations. One could, of course, take their assessments at face value: In 2002, the Operations Evaluation Department (OED) of the World Bank wrote that "at the project level, the outcomes of Bank-financed projects continue to improve", and more than 60% of all projects evaluated each year since 1990 have had satisfactory outcomes (World Bank, 2003a pg. xii); while the Asian Development Bank wrote that the proportion of successful projects/programs (by year of completion) has been more than 50% since 1997, and has been trending upwards since 1989 (Asian Development Bank, 2003 pg. 37).

The problem is that we do not really know what to make of the scale they use—what does it, at some absolute level, mean to say that the World Bank's OED feels that the project was satisfactory? How much of this assessment reflects, for example, what they expected (which we do not know) rather than some objective that we all share? Clearly there are many who would join Bill Easterly in his reluctant conclusion that the last fifty years of aid-giving by the Bank (which, for many years, employed Easterly) has achieved relatively little (Easterly, 2001; 2003). It seems safer, therefore, not to put too much weight on how the OED (or its equivalent elsewhere) feels about the project and to focus on how assessments vary across different projects or sets of projects for the same organization.

World Bank or ADB projects vary considerably in the degree of their involvement, as measured by the share of the project financed by these organizations. If we assume that putting more money into the project reflects a greater commitment to the cause, we can use this ratio as a measure of the bank's priorities.

What can we say about the World Bank's priorities? For each project approved (i.e., launched)

between 1994 and 2001, the World Bank reports the share of World Bank funding in total funding for that project.⁷ For the period 1987-2001, we also have the evaluation of projects by sector⁸, averaged over 3-year periods (1987-90, 1990-93, 1994-97 and 1998-2001).⁹ We label these four periods 1, 2, 3 and 4. We then regress the share of World Bank funding in a particular project approved in period t on the average evaluation in period $t - 2$ of the sector that it belongs to [Prevperf], and the improvement of its evaluation between $t - 1$ and $t - 2$ [Diffperf]. We control for fixed differences across sectors, countries and periods and cluster errors by sector.

The results in column 1 of Table 3.1 show that when a sector's performance improves, projects in that sector get a higher fraction of their financing from the Bank (the DiffPerf coefficient is positive). But it also helps to start at a low base (conditional on the same degree of improvement, projects in sectors that started with a worse record get more money: the coefficient on Prevperf is negative), which immediately implies that if two sectors have shown the same improvement, the one that is doing worse will get more money from the Bank.

A very similar pattern emerges when we look at the total amount of money allocated to each sector. For four sectors—Agriculture, Finance, Technical Assistance, and Water and Sanitation¹⁰—the World Bank provides data from 1974¹¹ to 1993¹² for every block of three years¹³ on the sector outcome measured by OED evaluation¹⁴, and the log of total expenditure on that sector.¹⁵ Using this data set, we regress¹⁶ $[\log(\text{expenditure}_t)]$ on $[\log(\text{expenditure}_{t-1})]$, $[\text{outcomet}-1]$, and $[\text{outcomet}-\text{outcomet}-1]$, correcting for period and sector effects. The results are shown in column 1 of Table 3.2. They show that increased spending for a particular sector is associated with an improving trend in sector performance over the immediate past. Once again, having started from

⁷Percentage figures from the Annual Reports of the World Bank: Summaries of Projects Approved for IBRD, IDA, and Trust Funds in each Fiscal year. For each project, the following information is provided: country name, sector, brief project description, World Bank contribution, total project cost.

⁸Sectors were determined by the classifications in World Bank (2000b, Annex 1), which conveniently classifies the historical performance data into the 1990-1993, 1994-1997 periods. From these sectors, the "Social" and "Environmental" sectors were dropped because zero/one evaluation was done between 1990 and 1993. We do not have 1987-1990 data for "Mining", "Multi-sector" and "Public Sector Management" due to a change in sector classification.

⁹The 1987-1990 performance data is derived by collating information from World Bank (1989) and World Bank (1993). The years 1987 and 1990 here refer to the year that the project was evaluated; hence they correspond to projects that had slightly earlier exit fiscal year groups. The 1990-1993 and 1994-1997 performance data is derived from World Bank (2000a), and correspond to projects that had exit fiscal years within the respective ranges. The 1998-2001 data is derived from collating information from World Bank (2000a; 2002a).

¹⁰These are the four selected sectors in Morra and Thumm (1997, Table 1.32).

¹¹Data starts from 1974.

¹²The Technical Assistance category was dropped in World Bank (1994).

¹³With the exception of the 1992-1993 year.

¹⁴Morra and Thumm (1997, Table 1.32).

¹⁵Annual Reports of the World Bank. Collecting data on the percentage funding for all these years requires an inordinate amount of work.

¹⁶The Arellano-Bond linear dynamic panel data estimator is used.

a lower initial level of performance helps, but the coefficient is not significant in this case.¹⁷

When we do the same exercise for the ADB, we get very different results. Both past performance and improvement in performance seems to have a negative impact on the allocation of its funding (column 2 of Table 3.1). However this result is very sensitive to the inclusion of the “Multi-sector/Others” sector, where there were only 2 evaluations between 1986 and 1989 (both successful). Once this sector is dropped, both the past level and the improvement in the level become insignificant (column 3 of Table 3.1). All that matters then are fixed cross-country differences and perhaps cross-sector differences.¹⁸

The results for the ADB seem very consistent with the view that its priorities are largely set by high level decisions and are not particularly subject to any short-term influences. This may be a good thing because it insulates the organization against the influence of fads and internal political shifts, but it clearly also prevents the organization from learning from its experience.

The case of the World Bank is more complicated. It clearly does not give priority to sectors that have been performing the best over the immediate past, which is what, under the (possibly brave) assumption that past performance is a reasonable index of what we might expect in the immediate future, would have been the way to maximize immediate impact. But it does favor the sectors that have been improving the fastest. One way to rationalize this may be to assume that the World Bank sees itself as a leader in the development community: As a leader, it would make sense for it to try to promote those sectors where the potential for improvement is the highest, rather than those where, on current record, the possibility of success is the highest. Sectors that have been improving fast over the past few years, but still have some distance to go, may therefore be exactly the sectors the Bank would want to favor.

It is, however, possible to take a more cynical view of the same evidence. In this view, the Bank is excessively influenced by shifts in current fashions in development thinking. The reason, in this view, that we see the Bank reacting to improvements is that these improvements shift fashions—when something does unexpectedly well, it is easy to get excited about it, even if, on balance, it is still doing worse than other, more well-established, options.

The question, in the end, comes down to whether the projects that are being given priority are

¹⁷Here, unlike in the results in the previous table, we are using the gap in performance between t and $t-1$ rather than that between $t-1$ and $t-2$, because of data limitations in the case of the World Bank. For the ADB we could use the gap between $t-1$ and $t-2$ —the results are very similar.

¹⁸Both for the ADB and the World Bank, the sector dummies as a group are significant at the 1% level (based on their joint F-Statistic) as are the country dummies, though the sector dummies become insignificant for the ADB when we drop the multi-sector/others category.

doing what they were intended to do. One way of looking at this is to look at the correlation between the fraction of planned project financing that was to come from the Bank and the performance of the project according to the Bank's evaluators, after controlling for fixed differences across sectors and countries, the length of the project, the year when it was approved, and the year of the evaluation. The results are shown in column 1 of Table 3.3. As already reported in Banerjee and He (2003), there is a negative and significant correlation between the priority that the Bank originally gave the project (measured by the fraction of financing that was supposed to come from the Bank) and its performance. Bank-favored projects seem to do worse from an *ex ante* point of view, relative to other projects in the same sector.

This negative relation goes away if we replaced the share of planned cost that the Bank was supposed to pay for with the share of actual cost (column 2 of Table 3.3). Basically, if projects are going really badly, the Bank cancels its promised contribution to them. But even with the help of this corrective procedure the correlation between performance and funding is nowhere near being positive and significant: Being a Bank priority does not help you perform better, even after the cancellations.

We repeat this exercise for the ADB, using projects evaluated¹⁹ between 1997 and 2002. Here percentage funding is defined as the [Loan Amount approved]/[Expected Project cost] in column 3 of Table 3.3, and [Loan Amount disbursed]/[Actual Project cost] in column 4 of Table 3.3. We find that the outcome rating is not significantly affected by the loan amount approved, but the amount disbursed has an effect that is negative and significant at the 10% level.

The lack of a positive correlation between funding and performance, in the case of the ADB, seems unsurprising, given everything else we seen about the way they target (or rather, do not target). In the case of the World Bank, these results are consistent with the view that the Bank is faddish. They can also be explained by assuming that the Bank is particularly bad at running its projects.

On the other hand the results are also quite consistent with any view that has the World Bank playing the role of a leader and prioritizing projects that others, more focused on immediate impact, would not choose. After all, we already knew that they have not given priority to the sectors that had performed best in the past—all that this evidence really shows is that this is true within sectors as well. Of course, as already pointed out, it is not clear that the World Bank is particularly effective as a leader, and to the extent that we take this as given, we may still want it

¹⁹Data from World Bank (1998; 1999; 2000a; 2001; 2002a; and 2003a).

to focus more on its rather limited direct impact.

In the end, there is very little that is reliable that we can say about donor performance. The most we can say is that we found no *prima facie* evidence of great effectiveness.

3.4 What limits effectiveness?

Donor organizations are in many ways very much like other organizations, and they share many of the standard organizational constraints. Organizations like the World Bank obviously need many people to act and take decisions on their behalf, and there is nothing to guarantee that they have the right incentives. In particular, in an organization that lives by doing projects and making loans, no one achieves prominence by rejecting projects and refusing loans. For this reason alone, most people on the implementation side of the Bank, either in their sector departments or in their country missions, are probably somewhat biased in favor of making something happen. Add to this the fact that they are the ones who deal with the potential recipients, and we have a recipe for a degree of over-enthusiasm and irresponsible lending.

There is also, obviously, the possibility of ideological conflict within the organization. We recently saw a public example of such a conflict in the World Bank, which ultimately led to the resignation of the person in charge of the 2000-01 World Development Report. There must also be other fights that have less to do with ideology than with personalities and individual ambitions. All those involved in these fights must be tempted to use the power to sanction projects to help their supporters and punish their enemies.

There is also the fact that being a donor organization involved in development makes one the target of influence from many people—ranging from the U.S. government to NGOs hostile to the U.S. government. These pressures are probably easier to resist for an organization that has an explicit ideology or an acknowledged political master—like the Salvation Army or USAID—than a nominally apolitical organization like the UNDP or the Bank. For such organizations the challenge is often in maintaining their reputation for being open to many different views while continuing to make the right choices about the allocation of funds. Resisting the temptation to placate the different sides by conceding some of their less merited demands, must be a challenge for these organizations.

However, one of the biggest problems and one that is discussed all too rarely, is the lack of an explicit scientific basis for their decision making. An eloquent exception is Lant Pritchett, a

long-term Bank employee:²⁰

Nearly all World Bank discussions of policies and project design had the character of ‘ignorant armies clashing by the night’—there was heated debate amongst advocates of various activities but rarely any firm evidence presented and considered about the likely impact of the proposed actions. Certainly in my experience there was never any definitive evidence that would inform decisions of funding one broad set of activities versus another (e.g., basic education versus roads versus vaccinations versus macroeconomic reform) or even funding one instrument versus another (e.g., vaccinations versus public education about hygiene to improve health, textbook reform versus teacher training to improve health, textbook reform versus teacher training to improve educational quality). How can this combination of brilliant well-meaning people and ignorant organization be a stable equilibrium? (Pritchett, 2001)

A recent World Bank publication, *Empowerment and Poverty Reduction: A Sourcebook*, provides an excellent case-study of the kinds of policies that result from these deliberations (Narayanan, 2000). The Sourcebook is meant to be a catalogue of what, according to the Bank, are the right strategies for poverty reduction. These are also, we presume, strategies into which the Bank is prepared to put its money. It provides a very long list of recommended projects, which include: computer kiosks in villages; cell phones for rent in rural areas; scholarships targeted towards girls who go to secondary school; schooling voucher programs for poor children; joint forest management programs; water users groups; citizen report cards for public services; participatory poverty assessments; internet access for tiny firms; land titling; legal reform; micro-credit based on group lending; and many others.

While many of these are surely good ideas, the book does not tell us how we know that they work. We now know that figuring out what works is not easy—here is a large literature documenting the many pitfalls of the usual, “intuitive”, approach to program evaluation. When we do something and things look like they are getting better, it is tempting to think that it was all because of what we did. The problem is that we have no way of knowing what would have happened in the absence of the intervention. The simplest and best way to avoid this problem is to do a randomized evaluation where we assign the intervention to a randomly selected subset of the set of potential locations, and compare those who got it with those who did not. This mimics the procedures used in trials

²⁰Bill Easterly, another long-term Bank employee, makes a very similar point in his recent article in the *Journal of Economic Perspectives* (Easterly, 2003).

of new drugs, which is the one place where, for obvious reasons, a lot of care has gone into making sure that only the things that really work are approved. In many ways social programs are very much like drugs—they have the potential of transforming the life prospects of people—it seems appropriate that they should be held to the same high standard.²¹

Of course, even randomized trials are not perfect. Something that may work in India may fail in Indonesia. Ideally, there should be multiple randomized trials in varying locations. There is also no substitute for thinking—there are often good and clear reasons why what works in Kenya will not work in Cameroon. And there are times when randomized experiments are simply not feasible (more on that later). However, with all that, it is hard to imagine a good reason for spending a lot of money without having done at least one successful randomized trial, assuming that a randomized trial is possible. When we talk of hard evidence, we will therefore have in mind evidence from a randomized experiment, or, failing that, evidence from a true natural experiment, by which we mean an accident of history that created a setting that mimics a randomized trial.²²

What is striking about the list of strategies offered by the Sourcebook is the lack of any distinction between strategies that can claim to be based on hard evidence and the rest. In fact, to the best of our knowledge, only one of these strategies—schooling vouchers for poor students in Colombia—has been subject to a randomized evaluation and that was because it was politically necessary to allocate the vouchers by lottery. Comparing those who won the lottery with those who did not provided the perfect experiment for studying the impact of the program, and the study by Angrist et al. (2002) takes advantage of it. Yet the results from this study receive no more weight than any of the other programs.

Indeed most of these programs are recommended on the basis of very little hard evidence. Legal reform, for example, is justified by asserting that, “The extent to which a society is law-bound affects its national income as well as its level of literacy and infant mortality (Narayanan, 2000).” This may be true, but the available evidence, which comes from comparing the more law-abiding countries with the rest, confounds too many things to warrant such a confident recommendation.

And some programs, it seems, no amount of negative evidence can stop: Our favorite example is the Gyandoot program in Madhya Pradesh in India which provides computer kiosks in rural areas. The Sourcebook acknowledges that this project was hit hard by lack of electricity and poor connectivity and that “Currently only a few of the Kiosks have proved to be commercially viable”.

²¹Those who are interested in the argument for (and against) randomized trials as a basis for social policy may want to see Duflo (2004) and Duflo and Kremer (2004).

²²See Angrist and Lavy (1999), for an example of a very convincing natural experiment.

It then goes on to say, entirely without irony: “Following the success of the initiative....(p. 80).”

3.5 Why do people resist evidence-based policy making?

Lant Pritchett, in the article mentioned above, goes on to argue that the resistance to hard evidence is, in part, a reflection of the mixed motives of those who give and receive aid. Even where there is no real corruption, as in the Bank, the problem is that many of these people are true believers and as such, see no intrinsic value in rigorously testing the policies that they are advocating. While they recognize that good evidence might help them win friends, they also worry that it might work against them—someone might misread the evidence or, as chance will have it, the evidence may just refuse to cooperate. Hard evidence is simply not worth the trouble, especially if eloquence and a few carefully chosen examples can carry the day.

We do not doubt that this is a piece of what goes on, but it only explains the motives of those who have things going their way. But the Bank, for example, is a contentious place—for every one person who likes where policy is currently headed, there is probably at least two who would like the tides to turn. These people have a strong incentive to look for hard evidence, since there is no other way they can upset the status quo, and it is hard to believe that they could not do a proper test of the intervention if they really wanted to. In other words, if the advocates do not provide the necessary hard evidence, we would expect their opponents to do so.

The fact that we see very little of this kind of “competition by evidence”, suggests to us that the deeper problem is not strategic resistance to evidence, but rather a view shared by most people in the development community that basing policy on hard evidence is simply not practical. This is consistent with our experience in talking to senior officials in donor organizations, who seem to genuinely believe that there is no real alternative to the current system of decision-making.

Their objections to the idea that policy should be based on evidence typically fall into one of two categories. First, there is the fear that requiring that every initiative be justified in terms of hard evidence will bias decisions in favor of what is measurable and easy to evaluate. Second, there is the conviction that at this point there is so little that can be justified in terms of randomized trials that to rely exclusively on this evidence is tantamount to considered inaction.²³

²³This is, in effect, how Stanley Fischer, a former Chief Economist of the Bank, put it, while commenting on our previous paper at the meetings of the American Economic Association.

3.6 The feasibility of evidence-based aid

We feel that both of the concerns articulated in the previous paragraph are substantially exaggerated. In terms of the first concern, we obviously recognize that there are things like macro-policy that are very hard to evaluate properly. The problem is that once something is big enough (“currency boards”, “democracy”) there is going to be no way to know what would have happened in its absence. And yet, there are clear examples of policies that, most people would agree, make very little sense (“over-valued fixed exchange rates”, “a pension plan that is headed for bankruptcy”, for example). Helping governments in their efforts to get out of these indefensible policy positions has to be a good use of donor money.

Once we go beyond these “obvious” interventions, there is still a range where we can expect to make an evidence-based decision. These are typically macro interventions that allow for the possibility of a limited micro evaluation. For example, while decentralizing political power is a macro reform, we could learn a lot about it by looking at the impact of an initial pilot, where the reform is only implemented for certain areas.²⁴ When we go beyond these, it is still useful to try to make use of the best available evidence: Evidence can at least help in choosing a model whose assumptions correspond better to the world. Obviously, how far the donor is prepared to go down this evidence “quality-ladder” will depend on the donor: What is key is that she has a sense of what she is giving up—the fact that there may be other projects out there for which we have much more reliable (and reassuring) evidence.

The other side of this same concern is that requiring evidence discriminates against projects that promote less measurable outcomes—such as female empowerment. Now it is true that historically the focus of economic measurement has been on concepts like consumption and income rather than empowerment, but when Chattopadhyay and Duflo (2001) needed a measure of female empowerment in the context of public action in Indian villages, they used the fraction of questions asked by women in village meetings. While this is not perfect, it is not obviously worse than using income to measure well-being, as we regularly do. We are therefore optimistic that once we commit ourselves to measurement, the interaction of the donors and the evaluators will generate a range of good measures of most things that are relevant.

To address the other main concern, that basing action on evidence will lead to paralysis, we carried out the following crude but useful exercise. We began by searching for interventions that

²⁴These would be chosen randomly from some larger set.

have been subjected to randomized evaluations and appear to work. To come up with a list, we asked researchers in the Bureau for Research in Economic Analysis of Development (BREAD) for references and used summary papers by Kremer (2003), Behrman and Knowles (2003), De Cock, et al. (2000), and Working Group 5 of the Commission of Macroeconomics and Health (2001) as starting points for a literature search. In addition, a web-based search was used. From these we deliberately left out regulations, such as Tobacco Taxes, Bans on Tobacco, etc. The Table of Interventions in Appendix C.2 lists all the papers that were eventually included in our list. It is meant to cover every category of micro-intervention that we would find that has been subjected to an evaluation.

From these papers, we highlight the subset of programs that, based on currently available evidence, look sufficiently good that it would be worth implementing them on a global scale. This cut was based on three criteria: the program must be sustainable without a strong intervention by the researchers;²⁵ the evidence must come from a run of the program where it was randomly placed; and finally, in that randomized trial it must have had a significant positive impact on at least one of the initially chosen objectives. If two experiments show different results, wherever there was a clear methodological difference, we favored the one that has the better experimental design. We also favored experiments that showed improvements in outcomes that have direct economic relevance—education, incidence of disease, weight—over outcomes that are only potentially correlated with economic outcomes—the presence of antibodies, etc. Finally, we favored experimental research by economists over experiments by clinical researchers, on the grounds that economists are more sensitive to problems relating to delivery. In the end, there were few cases where we had to exercise any judgment.²⁶

We left out all programs that simply gave away money, but included programs (like school vouchers) that give people money that can only be used for a specific purpose. Finally, we were not quite sure of how to deal with a program like PROGRESA in Mexico, which makes an income transfer to mothers who send their children to school. The problem is that we have no idea of how much we would need to pay to get children into schools, and how much of what was paid was a pure gift. We therefore treat it as a separate category.

The goal here is purely illustrative: We want to demonstrate that one can come up with a long

²⁵We feel that field workers ensuring that villagers comply with their daily supplement dosage constitutes overly strong intervention.

²⁶For example, this was the basis for choosing Miguel and Kremer's (2004) work on de-worming, and Bobonis et al. (2003) on iron supplementation.

list of interventions that have been shown to work based on a randomized evaluation. We are not at all suggesting that the interventions listed are the only ones that work or even that they are most efficacious among those that work. Nor do we suggest that the programs that we list but decided not to scale up are necessarily worthless. It is entirely possible (indeed we hope that it is true) that there are many other interventions that do work but are not included in our final list because we could not find a randomized evaluation applying to them. Other programs that were left out because there was a high degree of intervention by the researchers might have also worked with less intervention—we simply do not know. Yet others may really work in their non-experimental variant, but not in the form in which they were implemented during the experiment. Finally, there are probably many interventions yet to be thought of that have the potential to change the world.²⁷

Given this list of “successful” interventions, we ask the question: How much will it cost to more or less mechanically scale them up to a global level? Our definition of global covers only Low Income Countries (LICs), unless stated otherwise. We take the population of each country to be the average of the current population and the projected 2015 population.²⁸

The way we calculate costs is to take the point estimate of the per person cost for each program and then adjust it for each country. This adjustment involves converting the expenditure on goods (vitamins, drugs) using the standard Purchasing Power Parity conversion factor, and scaling up (or down) the expenditure on services (teachers, health-workers) in proportion to the GDP per capita at current exchange rates.²⁹ For each country, the size of the targeted population is derived from demographic information about the country. The cost spreadsheet is available online.³⁰ Figures are normalized to year 2000 U.S. dollars.

The results are shown in Appendix C.3. Our calculations show that a recurring annual expense of about \$11 billion could be justified by the hard evidence we already have, without including PROGRESA. In addition, if we accept the (non-experimental) results in Cutler and Miller (2003) showing that the health returns to improved water supply are enormous, we should consider investing in water supply infrastructure. This will cost an additional (one-time) \$15 billion.

²⁷We also recognize that researchers do not always have control over the experimental design, and in any case, that the research might have been done with different objectives in mind (for example, researchers might be interested in knowing the impact of supplements on growth if it could be assumed that compliance was not going to be a problem). Inclusion in the highlighted subset does not reflect any opinion of the quality of the research or the researcher. It merely reflects the suitability of the research for our purposes.

²⁸2015 is chosen because it is when the Millennium Development Goals are meant to be achieved and many of our programs could be ways to achieve the goals.

²⁹Under the plausible assumption that the PPP correction is unreliable for the salaries of skilled service providers, who tend to be scarce in developing countries.

³⁰<http://web.mit.edu/ruimin/www/whatworks/whatworksest.xls>

Eleven billion dollars is a substantial amount. It is more than the World Bank gives out as IDA loans (the main form of World Bank aid) in a given year (as previously mentioned, an average of US \$6,154 million between 1996 and 2001). If we add to this what countries absolutely need to make essential macro adjustments (of the kind discussed above) there may be very little left from the total available donor money.

To the extent that there is still money, it can be used to provide humanitarian aid: There are people in the world who are dying because they do not have enough money to buy food or medicines. Giving them money (or food, or medicines) may not promote development but it is hard to imagine that it would not be good thing.

Indeed once we decide that we are willing to make cash gifts to people, we could make that gift conditional on the recipients fulfilling certain conditions, such as sending their children to school. This is what PROGRESA does.

It is true that this does not deal with most of the objections against giving away money: The targeting problem was solved in Mexico by using the bureaucracy to make sure that money goes to the right person, which is part of what makes the program costly. On the other hand, it is not clear that perfect targeting is worth the effort, given that most people in developing countries are actually quite poor. It may be better to set up rather lax criteria for eligibility so that only the rich are ineligible, and then to randomly check claimants and impose harsh punishments on those who are caught cheating. It is usually quite easy to identify the rich in poor countries (ownership of a car, for example, might be used).

It is also true that we do not eliminate the possibility that the money may just go into consumption. But even this is less of a problem than it might appear: A number of studies (including those based on the PROGRESA experiment) have shown that more money in the hands of the female members of the family in poor countries does translate into better nutrition for children and better health care.

Ultimately, however, we should not need to argue that giving away money is without its problems. After all, it may still dominate trying to give away goods, which as we have seen, is fraught with problems.

Clearly being open to the idea of giving away money will make it much easier to find things to do. Just scaling up PROGRESA, by our calculations, will cost \$23 billion a year, which pushes up total annual expenditure on “good” programs to \$34 billion.

3.7 To conclude: A new challenge for the millennium

We live in an age of aid pessimism. There is a strong, if rarely completely articulated, presumption that aid can at best help people survive—it cannot promote development. The U.S. government's new initiative, the Millennium Challenge Account (MCA), is based on the idea that the whole idea of aid-giving needs to be rethought. In particular, they want to tie aid to country performance: Only countries that pursue economic policies that the U.S. government approves of will be eligible for aid from this account. The premise is that aid has not been working because the policy environment is not right. While it is clear that this is a problem—there are countries where the risk of the money ending up in a government official's pocket are substantial—the thrust of our argument is that the way the money is planned to be spent is also a very big problem, but a problem whose source lies in the way the donor organizations function. Combined with the fact that many of the world's neediest live in the countries that will not make it onto the MCA list, and that we expect the incentive effects of the MCA to be minimal, this suggests to us that the MCA approach amounts to abandoning a large part of the world's poorest through no fault of their own. A more effective and less unfair challenge may be to try to see if it is possible to design projects that work in the countries that have the biggest problems. If we could make that work, we would not only help those who need it the most, but what is perhaps even more valuable, we will raise expectations and build hope where there is none.

Table 3.1: OLS Regressions on Project Selection

	World Bank (1)	ADB [including multi/others] (2)	ADB [excluding [multi/others] (3)
<i>PrevPerf</i>	-.002553* [.0011]	-.006767* [.0033]	.0004254 [.0047]
<i>DiffPerf</i>	.001852* [.00051]	-.004361* [.0014]	-.001184 [.0020]
<i>N</i>	1513	519	468
<i>Adjusted R²</i>	.29	.33	.39

Notes — Dependent variable: % of individual projects that is funded by the World Bank. * Significant at the 5% level of significance. Robust standard errors reported.

Table 3.2: Cross-sectional Panel Regressions on Project Selection

	World Bank (1)	ADB (2)
$\text{Log}(\text{expenditure}_{t-1})$	.7677* [.028]	.1523 [.18]
outcome_{t-1}	-.002256 [.0026]	.000925 [.0089]
$\text{outcome}_t - \text{outcome}_{t-1}$	.001594* [.00075]	-.0004837 [.0073]
<i>Number of obs</i>	16	45
<i>Number of groups</i>	4	7

Notes — Dependent variable: $\text{Log}(\text{expenditure}_t)$. * Significant at the 5% level of significance. Robust standard errors reported.

Table 3.3: On Effectiveness of Fund Allocation

	World Bank (planned) (1)	World Bank (actual) (2)	ADB (planned) (3)	ADB (actual) (4)
<i>%Funding</i>	-.123** [.038]	.064 [.090]	-.585 [.54]	-.837* [.49]
<i>Length</i>	.0054 [.016]	.0100 [.016]	-.0705* [.038]	-.0722** [.035]
N	664	664	137	136
<i>Adj. R²</i>	.26	.24	.18	.20

Notes — Dependent variable: Outcome of individual projects. Regressions include sector dummies, country dummies, year of approval dummies, and year of closing dummies. ** Significant at the 5% level of significance. Robust standard errors reported. * Significant at the 10% level of significance. Robust standard errors reported.

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Chapter 4

Fiscal Virtual Representation: Evidence from Reapportionment in California (Joint with Stephen Ansolabehere and James Snyder)

4.1 Introduction

It is an axiom of politics that self-interested political agents always seek to maximize their own power. Yet history provides evidence that individual actors do not always behave in such a simplistic manner. Voting behavior is often driven by other factors, and this chapter explores one of them: fiscal virtual representation.

As a result of urbanization, migration and other demographic shocks, population did not grow at a constant rate across California in the 20th century. The state constitution made provisions for reapportionment based on population for both the Assembly and the Senate districts every decade. However, in response to a rapidly growing South, the legislature decided not to reapportion the districts in 1920. Hence, the number of state congressional representatives in the Assembly and the Senate from each county remained fixed at 1910 levels until 1928, when the Assembly was reapportioned. Between 1926 and 1962, California voters faced a number of ballot propositions

that sought to reapportion the houses. An examination of voting records reveals that a simple power explanation fails to account for voting behavior. Voters in counties that stood to gain power did not all vote in support of reapportionment, and those who were going to lose power did not all vote against reapportionment.

This chapter studies one possible reason for the “deviant” behavior. First, we develop a model that predicts that individual preferences for the size of government expenditure depends on whether a person is innately fiscally liberal/conservative, and on the size of the county that she resides in if power is malapportioned. We define an innately fiscally liberal person as one who, if given the same income and tax rate as everyone else, will choose a higher level of government expenditure compared to everyone else in the same county.

Next, we argue that when different structures of representation led to differences in the level of government expenditure, then depending on the individual propositions, some voters were willing to give up some of their individual power, hoping to shape the overall size of government spending by transferring their voting power to voters in other counties. Thus some of the voters who lived in over-represented counties supported a change to population-based apportionment because they found that their ideal amount of government expenditure better matched those who lived in under-represented areas. This form of virtual representation differs from the usual virtual representation story involving ideological preferences. More generally, voters could differ on other issues; but we find that “fiscal liberalism” seems to be the most important issue. We find statistically significant support for fiscal virtual representation at the individual, town, district, page¹ and county levels, which is consistent with a world where fair apportionment leads to greater government expenditure.

4.2 Background and Literature review

Between 1926 and 1962, voters in California faced a series of ballot propositions that would have altered the power distribution of individual voters. In this chapter, we use California as a case-study to propose a potential factor that could have driven voting behavior.

California’s legislative history is nicely summarized in a 1928 pre-election article in the Los Angeles Times:

The Constitution of 1879, the fundamental law of the State, created a Legislature of forty Senators and eighty Assembly-men, chosen from districts equally apportioned,

¹Elaborated later.

according to population, these districts to be revised after each Federal census. For many years the bulk of population, voting power and political weight rested in the North. The census of 1910 indicated this bulk was shifting toward the South. The 1920 census made this quite evident, with the results that the Legislature then and thereafter refused to follow the mandate of the Constitution and revise Senatorial and Assembly district representation. (LA Times, 21 Oct 1928)

The issue of reapportioning districts to equalize representation in the legislature was to re-emerge time and again for four decades after 1920, before *Baker v. Carr* resolved the issue by implementing court-ordered redistricting. Various authors have shown that power redistributions have tangible effects. Ansolabehere, Gerber & Snyder (2001) show the effects of reapportionment on the distribution of public expenditures. Lee (1998) shows the consequences on distribution of federal funds at the national level, while Erikson (1971) looks at effects on partisan behavior. Hence, voters had reason to want to increase their own representation. Indeed, Woon (2001) finds evidence that voters preferred more representation to less.

Yet voters did not all vote to increase their personal power when they had the chance to do so. For example, even though Los Angeles was going to be the largest beneficiary of a 1948 reapportionment initiative, only 41% of the population voted in favor of the initiative. The literature contains numerous explanations to account for this type of behavior.

A survey of pre-election articles in the San Francisco Chronicle and the Los Angeles Times reveals that the arguments for equal representation by population primarily appealed to notions of fairness. A typical argument sounded like, "I feel it is basically wrong when a county (Los Angeles) with 6 million residents has the same representation as an area with 15,000 persons" (LA Times, 25 Oct 1960). On the other hand, opponents of reapportionment argued that areas with lower populations should be given more representation than what a population based redistribution would allow because of the idea that rural areas accounted for more of the agricultural output and were the main tax bases. However, unless we can find evidence that indicate preferences for fairness among individual voters, we are unable to test this hypothesis.

Ansolabehere, Snyder & Woon (1999) argue for coalition politics. Other authors hypothesize that voters were willing to give up their individual voting power in order to shape the ideological liberal-conservative characteristics of the legislature (Lee, 1960). This motivated a flurry of research in 1960s/70s that sought evidence for and against the hypothesis that reapportionment

led to changes in public policy, primarily along the lines of liberal-conservative framework.² Key (1949) argues that virtual representation was at work in the South. However, their findings remain inconclusive.

We do not disprove the coalition argument, but show that virtual representation is possible in the form of fiscal preferences. Past research has missed this. While virtual representation could exist in other forms, such as ideological preferences, we could not find evidence supporting these other forms. In addition, instead of looking at reapportionment per se, we look at the specific implications of individual reapportionment propositions on the resultant fiscal slant of the legislature, over propositions spanning from 1926 to 1962.

4.3 The Model

4.3.1 Reapportionment schemes

Apportionment by population

Under a population-based apportionment, the number of representatives from each county in the legislature is proportional to the county's population. Larger counties hence receive a relatively larger voting share compared to what they would receive under the other apportionment schemes.

Apportionment by counties

A county-based apportionment means that all counties receive the same number of seats in the legislature, regardless of county population. Smaller counties receive a relatively larger voting share compared to a population-based apportionment.

The status quo in the 1920s

Under the status quo in the 1920s, both houses were apportioned by population based on the 1910 census. Different rates of population expansion in the 1910s and 1920s meant that the status quo distribution of legislative seats for each county did not accurately reflect the current (1926) proportion of population in each county, but it was still a better reflection than the distribution of seats under a county-based apportionment scheme. We thus expect the power distribution between large and small counties under the status quo in 1926 to be between the two extreme apportionment

²See references in Ansolabehere, Snyder & Woon (1999)

schemes. Consequently, individual and aggregate preferences over expenditure and utility under the status quo should be in between the two extremes, and we shall make no further explicit mention of the status quo in this section.

4.3.2 Individual ideal government expenditure preferences

In our model, voter preferences controlling the size of government expenditures are affected by their innate fiscal preferences, and by the voting power of their county. As we point out in Section 4.2, the effect of ideological heterogeneity is inconclusive, and we thus abstract away from it. There are obvious analogies to an indifference curve analysis for optimal consumption between two goods assuming the Weak Axiom of Revealed Preferences.

A voter's utility is an increasing function of both his amount of private consumption, X_i , and his consumption of public goods, G_i . We assume diminishing returns to each type of good, and a constant elasticity of substitution, α , between the two types. A higher α corresponds to a higher innate fiscal liberalism. For graphical purposes, we assume a Cobb-Douglas utility function in the form

$$U_i = G_i^\alpha X_i^{1-\alpha} \quad (4.1)$$

subject to a budget constraint, as a portion of a person's income is used to pay taxes. For the purposes of this chapter we assume a tax that is proportional to income I_i and public goods expenditure G_i , with proportionality constant β . Hence, we assume a budget constraint of the form

$$I_i = X_i + \beta G_i I_i. \quad (4.2)$$

Given income I_i and relative public/private preferences α , individual voters will thus want to maximize their utility functions U_i .

Solving the constrained optimization problem yields

$$G_i = \frac{\alpha}{\beta}, \quad (4.3)$$

i.e., that individual preferences for government expenditure increases with innate fiscal preferences, α , and decreases with tax incidence, β .³

If power were equally distributed across voters (*Apportionment by population*), then voters in

³This simple form is a result of the Cobb-Douglas utility specification. In general, an individual's income may also matter.

different counties should face the same tax incidence. Voters across different counties with the same α should then have the same ideal government expenditure. For simplicity assume a world with only two types of people, the fiscally liberal and the fiscally conservative. Then as shown in Table 4.1, we normalize the ideal government expenditure of the liberal and the conservative to be 2 and 1 respectively.

In a malapportioned legislature (*Apportionment by counties*), voters in the smaller counties have relatively greater voting power. This power difference can result in lower per capita tax incidences (smaller β) in smaller counties, and holding innate fiscal preferences constant, larger ideal government expenditures relative to the larger counties. Referring to Table 4.1, this difference in incidence means that regardless of innate fiscal preferences, those voters in small counties will want greater government expenditure under a county apportionment relative to a population apportionment ($\epsilon > 0$). The converse hold for voters in large counties.⁴

4.3.3 Aggregate government expenditure preferences

Aggregate expenditure is the weighted median of the individual expenditure preferences, and depends on the number of small/large counties, the relative power per voter in each small/large county, and the distribution of fiscally liberal/conservative people across counties. In other words, aggregate expenditure in an *Apportionment by county* scheme can be either higher or lower than in an *Apportionment by population* scheme. The empirical findings in this chapter suggest that the aggregate expenditure in a county-based apportionment is lower than in a population-based apportionment, at least in California over the period 1926-1962.

4.3.4 Utility under different apportionment schemes

Figures 4-1 to 4-3 show the relative utility for the different apportionment schemes by the two types of people living in different sized counties. The graphs differ depending on whether government expenditure is expected to increase or decrease under the different schemes.

First, we examine the situation in which government expenditure remains the same under either apportionment scheme. Support for reapportionment is then solely dependent on the expected tax incidence under the proposed apportionment scheme. Referring to Figure 4-1, since larger counties get more power under a population apportionment scheme, their utility should be higher under

⁴Alternatively, voters with larger right-to-vote might be able to skew government expenditure such that they receive higher utility for the same amount of spending. The same comparative statics hold for this alternative specification.

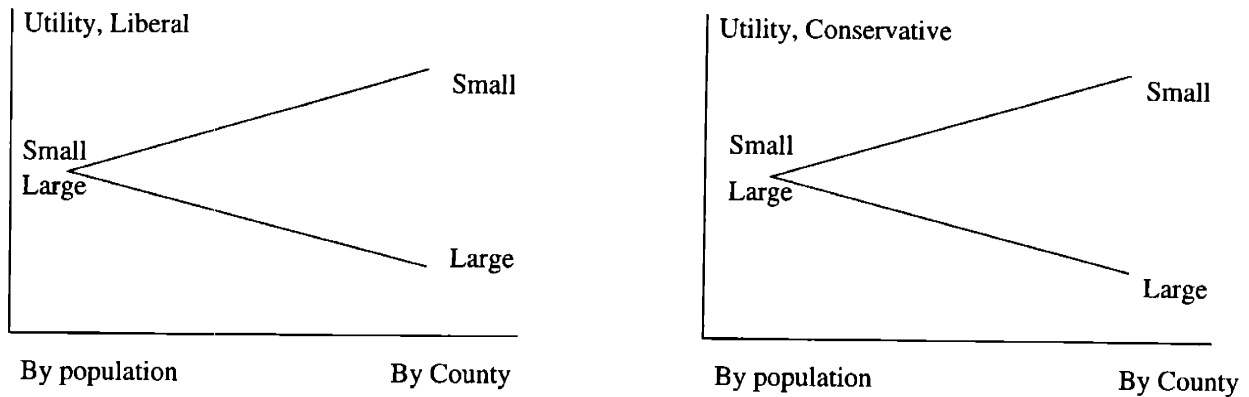


Figure 4-1: Graphs of relative utility if government expenditure remains the same

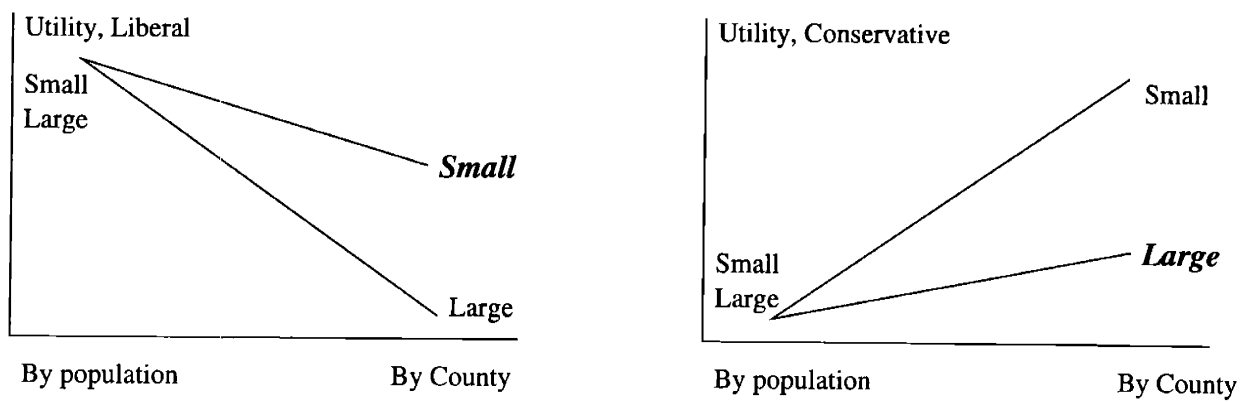


Figure 4-2: Graphs of relative utility if government expenditure is higher under apportionment by population

for the population-based rather than county-based apportionment. The converse is true for the small counties. The relative utilities should be independent of fiscal liberalism, as government expenditure remains the same. This independence implies complete support by large counties for population apportionment, and no support at all by small counties, a prediction that is directly rejected by the data. It also predicts no correlation between fiscal preferences and apportionment support, once we correct for county size. Everyone votes based on the direct power effect, and not because of fiscal virtual representation as expected future expenditure remains constant.

Next, we examine the situation in which government expenditure is higher under a population-based apportionment scheme. Regardless of county size, the liberals will have higher utility under a population-based apportionment, and the conservatives will have lower utility. Of course, the large county liberals will gain more from the switch, relative to small county liberals, due to the direct power effect. The small county conservatives will lose more, relative to the large county conservatives. Referring to Figure 4-2, differences in benefits from a switch imply incomplete

Table 4.1: Individual ideal government expenditure

	Apportionment by population		Apportionment by county	
	Small	Large	Small	Large
Liberal	2	2	$2 + \epsilon_1$	$2 - \epsilon_2$
Conservative	1	1	$1 + \epsilon_3$	$1 - \epsilon_4$

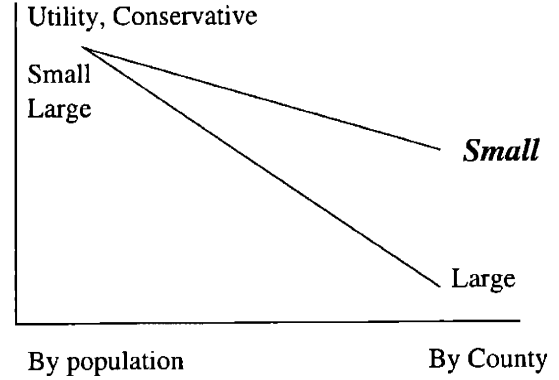
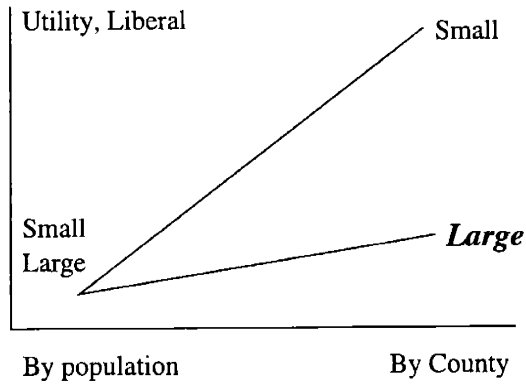


Figure 4-3: Graphs of relative utility if government expenditure is lower under apportionment by population

support from either sized county, a result that is consistent with the empirical evidence.

The figures also highlight the groups that might vote on the basis of virtual representation, rather than the direct power effect. These groups are highlighted in bold. Here we see support because of virtual representation by small county liberals (they support it because of the larger expenditure, even though they lose direct power) and no support because of virtual representation by large county conservatives (they vote against the proposition because they do not like the large expenditure implied, even though they gain power). Empirically, this model implies a positive correlation between fiscal preferences and support for population-based apportionment.

Relative to the previous case, the comparative statics are flipped for a situation in which expenditure is lower under a population apportionment scheme. As shown in Figure 4-3, the virtual representation voters are the large county liberals and the small county conservatives. Empirically, this model implies a negative correlation between fiscal preferences and support for population based apportionment.

4.4 Data

Voters in California were asked to vote on 6 reapportionment initiatives spanning the period between 1926 and 1962. The level of support for these initiatives is gauged by voting returns at the county, town, district and page levels, and an individual survey for the 1960 and 1962 initiatives. Voting returns and survey data from other ballot propositions are used to construct a government expenditure index. This section details the specifics of each initiative, highlights our data sources for each aggregation level, and describes the construction of the government expenditure index.

4.4.1 Reapportionment initiatives

Reapportionment initiatives appeared in the November general elections as ballot propositions. Appendix D.1 lists the exact propositions as they appeared in the elections. We detail the potential outcomes of each choice and the arguments presented in the leading newspapers of the day.

Choices in 1926

In 1926, voters faced two propositions involving reapportionment. Successful passage of Proposition 20 would have apportioned both houses by population as directed in the Constitution. Proposition 28 was known as the Federal plan, modeled after the national system, i.e., the Assembly was to

be reapportioned by population, while the Senate was to be reapportioned by counties, where each county was to have no more than 1 seat in the Senate. Failure of both propositions would propagate the status quo, in which both houses remained apportioned by population, based on the 1910 census.

The LA Times supported Proposition 20 but was neutral on Proposition 28. Their official recommendation stated that: "Nos. 20 and 28 are not necessarily contradictory and The Times makes no recommendation on the later" (La Times, 31 Oct 1926). Their recommendations are not surprising, considering that LA was going to be the biggest beneficiary in both houses if Proposition 20 was successful. On the other hand, Proposition 28 would have caused LA to win seats in the Assembly and to lose seats in the Senate. Their ambivalence towards Proposition 28 is thus understandable.

While San Francisco would also have gained seats in both houses under Proposition 20, and would have gained seats in the Assembly and lost seats in the Senate under Proposition 28, the San Francisco Chronicle supported Proposition 28 and opposed Proposition 20. As seen from the following, "No. 28 gives all hands a square deal, and that is the best kind to give in the long run. It is just to all concerned and assures a reasonable balance of power that will make it harder for politicians to manipulate things" (SF Chronicle, 15 Oct 1926). The Chronicle opposed Proposition 20 because it did not want San Francisco to lose power in both houses to Los Angeles. Ultimately, Proposition 28 passed while Proposition 20 failed.

Choices in 1928

Proposition 1 on the 1928 ballot sought to establish the legislative districts based on the principles of the 1926 Proposition 28. As emphasized by the LA Times, the proposition was just a particular implementation of the Federal plan. If it failed, the legislature would merely be compelled to devise another implementation/division of the Senate seats. The implication, therefore, is that this proposition had no real effect on the legislative system.

The newspapers reflected this sentiment. In contrast with 1926, there was only minimal coverage of the 1928 proposition in the San Francisco Chronicle. Furthermore, the LA Times was actually supportive of this proposition, in contrast to their ambivalence two years before. The editorial read: "The principles of the Boggs Act, that of balancing bulk of population against bulk of area, is in effect in thirty States. It is worth a trial. If found wanting, the populous areas will have the votes to change the Constitution again" (LA Times, 4 Nov 1928). With minimal cause for opposition,

1928 Proposition 1 passed.

Choices in 1948 and 1960

As a result of the 1928 elections, the Assembly was continually reapportioned as population changed, a practice that faced minimal opposition in later years. In contrast, the Senate was apportioned by counties. Proposition 13 in 1948 and Proposition 15 in 1960 were attempts to reapportion the Senate based on population. Failure of these propositions would have meant that the Senate would continue to be apportioned by counties.

Both the LA Times and the SF Chronicle were against these propositions, as they saw them as attempts by pressure groups to control the legislature (LA Times, 27 Oct 1948). The following editorial from the LA Times reflected the general newspaper sentiment:

Proposition No. 13 is an outright bid by organized labor and the professional political machine bosses of California large cities for control of the California Legislature. If approved the measure would change the method of electing State Senators by taking 16 Senatorial districts from rural and agriculture areas and allocating them to San Francisco, San Diego and Los Angeles, the latter county receiving a maximum of 10 members. The proposal is not in the public interest and would inaugurate an epoch of political bossism in this State. Vote NO. (LA Times, 29 Oct 1948)

Both measures failed.

Choices in 1962

Proposition 23 in 1962 should be seen as a compromise between apportionment by population and apportionment by counties. It proposed that the Senate was to be predominantly apportioned by counties, but LA was to get 5 extra seats, and 5 other counties were to get 1 extra seat each. Consequently, the Senate would have increased its size from 40 to 50 seats.

As reflective of the compromise, the LA Times supported the 1962 proposition. It editorized that:

It would be a grave error to believe that the proposed reapportionment would give the town a new and permanent ascendancy over the country in the California Senate: It is a play for justice, not power. It is designed to serve the needs, not the ambitions of the centers of population. (LA Times, 4 Oct 1962)

On the other hand, the Chronicle continued to maintain its opposition to the proposition. The 1962 proposition ultimately failed.

Summary statistics at the county level

County-level returns are readily available from the California State Archives. Table 4.2 provides summary statistics of the county level support for the various reapportionment initiatives, while Table 4.3 shows the correlations between the initiatives. Apportionment by population schemes (1926 Proposition 20, 1948 Proposition 48, 1960 Proposition 15 and 1962 Proposition 23) never had a level of support greater than 22% average (unweighted by county size). In contrast, the Federal plan proposal (1926 Proposition 28) and implementation (1928 Proposition 1) received much more support on average.

More than any other county, Los Angeles consistently supported county of population-based apportionment schemes; it was the least supportive of the Federal plan. The 4 counties (Alpine, Colusa, Tehama and Glenn) that occupied the other end of the spectrum for each proposition were all Northern counties, providing initial evidence of a regional effect, as alluded to in the LA Times (21 Oct 1928). Support for the four population-based propositions are positively correlated with each other, and negatively correlated with the Federal plan propositions, as expected.

Table 4.4 shows the demeaned support (in percentages) for the 4 population-based propositions for the 5 biggest counties (by population in 1962) and an aggregate number for the remaining north and south counties. Support levels in 1926 are remarkably similar to support levels in the 1960 and 1962. In contrast, support levels for the 1948 proposition are remarkably different from other propositions. The 1948 vote was seen as a organized labor issue, which might have introduced additional voting effects. The largest counties supported the population-based apportionment schemes more than the other counties, and we attribute this to the direct power effect. Again we see support for the regional effect as even the smaller counties of the South favored the population-based schemes more than the North.

Graphically, we plot the support levels for 4 representative propositions against population in Figure 4-4. Labels corresponding to the Southern counties are capitalized. Again we see a positive correlation between population and support for reapportionment for each of the 3 population-based proposals (1926 Proposition 20, 1948 Proposition 13, and 1960 Proposition 15), and a negative/zero correlation for the Federal reapportionment (1926 Proposition 28). While these correlations confirm our intuitions with regard to the direct power effect, the graphs also reveal huge anomalies that

beg to be explained: Why did many Los Angeles residents not vote for reapportionment? What is driving the outliers?

The graphs also give reasons to be suspicious of any analysis of fiscal reapportionment done at the County level. It is clear that out of the 58 counties, there are only about 10 big counties while the rest are very small, and we do not have a micro-founded functional form specification to correct for population size.⁵ Hence we use disaggregated data for our analysis.

4.4.2 Fiscal Variables

In order to support the virtual representation theory, we have to create a measure of a group's preference for the size of government expenditure. At this stage we do not distinguish between the two determinants that underlie this preference. Among the ballot propositions that voters faced every two years during the elections, some dealt with fiscal issues. These included propositions on bond issues and tax-exemptions. Support for either type of proposition would have indicated a preference for more government expenditure. Propositions that dealt with religious issues (tax exemptions to churches and religious schools) are not included here due to their more controversial nature, as well as laws governing bond issues (as opposed to bond issues themselves); the specifics of the law did not immediately translate into a government expenditure preference. Furthermore, some of the propositions in the 1920s dealt with tax increases and other tax redistribution issues. As it is unclear if people who prefer more government expenditure would have realized that greater spending/more tax exemptions required more taxes, and would have consequently voted for the tax propositions, we leave these propositions out.

Since individual propositions differ from each other, we aggregate over different spending propositions. For the district and county level analysis, we use all the propositions in the year of a reapportionment initiative, and in the 4 years preceding that. For example, the index *Fiscal* for 1928 uses bond measures from 1924, 1926 and 1928. The list of propositions that we use is found in Appendix D.2. For the page and town level data, due to data unavailability, we use only the propositions in the year of a reapportionment initiative. Each proposition is demeaned, and the average over all such propositions within the relevant time period used to smooth out proposition

⁵In fact, we did run County level regressions of Reapportionment support on $\log(\text{population})$ and fiscal preferences. Results from the Propositions in the 1920s and in 1948 conformed to the predictions of fiscal virtual representation, but propositions in the 1960s did not. However the coefficients of the fiscal variables lost significance when additional covariates were added for robustness checks, causing us to be even more suspicious of the results. On net, the County-level regressions provide ambiguous support for/against fiscal virtual representation. We also find, correcting for $\log(\text{population})$, a positive south fixed-effect.

idiosyncrasies.

Summary statistics at the county level

Table 4.5 shows summary statistics for the fiscal measures over the years, while Table 4.6 shows the correlations. In general, the fiscal measures are positively correlated with each other over time, although there is a slight negative correlation between the 1944-1948 measure and both the 1956-1960 and 1958-1962 measures. Perhaps support for veterans' bonds was affected by the on-going Second World War in the 1940s, but more research is needed to confirm this hypothesis.

4.4.3 Levels of Aggregation

Individual Survey data

We use individual poll data from the California Poll for the years 1960 and 1962.⁶ The lack of data for the individual surveys means that we have to use data from the single spending proposition that was asked for the survey-level regression. For some of the surveys, a question relating to a bond issue (*Fiscal*) and a question relating to reapportionment (*Reapport*) were asked. The bond question in 1960 (State water program) is different from the bond question in 1962 (Non-profit colleges). Other demographic variables such as the age, class, and political affiliation were recorded. The rural/urban characteristics of the region (*Urban*) were used as a proxy for county size. The survey contains information on the region of California in which those surveyed lived in. The 1962 survey has more region categories than that 1960 survey, which only differentiates between Northern and Southern California (*South*). The details of the primary regressors used are listed in Appendix D.3. Altogether data from 2711 individuals are used.

Town level

Aggregated town level ballot returns are obtained directly from Microfilm records of Election return statements from the California State Archives. The dataset contains the election returns for a random sample of 352 towns across 17 counties for the 1926 elections, and 158 towns across 11 counties for 1928. For the 1948, 1960 and 1962 elections, we have town level data for all the towns

⁶Data for this study, originally collected by the Field Research Corporation, were provided by the State Data Program of the Institute of Governmental Studies, with the assistance of the Survey Research Center, University of California, Berkeley. These organizations are not responsible for the analysis and interpretation of data appearing here.

in the Alameda and San Mateo counties. These counties were chosen because the microfilm data is recorded at the town instead of the precinct level, which made data collation easier.

The variable *Fiscal* is constructed using only bond issues and tax exemptions from the individual election year only (instead of including the previous two elections); detailed town level data for the preceding years were lacking.

District level

The district is the next level of aggregation after the town. Again the aggregated data are obtained in Microfilm from the California State Archives. San Francisco county is chosen because its records are the most legible amongst all counties, thus minimizing the risk of errors of transcription. Furthermore, it had one of the greatest number of districts among the counties, allowing for more observations.

The variable *Fiscal* is created for each election year with a reapportionment vote, using ballot propositions from the immediate 5 years preceding and including the particular election. Districts were re-shuffled after the 1960 elections, therefore the reapportionment of 1962 could not be studied.

“Page” level

The election records for Los Angeles spans many pages. Each page contains precinct level data for a subset of the precincts. These precincts are approximately divided across pages based on geographical proximity. In an attempt to study voting behavior in the largest county, we use page level data, i.e., each page contains the sum of the votes for all the precincts on that page. As page designations change every election, the variable *Fiscal* was constructed solely with the ballot propositions from the same election year (instead of the previous 5 years).

4.5 Method & Results

Based on sub-section 4.3.4, virtual representation exists if there is either a significant positive or negative correlation between innate fiscal liberalism and support for reapportionment, holding county size constant. Furthermore, a positive/negative correlation corresponds to a higher/lower government expenditure under an *Apportionment by population* scheme.

Innate fiscal liberalism is the residual of government expenditure preferences after we correct for county size through county fixed effects.

4.5.1 Hypotheses

If government expenditure increases under the population apportionment scheme, the virtual representation hypothesis would imply that innately fiscally liberal people would have voted for Proposition 20 in 1926. Both houses would then have been apportioned by population, and therefore they would both prefer a higher level of government expenditure.

On the other hand, there should not be a strong correlation between innate fiscal liberalism and votes for the 1926 Proposition 28. The Assembly would want to spend more, while the Senate would want to spend less. In a two-house system, the net effect is unclear.

Furthermore, as pointed out earlier, 1928 Proposition 1 merely dealt with a particular implementation of the 1926 Proposition 28; hence innate fiscal liberalism should not have a clear effect on voting behavior. Compared to innately fiscally conservative voters, innately fiscally liberal voters would have a stronger inclination to vote for all three of the propositions after 1930.

The signs on the coefficients are reversed if government expenditure decreases, and should be insignificantly different from zero if government expenditure remains the same.

Individual votes from survey data, and aggregated election returns at the town, district and page levels are examined. The results show coefficients that are consistent with government expenditures increasing under a population-based apportionment scheme.

4.5.2 Individual Survey

We run an Probit regression of *Reapport* on *Fiscal*, *Urban* and *South*. *Urban* is a dummy variable used as a proxy for malapportionment. We expect the urban counties to be the counties that were more malapportioned under an *Apportionment by County* scheme. *Fiscal* is used to measure innate fiscal preferences as we have partialled out the effect of malapportionment via the *Urban* variable. *South* is a dummy variable that takes on a value of 1 for the southern counties. *Urban * South* is the interaction dummy of *Urban* and *South*. We also correct for individual characteristics such as economic level, age and political party differences in the even columns. The regression is run for both the 1960 (Columns 1 and 2) and 1962 (Columns 3 and 4) elections.

The results of the regressions are shown in Table 4.7. The coefficients denote the increase in probability of voting for reapportionment if one moves from 0 to 1 for each of the dummy variables. The coefficient on the *Fiscal* variable across all the columns show that people who voted for the bond issue also tended to vote for reapportionment. The result is significant at the 1% level, and

holds for both the 1960 and 1962 polls, as predicted in the previous section. After correcting for individual characteristics, those who voted for the water bond in 1960 were 22% more likely to have voted for reapportionment, while those that voted for the tax exemption in 1962 were 8% more likely to have supported the reapportionment initiative. The differences between the two bond proposals makes it difficult to speculate on the causes for the different percentages.

The significantly positive coefficient on the *Urban* variable is indicative of the pure power effect (counties that are more negatively affected by malapportioned want to correct the malapportionment more). In 1960, the *PartyID* variable was not significant. In 1962, however, the significantly positive coefficient indicates that Democrats were more likely to support reapportionment.

Significantly positive coefficients on the *South* variable provides further evidence of a voting bloc effect that some authors have commented upon. The significant increase in power to Los Angeles county under a proposed apportionment scheme would have benefited Southern interests as a whole. The coefficients of the demographic variables and the interaction term were not significant.

The R^2 values for the individual level regressions are relatively small, and are attributed to the huge amounts of variation across heterogeneous individual actors. As the group size of the survey is increased, individual variation is reduced with aggregation and the R^2 value increases. The significance of the coefficients of all the variables were robust to the use of Ordered Probits (where a *Not Sure* reply is recorded as 0.5 instead of being dropped as previously), and the significance of the *Fiscal* coefficients are robust to the replacement of the *Urban* and *South* variables with dummy variables for each of the more detailed regions in the 1962 elections.

While the individual level data provides strong support for our hypothesis, one could argue that the single bond issues used for each of our *Fiscal* variables might not be representative of general fiscal preferences. It is with critique in mind that we turn to group level data so that we can construct measures of fiscal preferences via aggregating over different propositions.

4.5.3 Town level

Regressions of the demeaned reapportionment propositions against *Fiscal* are run. County fixed effects partial out the effects of county-size, implying that the remaining differences across towns are due to differences in innate fiscal liberalism. For robustness we run Ordinary Least Squares (OLS), Generalized Least Squares (GLS) and Robust Regressions. As the results in Table 4.8 show, the coefficients on *Fiscal* for 1926 Proposition 20, 1948 Proposition 13, 1960 Proposition 15, and 1962 Proposition 23 are all positive and significant at the 1% level. In contrast, the coefficients on

Fiscal for the 1926 Proposition 28 are not consistently positive and significant, but is consistently lower than the corresponding coefficients for the 1926 Proposition 20. These results are strongly consistent with the increase in government expenditure hypothesis.

4.5.4 San Francisco District level

Due to the small number of observations, only OLS regressions are reported. The results in Table 4.9 show that the coefficients on *Fiscal* for 1926 Proposition 20, 1948 Proposition 13, 1948 Proposition 13 and 1960 Proposition 15 are significantly positive. On the other hand, the coefficient on *Fiscal* for 1926 Proposition 28 is significantly negative, and not significantly different from 0 for 1928 Proposition 1.

These results are consistent with the earlier predictions, with the potential exception of 1926 Proposition 28. Two of the three coefficients on *Fiscal* in the town level regressions are significantly positive (albeit smaller than that for 1926 Proposition 20), while the coefficient is negatively significant in the district level regression. As previously explained, the Federal plan would have caused the Assembly to want to spend more and the Senate to want to spend less. It is possible that the net result swung one way for San Francisco as a district and the other way for California as a whole. Verification of this claim is made complicated by the presence of the 2 other options – 1926 Proposition 20 and the status quo.

4.5.5 Los Angeles Page level

We run regressions of the demeaned reapportionment propositions against *Fiscal*. Referring to Table 4.10, the coefficients of the *Fiscal* variable for the 1948 Proposition 13 and 1960 Proposition 15 are positive and significant. These results are consistent with the theoretical predictions.

While theory predicts that there should be positive and significant coefficients on the *Fiscal* variable for 1926 Proposition 20 and the 1962 Proposition 23, the empirical results show no significance and negative significance for these coefficients respectively. Furthermore, the empirical size of the 1926 Proposition 20 coefficients are smaller than those for the 1926 Proposition 28 coefficients, and we have positive significant (at the 10% level) coefficients for the 1928 Proposition 1 coefficients. These results do not conform with the theoretical predictions.

We have a couple of explanations for this deviation. Table 4.11 lists the standard deviations of the Reapportionment and *Fiscal* variables for each of the years. We find a very small standard deviation for the *Fiscal* variable in 1926, and for the 1926 Proposition 28 and the 1962 Proposi-

tion 23. This suggests that aggregation at the page level had absorbed most of the variation for these years, or alternatively, that Los Angeles had very homogeneous fiscal and reapportionment preferences for the respective years. Given that 75.8% of Los Angeles residents supported Reapportionment in 1962, it is not surprising that there is very little variation at the page level. Both these factors reduce the amount of variation that we can identify off, potentially causing counter-factual results. As will be mentioned later, for 1928 Proposition 1, the *Fiscal* coefficient is not robust to the inclusion of control variables.

4.5.6 Robustness checks

Apart from the government expenditure index, we aggregate propositions according into Communism, Criminal, Ideological liberalism and Liquor indices. Appendix D.4 lists the propositions used. These indexes are included in the regressions to partial out correlations between spending and the other potential voting issues. At the town and page level, all the indexes are simultaneously included with *Fiscal* in the regressions. However, due to the small number of observations for the district level regressions, we run only OLS, and use only one other index at any one time.

At the town and district levels, the significantly positive correlations between *Fiscal* and the reapportionment propositions remained for the population-based apportionment propositions, while remaining negative or insignificant for the other 2.

At the page level, the inclusion of the control variables did not alter the 1926, 1948, 1960 and 1962 results. For 1928, the inclusion of controls changed the coefficients from significantly positive to significantly negative, causing us to doubt the validity of the 1928 coefficient.

The coefficients on the control indexes were largely insignificant. In cases where they were significant, they were inconsistent across different aggregation levels and regression types (OLS, GLS, Robust).

As a further check, we constructed an alternative index of Fiscal preferences by dropping all the spending proposals that dealt with veterans. The presence/absence of significance for the respective town and district level coefficients were reassuringly invariant to this new index. For the page level coefficients, dropping the veteran bonds actually realigned the 1926 coefficients with theory, but concurrently caused the 1960 *Fiscal* coefficients to lose significance. The net strength of our findings is therefore unaffected by dropping Veteran bonds from our regressions.

4.5.7 Summary of results

In summary, except for some of the page level regressions (which we attribute to aggregation problems), the regressions are consistent with the virtual representation hypothesis, under the scenario where government expenditure increases in an *Apportionment by Population* scheme.

Unfortunately we do not have data on the counter-factual expenditure, and hence we can only postulate, based on the previous set of regressions, on the size of the expenditure, had the proposition passed.

4.6 Conclusion

The chapter presents integrated theory and empirical support for the virtual representation argument. In contrast to previous work, we find that innate fiscal preferences tell a more compelling story than support for propositions by the legislature in general. Voters in other states faced similar reapportionment decisions, and it will be interesting to find out if similar stories hold for the other states.

4.6.1 How significant is this effect?

To generate some numbers, we use the individual level data in 1960 to calculate the probabilities of voting for the reapportionment proposition under different combinations of *Urban*, *South* and *Fiscal*. We use the full range of interaction terms, do not display the individual characteristics that were included in the regression, and normalize the probability of supporting reapportionment for a Rural Conservative voter living in the North to be 0. As seen in table 4.12, for the 1960 survey data, the combination of Fiscal, Urban and South is sufficient to span the full probability range from 0 to 1 (and beyond). However, for 1962, the combination is only sufficient to span a probability range of only about 0.5.

4.6.2 Directions for future research

An explanation for voting patterns for the *Federal Plan* propositions (1926 Proposition 28, 1928 Proposition 1) seems to be the natural step forward. To this end, we present preliminary work of our findings in the next few paragraphs. In 1926, the Chronicle's arguments for Proposition 28 include, "No. 28 also insures immediate reapportionment. Then the Assembly and Senate would

be checks, each on the other. There could be no city boss control of the Legislature” (SF Chronicle, 26 Oct 1926). This suggests that the Federal plan was an effective way to “lock-in” the status quo.

The propositions for each of the elections have two distinct origins. More than half of them are proposed constitutional amendments that both houses had already passed, and voter approval would have been the final step before the constitution was accordingly amended. In contrast, the other propositions were initiatives that had been proposed by private citizens without passage through the legislature. We construct an index, *LegisBias*, which takes the average of all the non-bond propositions (demeaned) proposed by the legislature, subtracting away the average of all the non-bond initiatives (demeaned).

We then regress the reapportionment measure against *Fiscal* and *LegisBias*, and find support (at the town, district and county levels) that *LegisBias* is positively significant for the two Federal Plan propositions (except for the GLS regression at the county level for 1926 Proposition 28 “The Federal Plan”, where the coefficient on *LegisBias* was positive, but not significant), and negatively significant or insignificant for the other 4 propositions. The behavior of the *Fiscal* coefficients is robust to the inclusion of this additional variable. However, more thorough research must be done to examine the validity of this bias.

The reapportionment question in the United States House of Representatives was settled with *Baker v. Carr*. This is not the case in the United States Senate, or in other countries. The results presented in this chapter might provide a useful reference point to understand the underlying dynamics for these situations.

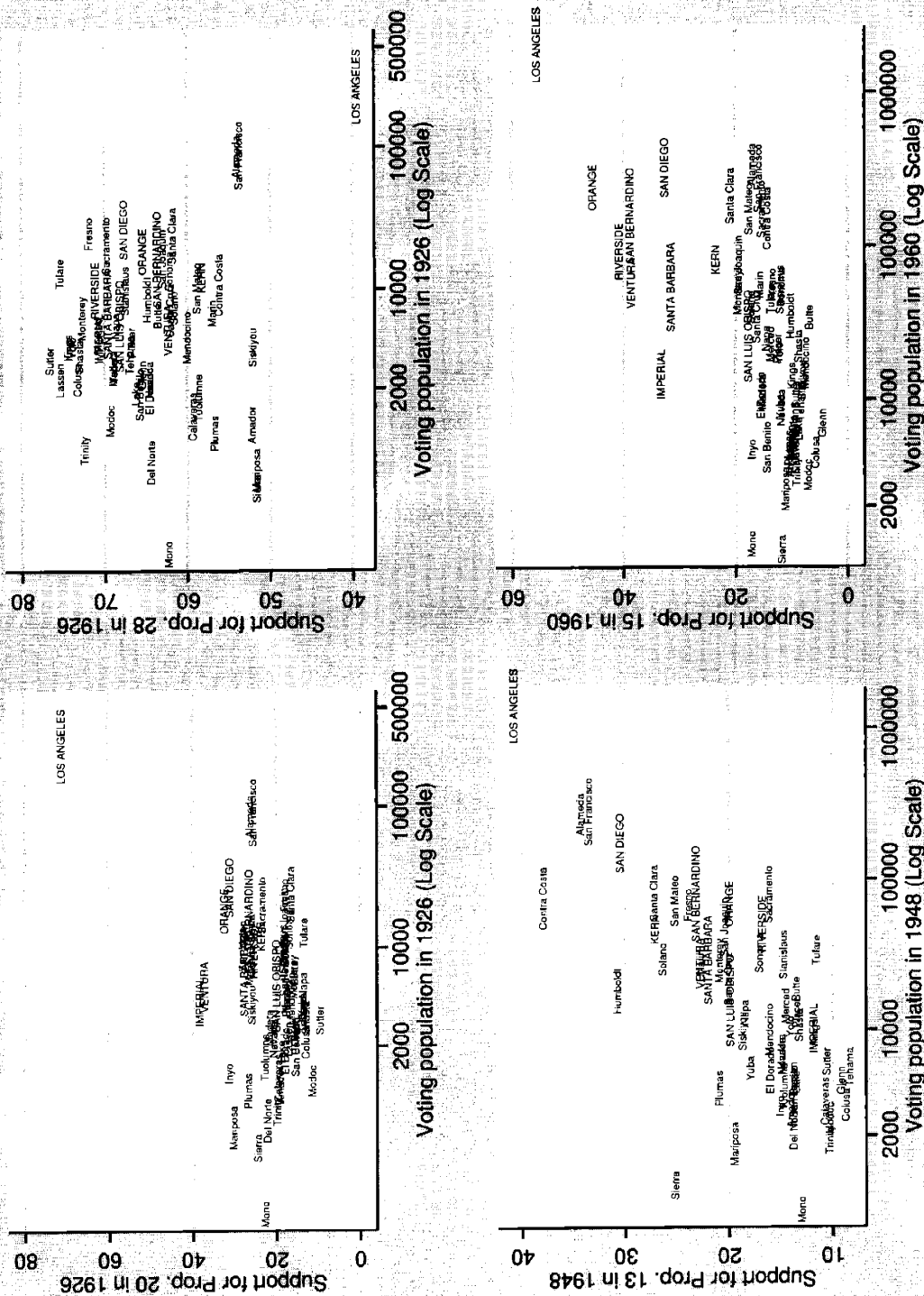


Figure 4-4: Graphs of County level support for Reapportionment

Table 4.2: Summary statistics of the reapportionment propositions

Variable	Obs	Mean	Std. Dev	Min	Min county	Max	Max county
1926 Prop 20	58	21.4	9.3	8.3	Alpine	71.1	Los Angeles
1926 Prop 28	58	64.9	7.9	39.4	Los Angeles	84.8	Alpine
1928 Prop 1	58	77.1	10.2	35.6	Los Angeles	90.7	Colusa
1948 Prop 13	58	18.6	7.4	8.4	Tehama	40.7	Los Angeles
1960 Prop 15	58	16.5	10.5	4.5	Glenn	55.5	Los Angeles
1962 Prop 23	58	21.2	12.7	9.2	Glenn	75.8	Los Angeles

Table 4.3: Cross correlations of the Reapportionment propositions

	1926 P.20	1926 P.28	1928 P.1	1948 P.13	1960 P.15	1962 P.23
1926 Prop 20	1.0					
1926 Prop 28	-0.59	1.0				
1928 Prop 1	-0.73	0.71	1.0			
1948 Prop 13	0.54	-0.55	-0.67	1.0		
1960 Prop 15	0.74	-0.19	-0.42	0.48	1.0	
1962 Prop 23	0.74	-0.25	-0.46	0.49	0.96	1.0

Table 4.4: Relative support for population apportionment for various counties

County	Relative support				Population			
	1926	1948	1960	1962	1926	1948	1960	1962
Los Angeles (S)	49.7	22.1	39.0	54.6	274,553	1,421,388	2,081,304	1,786,036
San Diego (S)	9.7	11.8	16.1	13.2	27,044	174,312	326,124	302,226
Alameda (N)	4.8	15.7	.6	3.3	86,006	271,097	339,491	288,405
Orange (S)	10.9	1.3	29.0	40.6	18,580	68,770	245,552	240,635
San Francisco (N)	4.1	14.9	-.6	4.1	91,359	281,477	276,887	221,035
Other (S)	6.9	1.8	15.5	16.5	60,344	260,054	486,772	435,943
Other (N)	-2.8	-1.7	-4.2	-5.0	256,928	843,738	1,528,145	1,402,918

Notes — Population refers to the number of voters who voted for the particular proposition. (N): North, (S): South

Table 4.5: Summary statistics of the fiscal measures

Variable	Obs	Mean	Std. Dev	Min	Min county	Max	Max county
1922-1926	58	0	5.1	-14.5	Modoc	13.2	Sacramento
1924-1928	58	0	5.1	-13.7	Modoc	11.3	Sacramento
1944-1948	58	0	2.7	-5.1	Orange	5.5	Mariposa
1956-1960	58	0	2.9	-5.5	Modoc	6.4	Imperial
1958-1962	58	0	3.2	-6.0	San Francisco	6.1	Santa Barbara

Table 4.6: Cross correlations of the fiscal measures

	1922-1926	1924-1928	1944-1948	1956-1960	1958-1962
1922-1926	1.0				
1924-1928	0.89	1.0			
1944-1948	0.11	0.20	1.0		
1956-1960	0.39	0.27	-0.07	1.0	
1958-1962	0.47	0.36	-0.01	0.86	1.0

Table 4.7: Probit regressions at the Individual level

	1960 (1)	1960 (2)	1962 (3)	1962 (4)
<i>Fiscal</i>	.225** [.03]	.219** [.03]	.085** [.03]	.081** [.03]
<i>PartyID</i>	-	.018 [.02]	-	.041** [.01]
<i>Urban</i>	.218** [.06]	.221** [.06]	.069 [.05]	.085* [.05]
<i>South</i>	.433** [.07]	.432** [.07]	.219** [.08]	.227** [.08]
<i>Urban * South</i>	-.054 [.07]	-.050 [.07]	.015 [.08]	.006 [.08]
Survey dummies?	Yes	Yes	Yes	Yes
Individual characteristics? ⁴	No	Yes	No	Yes
R^2	.20	.21	.07	.08
OBS	1293	1293	1418	1418

Notes — Robust standard errors are in parentheses. *, ** Variables significant at the 10% and 1% level respectively. Coefficients show the increase in probability in voting for the the proposition when the dummy variable increases from 0 to 1. ⁴Economic level, age, and political party affiliation.

Table 4.8: Regressions at the Town level against innate fiscal preferences

	1926 Proposition 20			1926 Proposition 28			1928 Proposition 1		
	OLS	GLS	Robust	OLS	GLS	Robust	OLS	GLS	Robust
<i>Fiscal</i>	.867** [.11]	.652** [.11]	.666** [.06]	.250* [.13]	-.398 [.37]	.259** [.07]	-.116 [.12]	-.402 [1.2]	-.113 [.10]
CONS	.000 [.01]	-.022** [.00]	-.051 [.09]	.000 [.01]	-.029** [.01]	-.063 [.11]	.101** [.03]	.110 [.08]	.110** [.03]
County Dummies?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	.70	-	-	.31	-	-	.60	-	-
OBS	352	352	352	352	352	352	163	163	163

	1948 Proposition 13			1960 Proposition 15			1962 Proposition 23		
	OLS	GLS	Robust	OLS	GLS	Robust	OLS	GLS	Robust
<i>Fiscal</i>	.819** [.05]	.767** [.09]	.826** [.05]	.987** [.20]	.678** [.18]	1.033** [.21]	.811** [.06]	.699** [.07]	.777** [.06]
CONS	-.006 [.006]	.006 [.008]	-.005 [.006]	.0026 [.008]	.0008 [.007]	.0028 [.008]	.0036 [.005]	-.0008 [.005]	.0047 [.005]
County Dummies?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	.83	-	-	.32	-	-	.79	-	-
OBS	57	57	57	58	58	58	59	59	59

Notes — Robust standard errors are in parentheses. *, ** Variables significant at the 10% and 1% level respectively.

Table 4.9: OLS Regressions at the San Francisco district level against innate fiscal preferences

Year Proposition	1926 20	1926 28	1928 1	1948 13	1960 15
<i>Fiscal</i>	1.49** [.22]	-2.19** [.60]	1.05 [2.1]	2.11** [.07]	1.46** [.14]
CONS	.000 [.00]	.000 [.01]	.000 [.03]	.000 [.01]	.000 [.01]
R^2	.78	.49	.02	.97	.78
OBS	13	13	13	8	6

Notes — Robust standard errors are in parentheses. *, ** Variables significant at the 10% and 1% level respectively.

Table 4.10: Regressions at the Los Angeles page level against innate fiscal preferences

	1926 Proposition 20			1926 Proposition 28			1928 Proposition 1		
	OLS	GLS	Robust	OLS	GLS	Robust	OLS	GLS	Robust
<i>Fiscal</i>	.017 [.53]	-.062 [.57]	.106 [.50]	.195 [.22]	.151 [.23]	.232 [.21]	.578* [.31]	.618** [.30]	.878** [.22]
CONS	.000 [.01]	.005 [.01]	-.002 [.01]	.000 [.003]	.000 [.003]	.000 [.003]	.000 [.01]	.001 [.01]	-.007 [.01]
R^2	.00	-	-	.02	-	-	.12	-	-
OBS	45	45	45	45	45	45	47	47	47

	1948 Proposition 13			1960 Proposition 15			1962 Proposition 23		
	OLS	GLS	Robust	OLS	GLS	Robust	OLS	GLS	Robust
<i>Fiscal</i>	.913** [.05]	.907** [.05]	.902** [.04]	1.285** [.12]	1.357** [.14]	1.237** [.11]	-.065* [.03]	-.056* [.03]	-.083** [.03]
CONS	.000 [.004]	.000 [.004]	-.006 [.004]	.000 [.003]	.001 [.003]	.001 [.003]	.000 [.002]	.001 [.002]	.000 [.002]
R^2	.79	-	-	.45	-	-	.03	-	-
OBS	103	103	103	153	153	153	140	140	140

Notes — Robust standard errors are in parentheses. *, ** Variables significant at the 10% and 1% level respectively.

Table 4.11: Comparison of Standard Deviations

County	Year	Proposition	Reapp. Stdev	<i>Fiscal</i> Stdev	No. obs
Los Angeles	1926	20	.0519	.01667	45
Los Angeles	1926	28	.0212	.01667	45
Los Angeles	1928	1	.0619	.0374	47
Los Angeles	1948	13	.097	.0942	103
Los Angeles	1960	15	.0515	.0268	153
Los Angeles	1962	23	.0244	.0666	140

Table 4.12: Voting probabilities for different demographic combinations

North			South		
	Urban	Rural		Urban	Rural
Fiscal Liberal	.7244	.5324	Fiscal Liberal	1.0461	.8578
Fiscal Conservative	.4108	0	Fiscal Conservative	.9084	.6595
1960 Proposition 15					
North			South		
	Urban	Rural		Urban	Rural
Fiscal Liberal	.0666	.0686	Fiscal Liberal	.4186	.5146
Fiscal Conservative	-.0019	0	Fiscal Conservative	.1942	.2840
1962 Proposition 23					

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Appendix A

Appendix to Chapter 1

A.1 Proofs of Lemmas

Lemma 1.

$$\frac{dF}{d\tilde{X}} = \frac{\frac{\gamma\lambda_2 N}{M} \cdot \frac{\partial^2 D}{\partial R^2} - \frac{N\lambda_2}{M(T-F)} \frac{\partial D}{\partial R}}{\frac{\partial^2 C}{\partial F^2} - \gamma\lambda_1 \frac{\partial^2 D}{\partial R^2} + \frac{N\lambda_2 \tilde{X}}{M(T-F)^2} \frac{\partial D}{\partial R}}.$$

where

$$\gamma = \lambda_1 - \lambda_2 \frac{N\tilde{X}}{M(T-F)} > 0.$$

Proof. Differentiating Equation 1.11 by $\tilde{X}$ ($\tilde{X}$ does not enter Equation 1.4 directly except through its effect on F and H_2) yields

$$(\lambda_1 - \lambda_2 \frac{N\tilde{X}}{M(T-F)}) \frac{\partial^2 D}{\partial R^2} (\lambda_1 \frac{dF}{d\tilde{X}} + \lambda_2 \frac{N}{M}) + \frac{\partial D}{\partial R} \cdot \frac{-N\lambda_2}{M} \cdot \frac{(T-F) + \tilde{X} \frac{dF}{d\tilde{X}}}{(T-F)^2} = \frac{\partial^2 C}{\partial F^2} \cdot \frac{dF}{d\tilde{X}}.$$

Let

$$\gamma = \lambda_1 - \lambda_2 \frac{N\tilde{X}}{M(T-F)} > 0.$$

I have $\lambda_1 > \lambda_2$, and $\frac{N\tilde{X}}{M(T-F)}$ is the fraction of corrupt deals amongst strangers out of the total number of deals with strangers, a fraction that is smaller than 1. Hence $\gamma > 0$. Collecting terms proves the lemma. $\square$

Lemma 2. *I have*

$$\ln(S) \cdot G_P + B \cdot G_{PP} < 0.$$

Proof. $0 < S < 1, \ln(S) < 0, G_P > 0, B > 0$ and $G_{PP} < 0$ yields the result directly. $\square$

A.2 Proofs of Propositions

Proof of Proposition 1:

Proof. Solving the maximization Equation 1.12 yields

$$\frac{\partial L}{\partial Q} \cdot \frac{dQ}{dCut} = 0.$$

Since $\frac{\partial L}{\partial Q} > 0$, this implies

$$Cut \cdot S^X \cdot \ln(S) \cdot \frac{dX}{dCut} + S^X = 0,$$

i.e.

$$\ln(S) \cdot Cut \frac{dX}{dCut} = -1.$$

Deriving $\frac{dX}{dCut}$ requires differentiating Equation 1.1 with respect to Cut , yielding

$$(\ln(S)G_P + B \cdot G_{PP}) \frac{dP}{dCut} = 0.$$

Hence from Lemma 2,

$$\frac{dP}{dCut} = 0 = -E + B \frac{dX}{dCut},$$

and $E = 1$ for officials under corrupt supervisors (the only people that this decision applies to in the first place),

$$\frac{dX}{dCut} = \frac{1}{B}.$$

Together,

$$Cut = -\frac{B}{\ln(S)}$$

□

Proof of Proposition 2:

Proof. Consider a shock $Shock$ that lowers the probability that an official will engage in a corrupt deal with a stranger. Then $\frac{dX}{dShock} < 0$. The effect on net corruption is

$$\frac{d(X+Y)}{dShock} = (1 + \frac{\partial Y}{\partial F} \cdot \frac{\partial F}{\partial \tilde{X}} \cdot \frac{\partial \tilde{X}}{\partial X}) \frac{dX}{dShock}.$$

$\frac{\partial \tilde{X}}{\partial X} = 1$ from rational expectations. The requirement that $\frac{d(X+Y)}{dShock}$ has the same sign as $\frac{dX}{dShock}$

implies

$$1 + \frac{\partial Y}{\partial F} \cdot \frac{\partial F}{\partial \tilde{X}} > 0,$$

which leads to

$$-1 < \frac{M}{N} \cdot \frac{\partial F}{\partial \tilde{X}}.$$

From $D_{RR} < 0$, $D_R > 0$, $C_{FF} > 0$, and Lemma 1, I know that each of the individual terms in the denominator of $\frac{dF}{d\tilde{X}}$ is positive, and I substitute Equation 1.11 to get the condition

$$-\frac{d^2C}{dF^2} + \frac{1}{T-F} \frac{dC}{dF} + \gamma\lambda_1 \frac{d^2D}{dR^2} - \frac{\lambda_1}{T-F} \frac{dD}{dR} < \gamma\lambda_2 \frac{d^2D}{dR^2} - \frac{\lambda_2}{T-F} \cdot \frac{dD}{dR}$$

or

$$\frac{d^2C}{dF^2} - \frac{1}{T-F} \frac{dC}{dF} + \gamma(\lambda_2 - \lambda_1) \frac{d^2D}{dR^2} + \frac{\lambda_1 - \lambda_2}{T-F} \frac{dD}{dR} > 0.$$

Since $\lambda_1 > \lambda_2$, a sufficient condition for this inequality to be satisfied is $\frac{d^2C}{dF^2} - \frac{1}{T-F} \cdot \frac{dC}{dF} > 0$, i.e. there is sufficient curvature in the cost function. A D_{RR} that is hugely negative will also aid in the satisfaction of this inequality - i.e. if the demand function is sufficiently curved. $\square$

Proof of Proposition 3:

Proof. From Lemma 1,

$$\frac{\partial F}{\partial \tilde{X}} = \frac{\frac{\gamma\lambda_2 N}{M} \frac{\partial^2 D}{\partial R^2} - \frac{N\lambda_2}{M(T-F)} \frac{\partial D}{\partial R}}{\frac{\partial^2 C}{\partial F^2} - \gamma\lambda_1 \frac{\partial^2 D}{\partial R^2} + \frac{N\lambda_2 \tilde{X}}{M(T-F)^2} \frac{\partial D}{\partial R}}.$$

Since $C_{FF} > 0$, $D_R > 0$, $D_{RR} < 0$, $\lambda_1 > \lambda_2 > 0$, and $\gamma > 0$,

$$\frac{\partial F}{\partial \tilde{X}} < 0.$$

$\square$

Proof of Proposition 4:

Proof. Differentiating Equation 1.3 by Y yields

$$\frac{dG}{dY} \cdot \ln(S) + B \frac{d}{dY} \left(\frac{dG}{dP} \right) = 0,$$

and since exogenously determined W implies $\frac{dW}{dY} = 0$,

$$(\ln(S) \cdot G_P + B \cdot G_{PP})(A + B \frac{dX}{dY}) = 0.$$

From Lemma 2,

$$\frac{dX}{dY} = -\frac{A}{B}.$$

Given $0 < A < B$,

$$-1 < \frac{dX}{dY} = -\frac{A}{B} < 0.$$

□

Proof of Proposition 5:

Proof. Total revenue over all merchants is

$$\lambda_1 M F + \lambda_2 N \tilde{X}.$$

Given 4, differentiating the previous equation by F yields

$$\lambda_1 M + \lambda_2 N \frac{d\tilde{X}}{dF} = \lambda_1 M - \lambda_2 \frac{M A}{B},$$

which is the marginal revenue of increasing F by 1 for all M merchants. The marginal revenue per friendship created is then (divide by M)

$$\lambda_1 - \lambda_2 \frac{A}{B}.$$

□

Proof of Proposition 6:

Proof. Differentiating Equation 1.3 by S implies

$$\frac{dG}{dS} \cdot \ln(S) + \frac{G}{S} + B \frac{d}{dS} \left(\frac{dG}{dP} \right) = 0,$$

or

$$\frac{G}{S} + (B \cdot G_{PP} + \ln(S) G_P) \left(-E \frac{dCut}{dS} + A \frac{\partial Y}{\partial \tilde{X}} \frac{\partial \tilde{X}}{\partial X} \frac{dX}{dS} + B \frac{dX}{dS} \right) = 0.$$

Using Proposition 1,

$$\frac{dCut}{dS} = \frac{B}{S(\ln(S))^2}.$$

Together with Proposition 2 and rational expectations ($\frac{\partial \tilde{X}}{\partial X} = 1$)

$$\frac{dX}{dS} = -\frac{G - (B \cdot G_{PP} + \ln(S)G_P) \frac{EB}{(\ln(S))^2}}{S(B \cdot G_{PP} + \ln(S)G_P)(B + A \frac{M}{N} \cdot \frac{\partial F}{\partial \tilde{X}})} > 0$$

since $0 < S < 1, \ln(S) < 0, B > A > 0$ and $0 > \frac{M}{N} \cdot \frac{\partial F}{\partial \tilde{X}} > -1$ by assumption. With the assumptions in Proposition 2, total corruption also rises. $\square$

Proof of Proposition 7:

Proof. Differentiating Equation 1.3 by W implies

$$\frac{dG}{dW} \cdot \ln(S) + B \frac{d}{dW} \left(\frac{dG}{dP} \right) = 0,$$

or

$$(B \cdot G_{PP} + \ln(S) \cdot G_P) \left(1 + B \frac{dX}{dW} + A \frac{\partial Y}{\partial \tilde{X}} \frac{\partial \tilde{X}}{\partial X} \frac{dX}{dW} \right) = 0.$$

Using Lemma 1, Proposition 2 and rational expectations ($\frac{\partial \tilde{X}}{\partial X} = 1$)

$$\frac{dX}{dW} = -\frac{1}{B + A \cdot \frac{M}{N} \cdot \frac{\partial F}{\partial \tilde{X}}} < 0$$

based the inequalities used to prove Proposition 6. The assumptions in Proposition 2 imply that total corruption also decreases. $\square$

Proof of Proposition 8:

Proof. Compare two scenarios, $\mathcal{A}$ and $\mathcal{B}$ with $\tilde{S}_{\mathcal{B}} < S_{\mathcal{B}} = S_{\mathcal{A}} = \tilde{S}_{\mathcal{A}}$. In Scenario $\mathcal{B}$, before the actual government policy is revealed, the probability that merchants do not squeal is lower than in Scenario $\mathcal{A}$. However the revealed probabilities are the same. As seen from Section 1.3.2, X is a function of W , S and F . $\tilde{X}$ denotes the Period one anticipated X that depends exogenously on $\tilde{W}$ and $\tilde{S}$, and endogenously by F . The analogous results of $\frac{dX}{dS} > 0$ and $\frac{dX}{dW} < 0$ hold in anticipation, i.e.

$$\frac{d\tilde{X}}{d\tilde{S}} > 0, \frac{d\tilde{X}}{d\tilde{W}} < 0,$$

i.e.

$$\tilde{X}_B < \tilde{X}_A,$$

while Proposition 3 implies

$$F_B > F_A,$$

and given actual $S_A = S_B$, Proposition 4 yields

$$X_B + Y_B > X_A + Y_A,$$

that there is more corruption if the government promised to crack-down on corruption but did not do so, compared to a scenario where the government did not make any promise at all. The same results hold for an anticipated but un-executed increase in official wages, W . $\square$

Proof of Proposition 9:

Proof. Differentiating Equation 1.11 with respect to N yields (N does not enter Equation 1.4 directly except through its effect on F and H_2 via Equation 1.5)

$$\gamma \frac{\partial^2 D}{\partial R^2} \left(\lambda_1 \frac{dF}{dN} + \frac{\lambda_2}{M} \cdot \frac{d(N\tilde{X})}{dN} \right) + \frac{\partial D}{\partial R} \cdot \frac{-\lambda_2}{M} \cdot \frac{(T-F) \frac{d(N\tilde{X})}{dN} + N\tilde{X} \frac{dF}{dN}}{(T-F)^2} = \frac{\partial^2 C}{\partial F^2} \cdot \frac{dF}{dN}.$$

Rearranging and using Lemma 1,

$$\frac{dF}{dN} = \frac{d(N\tilde{X})}{dN} \cdot \frac{\frac{\gamma\lambda_2}{M} \cdot \frac{\partial^2 D}{\partial R^2} - \frac{\partial D}{\partial R} \cdot \frac{\lambda_2}{M(T-F)}}{\frac{\partial^2 C}{\partial F^2} - \gamma\lambda_1 \frac{\partial^2 D}{\partial R^2} + \frac{\partial D}{\partial R} \cdot \frac{\lambda_2 N\tilde{X}}{M(T-F)^2}} = \frac{d(N\tilde{X})}{dN} \cdot \left(\frac{1}{N} \frac{\partial F}{\partial \tilde{X}} \right).$$

Then

$$\frac{dF}{dN} = \left(N \frac{d\tilde{X}}{dN} + \tilde{X} \right) \cdot \left(\frac{1}{N} \frac{\partial F}{\partial \tilde{X}} \right) = \left(N \frac{\partial \tilde{X}}{\partial X} \cdot \frac{\partial X}{\partial F} \cdot \frac{\partial F}{\partial N} + \tilde{X} \right) \cdot \left(\frac{1}{N} \frac{\partial F}{\partial \tilde{X}} \right),$$

and rearranging,

$$\frac{dF}{dN} = \frac{\frac{\tilde{X}}{N} \frac{\partial F}{\partial \tilde{X}}}{1 - \frac{\partial F}{\partial \tilde{X}} \cdot \frac{d\tilde{X}}{dF}} = \frac{\frac{\tilde{X}}{N} \frac{\partial F}{\partial \tilde{X}}}{1 + \frac{M}{N} \cdot \frac{\partial F}{\partial \tilde{X}} \cdot \frac{A}{B}} < 0,$$

given $-1 < \frac{M}{N} \cdot \frac{\partial F}{\partial \tilde{X}} < 0$ and $B > A > 0$. $\square$

A.3 Choice of Functional Form

This Appendix section explains the rationale behind the functional form choice used for this analysis. First, additively separable wealth and diminishing marginal utility over total wealth seem to be plausible, commonly-used assumptions.

Second, a more general functional form specification of Equation 1.4 might take the format of

$$U_M = \max_{0 < F < T} D'(H_1(F), H_2(F), W^a, S^a, F') - C(F)$$

in which I only assume $D_{H_1} > 0, D_{H_1 H_1} < 0, D_{H_2} > 0, D_{H_2 H_2} < 0, D_{H_1 H_2} < 0$. The first-order condition of this functional form yields

$$\frac{\partial D}{\partial H_1} \cdot \frac{dH_1}{dF} + \frac{\partial D}{\partial H_2} \cdot \frac{dH_2}{dF} = \frac{dC}{dF},$$

and plugging in $H_1 = F$ and Equation 1.9 gives

$$\frac{\partial D}{\partial H_1} + \frac{\partial D}{\partial H_2} \cdot \frac{-N\tilde{X}}{M(T-F)} = \frac{dC}{dF},$$

where I substitute F for F' in the Nash Equilibrium, which still looks tractable.

However, the next crucial step of differentiating by $\tilde{X}$ makes the equation unmanageable.

$$\begin{aligned} \frac{\partial^2 C}{\partial F^2} \cdot \frac{dF}{d\tilde{X}} &= \frac{\partial^2 D}{\partial H_1^2} \cdot \frac{\partial H_1}{\partial F} \cdot \frac{dF}{d\tilde{X}} + \frac{\partial^2 D}{\partial H_1 \partial H_2} \cdot \frac{\partial H_2}{\partial \tilde{X}} + \frac{\partial^2 D}{\partial H_2^2} \cdot \frac{\partial H_2}{\partial \tilde{X}} \cdot \frac{-N\tilde{X}}{M(T-F)} \\ &\quad + \frac{\partial^2 D}{\partial H_1 \partial H_2} \cdot \frac{-N\tilde{X}}{M(T-F)} + \frac{\partial D}{\partial H_2} \cdot \frac{-N}{M} \cdot \frac{(T-F) + \tilde{X} \frac{dF}{d\tilde{X}}}{(T-F)^2} \end{aligned}$$

with $\frac{dH_2}{dF} = \frac{-N}{M}$. A necessary condition for $\frac{\partial F}{\partial \tilde{X}} < 0$ is

$$\frac{\partial^2 D}{\partial H_1 \partial H_2} > \frac{-1}{\tilde{X}} \frac{\partial D}{\partial H_2}.$$

An increase in $\tilde{X}$ increases H_2 , the number of corrupt deals with official-strangers for each merchant. If $\frac{\partial^2 D}{\partial H_1 \partial H_2}$ were too negative, $\frac{\partial D}{\partial H_1}$ switches sign from positive to negative as H_2 increases. This sign switch alone violates the initial $D_{H_1} > 0$ assumption previously made. In such a situation, to maintain the equality of the first-order condition expressed in Equation A.3, F has to increase, implying $\frac{\partial F}{\partial \tilde{X}} > 0$ which is rather implausible. Returning to the traffic congestion example, a

decreased cost of using one road should not lead to an increased usage of the other road.

In general though, the huge level of arbitrariness over the relative size of cross-partial, first and second order derivatives makes it impossible to draw any policy conclusion from this functional form. Hence the original formulation was chosen instead.

Appendix B

Appendix to Chapter 2

B.1 Proofs of Propositions

Proof of Proposition 10:

Proof.

$$\begin{aligned}\Pr(\beta_{i,t+1} > \beta_{c,t+1}) &= \Pr(\beta_{w,t} > \beta_{l,t} + \epsilon) \\ &= \Pr(\epsilon < \beta_{w,t} - \beta_{l,t}) \\ &= \frac{\beta_{w,t} - \beta_{l,t}}{c} + \frac{1}{2}\end{aligned}$$

for β not too different. The case (never seen empirically) that the incumbent/competitor is so far ahead that he wins with a 100% vote share is ruled out. Hence we assume $c > 2|\beta_{w,t} - \beta_{l,t}|$. $\square$

Proof of Proposition 11:

Proof. For $\gamma = 0$, the indifferent voter $\bar{j}$ finds that

$$\beta_{i,t} - G\theta_{\bar{j}} = \beta_{c,t} - G(1 - \theta_{\bar{j}}),$$

yielding an incumbent and challenger vote share of

$$\alpha_{i,t} = \theta_{\bar{j}} = \frac{1}{2} + \frac{\beta_{i,t} - \beta_{c,t}}{2G}, \alpha_{c,t} = \frac{1}{2} - \frac{\beta_{i,t} - \beta_{c,t}}{2G}$$

respectively, assuming that the β s are not too different, i.e. $|\beta_{i,t} - \beta_{c,t}| < G$. The above equation

implies that if $\gamma = 0$,

$$\Pr(\alpha_{p,t+1} > \frac{1}{2}) = \Pr(\beta_{p,t+1} > \beta_{p,t+1}),$$

i.e. a person p will win election $t + 1$ only if he is more able than the other person competing in the election $t + 1$ (as shown later, this result is not true once $\gamma > 0$). Equivalently, investors look at the vote share of the winning party for election t , and figure out that

$$\beta_{w,t} - \beta_{l,t} = G(2\alpha_{w,t} - 1).$$

Combining Proposition 10 and Equation 2.5 then yields

$$\begin{aligned} I(\alpha_{w,t}) &= kE\{\Pr(\alpha_{i,t+1} > \frac{1}{2}) + (1 - \Pr(\alpha_{i,t+1} > \frac{1}{2}))D|\alpha_{w,t}\} \\ &= kE\{\Pr(\beta_{i,t+1} > \beta_{c,t+1}) + (1 - \Pr(\beta_{i,t+1} > \beta_{c,t+1}))D|\alpha_{w,t}\} \\ &= k\left(\left(\frac{\beta_{w,t} - \beta_{l,t}}{c} + \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{\beta_{w,t} - \beta_{l,t}}{c}\right)D\right) \\ &= k\left(\frac{\beta_{w,t} - \beta_{l,t}}{c}\right)(1 - D) + \frac{k(1 + D)}{2} \\ &= k\left(\frac{\beta_{w,t} - \beta_{l,t}}{c}\right)(1 - D) + I' \\ &= k\left(\frac{G(2\alpha_{w,t} - 1)}{c}\right)(1 - D) + I' \end{aligned}$$

where I' corresponds to the amount of investment when the winning vote-share is 50%. $\square$

Proof of Proposition 12:

Proof. Consider the equilibrium where the incumbent is more able ($\beta_{i,t} > \beta_{c,t}$) and wins ($\beta_{w,t} = \beta_{i,t}$). Following Equation 2.6, the indifferent voter, $\bar{j}$ in election t satisfies

$$\beta_{i,t} - G\theta_{\bar{j}} + \gamma I(\alpha_{i,t}) = \beta_{c,t} - G(1 - \theta_{\bar{j}}) + \gamma I(\alpha_{i,t} - \frac{1}{n})$$

given from Proposition 11 that

$$I(\alpha_{i,t}) - I(\alpha_{i,t} - \frac{1}{n}) = \frac{2kG(1 - D)}{nc}$$

which leads to

$$\alpha_{w,t} = \alpha_{i,t} = \theta_{\bar{j}} = \frac{1}{2} + \frac{\beta_{i,t} - \beta_{c,t}}{2G} + \frac{\gamma k(1 - D)}{nc}.$$

Compared to the parallel expression in the proof of Proposition 11, signal jamming causes a *portion* of voters to over-vote, OV , by

$$OV = \alpha_{w,t}|_{actual\gamma} - \alpha_{w,t}|_{\gamma=0} = \frac{\gamma k(1-D)}{nc}$$

which is independent of the relative β s. Of course, it is not possible for more people to over-vote if everyone is already voting for the incumbent, hence we impose yet another condition such that the relative β s are not too different, i.e.

$$\frac{1}{2} > \frac{\beta_{i,t} - \beta_{c,t}}{2G} + \frac{\gamma k(1-D)}{nc}.$$

Now consider the other equilibrium where the incumbent is more able ($\beta_{i,t} < \beta_{c,t}$) but loses because voters self-fulfillingly expect the challenger to win ($\beta_{c,t} = \beta_{w,t}$). Following Equation 2.7, the indifferent voter, $\underline{j}$ satisfies

$$\beta_{i,t} - G\theta_{\underline{j}} + \gamma I(\alpha_{i,t} - \frac{1}{n}) = \beta_{c,t} - G(1 - \theta_{\underline{j}}) + \gamma I(\alpha_{i,t})$$

which leads to

$$\alpha_{i,t} = \theta_{\underline{j}} = \frac{1}{2} + \frac{\beta_{i,t} - \beta_{c,t}}{2G} - \frac{\gamma k(1-D)}{nc}$$

or equivalently,

$$\alpha_{w,t} = \alpha_{c,t} = 1 - \theta_{\underline{j}} = \frac{1}{2} - \frac{\beta_{i,t} - \beta_{c,t}}{2G} + \frac{\gamma k(1-D)}{nc}$$

which is the vote share of the winning challenger. The amount of over-voting for the winning candidate c here is the same as what candidate i received in the previous equilibrium, even though $\beta_{i,t} > \beta_{c,t}$ in both cases. This can only happen if

$$\frac{\gamma k(1-D)}{nc} > \frac{\beta_{i,t} - \beta_{c,t}}{2G}$$

so that there is enough over-voting for the lower β candidate to win.

□

Proof of Proposition 13:

Proof. For mixing to occur, the mixing voter, θ_m , has to be indifferent between

$$\beta_{i,t} - G\theta_m + \Pr(\alpha_{i,t} > \frac{1}{2})\gamma I(\alpha_{i,t} | (\alpha_{i,t} > \frac{1}{2})) + \left(1 - \Pr(\alpha_{i,t} > \frac{1}{2})\right) \gamma I(\alpha_{c,t} - \frac{1}{n} | (\alpha_{i,t} < \frac{1}{2}))$$

and

$$\beta_{c,t} - G(1 - \theta_m) + \Pr(\alpha_{i,t} > \frac{1}{2})\gamma I(\alpha_{i,t} - \frac{1}{n} | (\alpha_{i,t} > \frac{1}{2})) + \left(1 - \Pr(\alpha_{i,t} > \frac{1}{2})\right) \gamma I(\alpha_{c,t} | (\alpha_{i,t} < \frac{1}{2})).$$

Since the probabilities are ex ante known, every voter with $\theta_j < \theta_m$ will vote for the incumbent, while every voter with $\theta_j > \theta_m$ will vote for the challenger. Hence the only voter who can mix and remain pivotal has $\theta_m = \frac{1}{2}$. This is the median voter for odd n , and no such voter exists for even n . Then

$$\alpha_{w,t} = \frac{1}{2} + \frac{1}{2n}.$$

Solving, we get

$$\Pr(\alpha_{i,t} > \frac{1}{2}) = \frac{1}{2} - \frac{\beta_{i,t} - \beta_{c,t}}{2\gamma(I(\frac{1}{2} + \frac{1}{2n}) - I(\frac{1}{2} - \frac{1}{2n}))} = \frac{1}{2} - (\beta_{i,t} - \beta_{c,t}) * \frac{nc}{4\gamma k G(1 - D)}$$

The comparative statics, at least on the relative β , make sense, as to mix, you need to assign a lower probability to the stronger event. The constraint that probabilities lie between 0 and 1 imply that

$$|\beta_{i,t} - \beta_{c,t}| < \frac{2\gamma k G(1 - D)}{nc}$$

for mixing to occur. □

Proof of Proposition 14:

Proof. Investors once again make investment I based on their best prediction of relative β s, predicting $\beta_{w,t} - \beta_{l,t} = f(\alpha_{w,t})$. The dependence of I on predicted $\beta_{w,t} - \beta_{l,t}$ will be discussed later in this proof, but for now

$$I(\alpha_{w,t}) = k \cdot H(E(\beta_{w,t} - \beta_{l,t} | \alpha_{w,t})) = k \cdot H(f(\alpha_{w,t})),$$

where H is the net probability that an investment is not expropriated by the government after the subsequent election for an expected winning probability of the current election. Then the increased

investments as a result of increasing $\alpha_{w,t}$ by $\frac{1}{n}$ is thus

$$I(\alpha_{w,t}) - I(\alpha_{w,t} - \frac{1}{n}) = \frac{k f'(\alpha_{w,t})}{n} H'(f(\alpha_{w,t}))$$

which analogously to Proposition 12, leads to

$$\alpha_{w,t} = \frac{1}{2} + \frac{\beta_{w,t} - \beta_{l,t}}{2G} + \frac{\gamma k f'(\alpha_{w,t})}{2Gn} H'(f(\alpha_{w,t})).$$

Since $f'(\alpha_{w,t})$ is a rule followed by investors when they do not know $\beta_{w,t} - \beta_{l,t}$, f' cannot depend on the relative β s. Given

$$\frac{d(\alpha_{w,t})}{d(\beta_{w,t} - \beta_{l,t})} = \frac{1}{2G}$$

which implies that investors think that an increase in $\alpha_{w,t}$ implies a $2G$ increase in $\beta_{w,t} - \beta_{l,t}$, i.e.

$$f'(\alpha_{w,t}) = 2G,$$

then the amount of over-voting is

$$OV = \frac{\gamma k}{n} H'(f(\alpha_{w,t})).$$

$H(f(\alpha_{w,t}))$ is the risk of government expropriation after the subsequent election, where

$$H(f(\alpha_{w,t})) = E(\Pr(\alpha_{i,t+1} > \frac{1}{2}) + (1 - \Pr(\alpha_{i,t+1} > \frac{1}{2})) D | f(\alpha_{w,t})) = D + E(\Pr(\alpha_{i,t+1} > \frac{1}{2}) | f(\alpha_{w,t})) (1 - D).$$

Differentiation implies the amount of over-voting is

$$OV = \frac{\gamma k (1 - D)}{nc} E(\Phi(\beta_{i,t} - \beta_{c,t} + \frac{c}{2}) - \Phi(\beta_{i,t} - \beta_{c,t} - \frac{c}{2})).$$

□

Proof of Proposition 15:

Proof. Since $-W$ is the minimum point where the incumbent could win, indifferent voter over-votes in favor of the incumbent, and his vote is just sufficient for the incumbent to win. Hence

$$\alpha_{i,t} = \frac{1}{2} = \frac{1}{2} + \frac{\beta_{i,t} - \beta_{c,t}}{2G} + \frac{\gamma k (1 - D)}{nc} E(\Phi(\beta_{i,t} - \beta_{c,t} + \frac{c}{2}) - \Phi(\beta_{i,t} - \beta_{c,t} - \frac{c}{2}) | f(\alpha_{i,t}) > \frac{1}{2}),$$

evaluated at $\beta_{i,t} - \beta_{c,t} = -W$. Since all points to the left of $-W$ are pure vote-level equilibria where the challenger definitely wins,

$$\Phi(-W - \frac{c}{2}) = 0$$

we get

$$W = \frac{2G\gamma k(1-D)}{nc} \Phi(-W + \frac{c}{2}).$$

Since we assume that $c > 2|\beta_{i,t} - \beta_{c,t}| = 2W$, then $0 < -W + \frac{c}{2}$, and thus $\Phi(-W + \frac{c}{2}) = r$ or $\Phi(-W + \frac{c}{2}) = 1$.

□

Appendix C

Appendix to Chapter 3

C.1 Evaluation Organizations

Organization (Evaluation Office website)	Evaluation Office	Year	% of projects independently evaluated	Level of evaluation detail available online for independent evaluations	Rating scale for independent evaluations	Evaluation categories for independent evaluations
National Organizations						
Department for International Development DFID http://62.189.42.51/DFIDstage/policiesandpriorities/files/ev_home.htm	Evaluation Department (EVD)	1968 ⁱ onwards	Very few: An average of 9 projects were independently evaluated annually between 1993-1999. ⁱⁱ Only 25% of bilateral expenditure is covered by a Project Completion Report (PCR) done by program managers ⁱⁱⁱ , "there is no system of independent verification, although one has been mooted", and "more significantly, PCRs are not used" ^{iv} .	All evaluation reports and their summaries are either online or can be ordered online.	Numerical rating scales for numerous dimensions for the PCRs, but these are not independent evaluations. Independent observations are mostly prose-based. Since independent evaluations are carried out by different evaluators, no standardized rating scale/category exists.	
U.S. Agency for International Development USAID http://www.dcc.org/partners/eval.cfm	Bureau for Policy and Program co-ordination, ^v Center for Development Information and Evaluation (CDIE)	Before 1995 ^{vi}	All projects were evaluated, ^{vii} but how many are independent?			
		After 1995 to 2000	Regarded poorly. See footnote. ^{viii}			
		After 2000 ^x	Searching through the USAID evaluation website shows that most project evaluations continue to be institutionally co-authored by the USAID mission to a specific country, although CDIE has some evaluations that do not involve the country mission as a co-author. ^x	Individual project evaluations. Mostly joint authored with country mission, raising questions about independence. Annual Report uses country macro-economic values as performance indicators. ^{xi}	Qualitative measures used in its performance and accountability report. ^{xii}	
Regional Development Banks						
African Development Bank. AfDB http://www.afdb.org/about_aadb/OPEV.htm	Operations Evaluation Office	1964(est.) – 2001				
	Operations Evaluation Department (OPEV) ^{xiii}	2001 onwards ^{xiv}	Annual Reviews of Development Effectiveness starting in 2003. ^{xv} Annual Review of Results of Operations Evaluation apparently published, according to http://www.afdb.org/about_aadb/OPEV_evaluation_guidelines.htm , however it is not available online.	Project evaluations prior to 1997 are available at the DAC website. ^{xvi} Qualitative abstracts of Project Performance Audit Reports from 1994 to 1997 available online or by request to the web-master. However, it is not clear if these were independent evaluations.	- ARDE 2003 not currently available online yet. -	
Asian Development Bank ADB http://www.adb.org/Evaluation/	Operations Evaluation Department	1974 onwards ^{xvii}	40% evaluated ^{xviii}	Individual project reports (from 1995 onwards). ^{ix} Quantitative panel data grouped by sector and region available. ^{xx} Summary tables collating evaluation details per project evaluated. ^{xxi}	Overall project rating: Highly Successful, Successful, Partly Successful, Unsuccessful ^{xxii}	Relevance, Effectiveness, Efficiency, Sustainability, Institutional Development (REESI) ^{xxiii}
European Bank for Reconstruction and Development	Project Evaluation Department	1991 (year of establishment) ^{xix} to 2003	44% (1991 to 1998) ^{xxv}	Examples of successful and less successful projects. Time series quantitative data of performance for all projects grouped together. ^{xxvi} 40 project summaries and lessons		

EBRD http://www.ebrd.org/projects/eval/index.htm				learned (each 1 page long) representing "a cross-section of EBRD investment operations" published online. Previous year overview reports and all of the other Project Completion Reports are internal documents. ^{xxvii}		
		2003 onwards	76% (1993 to 2002) ^{xxviii}	In 2003, for the first time, Annual Evaluation Overview Report available online, giving quantitative evaluation data for each project evaluated since 1993.	Quantitative rating scales (> 3 possible ratings) for each category ^{xxix}	Transition impact, Environmental performance of sponsor and bank, extent of environmental change, overall rating ^{xxx}
Inter-American Development Bank IADB http://www.iadb.org/cont/evo/evo_eng.htm	Evolve over time. (i) Group of Three Controllers, (ii) Office of External Review and Evaluation and the Operations Evaluation Office, (iii) Office of Evaluation ^{xxxi}	1959 – 1998	Seldom undertaken, even then, only by borrower. ^{xxxii} Exact figures unknown, but we find that no ex-post evaluation was carried out for Mexico throughout the 1990s, ^{xxxiii} which gives cause for worry.			
	Office of Evaluation and Oversight ^{xxxiv} (OVE)	1999 onwards ^{xxxv}	6 to 7 country programs evaluated each year (at least from 1999 to 2001). However, the lack of ex-post program evaluation in the 1990s means that countries are evaluated on macro-performance. The presence of multiple donors makes assigning credit hard. ^{xxxvi} Very few project evaluations are done independently. ^{xxxvii}	Country program evaluations, each covering over a ten year period. ^{xxxviii} No cross-country or cross-sector comparisons.	Quantitative figures, but these are based on project evaluations that were primarily self-evaluated, ^{xxxix} and sometimes based on the most recent Project Performance Monitoring Reports (as opposed to the Project Completion Reports which were seldom done ^{xl}).	Relevance, coherence, efficiency and effectiveness ^{xli}
International Organizations						
International Monetary Fund IMF http://www.imf.org/external/np/ieo/index.htm	Office of Internal Audit and Inspection	Before 1996	Insignificant numbers of external evaluation			
	Evaluation Group of Executive Directors ^{xlii}	Between 1996 to 2000	Trial run which covered a range of topics and were of different scale. ^{xliii}			
	Independent Evaluation Office (EVO)	After 2000	At capacity, 5 projects to be undertaken each year, including both country-specific cases ^{xliv} and broader thematic questions ^{xlv} .	Commitment to promptly publish all reports, unless under exceptional circumstances ^{xlvi} No cross-sector (for example, comparing the performance of capital account crisis intervention versus the prolonged use of IMF resources) or time series comparisons available yet.	While not explicit, key cross-country macro-economic indicators are provided, and implicitly used as indicators of performance. Within the 2 reports published thus far, there is an attempt to	As at Jan 2004, 4 reports have been published. ^{xlvii} These have looked at surveillance, program effectiveness and IMF governance. The medium program is

					distinguish between different levels of "success".	given in its annual report. ^{xlviii}
United Nations Development Program UNDP http://www.un.org/oe/	Office of Evaluation and Strategic Planning (OESP) ^{xlix}	Before 1996	For programs <US\$1 million, country managers decide if project should be evaluated. Mandatory evaluations for projects >US\$1 million. ^l Even then, the compliance for mandatory evaluations was less than 80%. ^{li}			
	Evaluation Office (EO)	1996-1999				
	Evaluation Office (EO)	After 1999	Requirement that all projects over US\$1 million be evaluated is abolished. ^{lii} Decentralized evaluation process, with country managers selecting the evaluation teams and designing the term of reference. ^{liii} The EO concentrates on independent, country-level Assessment of Development Results (5 to 10 countries a year), and sector-thematic level evaluations. ^{liv}	Individual project reports available online. ^{lv} Quantitative time series data for each performance indicator ^{lvi} (all projects lumped together), and cross-sector (for projects evaluated from 1999 to 2000) performance impact indicators. ^{lvii}	Yes/Partial/No and Significant/Satisfactory/Poor ^{lviii}	Typically covers Relevance, Performance and Success (Impact, Sustainability, and Contribution to capacity building). ^{lx} However, this criterion is different from the previous year, which raises some questions. ^{lxi}
World Bank http://www.worldbank.org/oe/	Operations Evaluation Department	1973 ^{lii} onwards	Independent evaluation at the Country, Sector and Project levels. Project Performance Assessment Reports for 25 percent of all completed projects. ^{liii}	Individual Project reports. ^{liiii} Quantitative panel data grouped by sector and region available. ^{liv} Summary tables collating evaluation details per project evaluated. ^{lv}	Outcome: Highly Satisfactory (Sat), Sat, Moderately Sat, Moderately UnSat, UnSat, Highly UnSat. Sustainability: Highly Likely, Likely, Highly Unlikely, Unlikely Institutional Development: Substantial, Modest, Negligible. ^{lvii}	Outcome, Sustainability, Institutional Development. ^{lviii}

We only look at the level of evaluation detail available online for the latest incarnation of the evaluation office.

ENDNOTES FOR APPENDIX C.

ⁱ Year of first independent evaluation as given by: Department of International Development. 2003. *Catalogue of DFID Evaluation Studies*, DFID Evaluations Department. Available online at (http://www.dfid.gov.uk/Pubs/files/eval_studies_catalogue.pdf)

ⁱⁱ Michael Flint et al. 2002. "How effective is DFID? Development effectiveness report 2001", 2nd Draft, DFID, March 30th, 2002, Paragraph 118.

ⁱⁱⁱ Ibid, paragraph 115.

^{iv} Ibid, paragraph 116.

^v United States Agency for International Development. "Agency Reorganization: Generic Functional Statements." Cited 11 Feb 2004. Available online (<http://www.usaid.gov/about/reform/functions.html>)

^{vi} Clapp-Wincek, Cynthia and Richard Blue. 2001. "Evaluation of Recent USAID Evaluation Experience", Washington: Center for Development Information and Evaluation, Working paper No. 320, June, page iii.

^{vii} Ibid, page iii.

^{viii} Ibid, page 37. Here we face the related problems of:

(a) Few evaluations done: "Of most concern is the very limited number of in-depth, program evaluations."

(b) Lack of learning even if evaluations are carried out: "Most of the evaluation work that is being done is being done by partners. The partner organizations are learning from the experience; USAID is not".

(c) Lack of independence: The USAID managers decide if they want to do the evaluations.

(d) The Fly-in approach: "Scopes ask a team to come for 4-6 weeks and interview the mission, the activity staff, and 'representatives' of the local people. There isn't enough time to get any kind of representative sample. The team frequently tells the USAID manager pretty much what he already knows."

^{ix} Quote from *ibid*, page 1, "The ADS 200 series (Sept. 2000) added a new dimension to evaluations. The 'Reform Vision' in ADS 200 states these expectations: 'Applying the lessons of successes and failure systematically and providing leadership in tackling complex problems that demand multi-agency or multi-donor responses.'"

^x DuRette, Jean and Glenn Slocum. 2001. "The Role of Transitional Assistance: The Case of East Timor". Washington DC: U.S. Agency for International Development.

^{xi} Center for Development Information and Evaluation. 2001. *FY 2000 Performance Overview*. Washington DC: United States Agency for International Development. See Page 29 for example.

^{xii} United States Agency for International Development. 2002. *USAID Performance and Monitoring Report, FY 2002*. Washington DC: United States Agency for International Development. See Appendix 2.

^{xiii} African Development Bank Group. "The ADB Group – In Brief" Cited 11 Feb 2004. Available online (http://www.afdb.org/knowledge/documents/ADB_in_brief.htm)

^{xiv} United States Department of the Treasury. 2003. "African Development Bank Group". Cited 11 Feb 2004. Available online (http://www.ustreas.gov/offices/international-affairs/intl/fy2003/tab10_afdbg_afdb.pdf).

^{xv} African Development Bank Group. 2002. "Strategic Plan 2003-2007". Available online (http://www.afdb.org/knowledge/publications/pdf/adb_strategic_plan2003-2007e.pdf), See page 52.

^{xvi} See <http://www.dac-evaluations-cad.org/dac/>.

^{xvii} Asian Development Bank, Operations Evaluation Department. 2003. "Annual review of evaluation activities in 2002", Asian Development Bank. See page 61.

^{xviii} Asian Development Bank, "Frequently Asked Questions". Cited 11 February 2004. Available online (<http://www.adb.org/Evaluation/faqs.asp>)

^{xix} All reports online since 1995, at Asian Development Bank, <http://www.adb.org/Evaluation/reports.asp>.

^{xx} Asian Development Bank, Operations Evaluation Department. 2003. "Annual review of evaluation activities in 2002", Asian Development Bank. See page 67.

^{xxi} Ibid, pg. 47.

^{xxii} Ibid, pg. 47.

^{xxiii} Asian Development Bank, "PPMS Project Framework and Performance Indicators 2", Slide Show. Available online (http://www.adb.org/Documents/Slideshows/PPMS/4b_PPMS_Indicators.pdf)

^{xxiv} Project Evaluation Department. 1999. "Project Evaluation Department" (Brochure) United Kingdom: European Bank for Reconstruction and Development. See page 6.

^{xxv} Ibid. See pg 6.

^{xxvi} European Bank for Reconstruction and Development. 2003. *Annual Report 2002: Annual Review and Financial Report*, London: European Bank for Reconstruction and Development. See page 70.

^{xxvii} Private Communication, Fredrik Korfker, Corporate Director, Evaluation, EBRD. September 9, 2003.

- ^{xxviii} European Bank for Reconstruction and Development, Project Evaluation Department. 2003. *Annual Evaluation Overview Report 2003*, United Kingdom: European Bank for Reconstruction and Development. See page 1.
- ^{xxix} Ibid. See Appendix 2.
- ^{xxx} Ibid. See Appendix 2.
- ^{xxxi} Inter-American Development Bank, Office of Evaluation and Oversight, OVE. 2001. *Annual Report of the Evaluation Office 2000*, Washington D.C.: Inter-American Development Bank, See Chapter 2.1
- ^{xxxii} Inter-American Development Bank, Office of Evaluation and Oversight, OVE. 2002. *Annual Report of the Evaluation Office 2001*, Washington D.C.: Inter-American Development Bank, See pg 25.
- ^{xxxiii} Inter-American Development Bank, Office of Evaluation and Oversight, OVE. 2003. *Country Program Evaluation: Mexico. 1990-2000*. Washington D.C.: Inter-American Development Bank, See pg 29.
- ^{xxxiv} See <http://www.iadb.org/cont/evo/ovedocs.htm>
- ^{xxxv} Inter-American Development Bank, Office of Evaluation and Oversight, OVE. 2001. *Annual Report of the Evaluation Office 2000*, Washington D.C.: Inter-American Development Bank, See Chapter 2.1
- ^{xxxvi} Inter-American Development Bank, Office of Evaluation and Oversight, OVE. 2002. *Country Program Evaluation: Peru. 1990-2000*. Washington D.C.: Inter-American Development Bank, See pg 13.
- ^{xxxvii} Inter-American Development Bank, Office of Evaluation and Oversight, OVE. 2003. *Country Program Evaluation: Costa Rica. 1990-2001*. Washington D.C.: Inter-American Development Bank, See footnote 20.
- ^{xxxviii} Inter-American Development Bank, Office of Evaluation and Oversight, OVE. 2002. *Annual Report of the Evaluation Office 2001*, Washington D.C.: Inter-American Development Bank. See pg. 7
- ^{xxxix} Inter-American Development Bank, Office of Evaluation and Oversight, OVE. 2003. *Country Program Evaluation: Costa Rica. 1990-2001*. Washington D.C.: Inter-American Development Bank, See footnote 20.
- ^{xl} Ibid. See pg 40. Out of the 4 categories of projects, only the Public Reform Sector has Project Completion Reports.
- ^{xli} Inter-American Development Bank, Office of Evaluation and Oversight, OVE. 2002. *Annual Report of the Evaluation Office 2001*, Washington D.C.: Inter-American Development Bank. See pg. 7.
- ^{xlii} International Monetary Fund. Evaluation Group of Executive Directors. 2000. "Review of Experience with Evaluation in the Fund". Available online at (<http://www.imf.org/external/np/eval/2000/031400.HTM>)
- ^{xliii} Ibid. See chapter 4.
- ^{xliv} Unlike the other agencies, the IMF looks after Macro-economic performance, hence country-level evaluations make sense here.
- ^{xlv} International Monetary Fund, Evaluation Group. 2001. "Progress in making the independent evaluation office (IEO) operational." Available online at (<http://www.imf.org/external/np/eval/2001/103101.htm>)
- ^{xlvi} International Monetary Fund, Executive Board. 2000. "IMF Executive Board Report to the IMFC on the Establishment of the Independent Evaluation Office (EVO) and its Terms of Reference". Available online at (<http://www.imf.org/external/np/eval/2000/091200.htm>)
- ^{xlvii} Independent Evaluation Office, International Monetary Fund: "Draft Issues Paper for an Evaluation of Technical Assistance Provided by the IMF", Nov 2003, "Fiscal Adjustment in IMF-Supported Programs", Sept 2003, "IMF and the Recent Capital Account Crises: Indonesia, Korea, Brazil", July 2003 and "Evaluation of Prolonged Use of IMF Resources", 2002
- ^{xlviii} International Monetary Fund, Independent Evaluation Office. 2003. *IEO Annual Report 2003*. Washington DC, International Monetary Fund. See pg 10.
- ^{xlix} United Nations Development Program and the United Nations Population Fund, Executive Board. 2000. "Evaluation: Report of the Administrator". Available online at (<http://www.undp.org/execbrd/pdf/dp00-34e.pdf>) See pg 2.
- ⁱ United Nations Development Program, Evaluation Office. 2001. "Development effectiveness: Review of evaluative evidence". New York: United Nations Development Program. See Page 15.
- ⁱⁱ United Nations Development Program and the United Nations Population Fund, Executive Board. 2001. "Evaluation: Report of the Administrator". Available online at (<http://www.undp.org/execbrd/pdf/dp01-26e.pdf>) See Table 6.
- ^{hi} United Nations Development Program, Evaluation Office. 2002. "Handbook on monitoring and evaluating for results." New York: United Nations Development Program. See Part 1, Table 3.
- ^{liii} Ibid. See Part 2, pg. 45.
- ^{liv} United Nations Development Program, Evaluation Office. 2001. "Development effectiveness: Review of evaluative evidence". New York: United Nations Development Program. See Page 15.

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- ^{lv} See United Nations Development Program Evaluations Database. <http://www.undp.org/eo/database/index.htm>
- ^{lvi} United Nations Development Program, Evaluation Office. 2001. "Development effectiveness: Review of evaluative evidence". New York: United Nations Development Program. See Page 15 to 19.
- ^{lvii} Ibid. See Page 35.
- ^{lviii} Ibid. See Page 19.
- ^{lix} Ibid. See Page 15.
- ^{lx} Ibid. See Page 10.
- ^{lxi} World Bank, Operations Evaluation Department. 2002. *OED: The first 30 years*. Conference Proceedings: September 23, 2002. Available online at (http://www.worldbank.org/wbi/B-SPAN/docs/oed_thirty.pdf)
- ^{lxii} World Bank, Operations Evaluation Department. 2002. *2001 Annual Review of Development Effectiveness*. Washington DC: World Bank. See Annex D.
- ^{lxiii} See <http://lnweb18.worldbank.org/servlet/OEDSearchServlet?SearchType=byField&PerPage=20&DbURL=oed/oeddoclib.nsf&Series=OED%20Project%20Evaluation>
- ^{lxiv} World Bank, Operations Evaluation Department. 2003. *2002 Annual Review of Development Effectiveness*. Washington DC: World Bank. See Table 5.
- ^{lxv} Ibid. See Table 14.
- ^{lxvi} Ibid. See Table 14.
- ^{lxvii} Ibid. See Table 14.

C.2 Table of Interventions

The following abbreviations are used in this table

C: Intervention Category

E: Education

P: Direct poverty reduction

H: Health

O: Others

S: Is the program self-sustainable? We claim that a proposed program is not self-sustainable if the researchers ensured compliance amongst the test subjects. This is an issue for the clinical evaluations of nutritional supplements, which look at the impact of a supplement assuming compliance. In real life, compliance cannot be assumed.

R: Randomized?

Y: Yes

N: No

Intervention	C	Evaluation cited	S	R	Benefits	Incl. cost?
Water & Sanitation						
Clean water through a water container with a cover and a sprout	H	Lee, Roberts et al. 2001. "Keeping clean water clean in a Malawi refugee camp: a randomized intervention trial", <i>Bulletin of the World Health Organization</i> , 79:4, pp. 280-287. Self-sustainable because Malawian field worker used, and "this [how to use the bucket] educational message generally took less than one minute and was never reinforced or restated during the study". Not scaled up because it was carried out in a specific instance of a refugee camp that had experienced a cholera outbreak.	Y	Y	Reduced diarrhea	No
Latrine provision	H	Daniels, D.L. et al. 1990. "A case-control study of the impact of improved sanitation on diarrhoea morbidity in Lesotho", <i>Bulletin of the World Health Organization</i> , 68:4, pp. 455-63. Selection bias: Ownership depends on constituency, and distance to the recruiting health facility.	Y	N	Reduced diarrhea	No. Not random.
Latrine provision	H	Esrey S.A. 1996. "Waste, waste, and well-being: a multicountry study", <i>American Journal of Epidemiology</i> , March, 143:6, pp. 608-623. Potential non-random location of latrines.	Y	N	Reduced diarrhea, taller and heavier children	No. Not random
Education for water-sanitation behavior	E H	Stanton, B.F. 1987. "An educational intervention for altering water-sanitation behaviors to reduce childhood diarrhea in urban Bangladesh. II. A randomized trial to assess the impact of the intervention on hygienic behaviors and rates of diarrhea", <i>American Journal of Epidemiology</i> , February, 125:2, pp. 292-301.	Y	Y	Reduced diarrhea	Yes. (We assume that if everyone goes to school, the schools will do the education)
Historical clean water interventions in America	H	Cutler, David and Grant Miller. 2003. "Clean water measures in American History", power-point slides, private communication. Attempts to deal with potential endogenous placement.	Y	N	Clean water reduces mortality	Yes (although not random).
Health-care services		Summary: Gelband, H. and S. Stansfield. 2001. "The Evidence Base for Interventions to Reduce Under Five Mortality in Low and Middle-income Countries", Commission on Macroeconomics and Health, Working Group 5, Working Paper 9.				
Midwife services	H	Walker, Damien et al. 2002. "An economic analysis of midwifery training programmes in South Kalimantan, Indonesia", <i>Bulletin of the World Health Organization</i> , 80:1, pp. 47-55.	Y	N	Better skills	No. Not random.

		Not random due to participation selection bias.				
Midwife services	H	Frankenberg, Elizabeth and Duncan Thomas. 2001. "Women's health and pregnancy outcomes: Do services make a difference?", <i>Demography</i> , May, 38:2, pp. 253-265.	Y	N	Increased women Body Mass Index	No. Not random
		Actual program, but selective non-random midwife placement.				
Reduction in antenatal care	H	Munjanja S.P., G. Lindmark and L. Nystrom. 1996. "Randomized controlled of a reduced-visits programme of antenatal care in Harare, Zimbabwe." <i>Lancet</i> , Aug, 348:9034, pp. 364-369.	Y	Y	No effect on maternal/fetal outcomes	Cost reduction, not included
		Actual reduction in number of visits, clearly sustainable.				
Reduction in antenatal care	H	Villar, J. et al. 2001. "WHO antenatal care randomized trial for the evaluation of a new model of routine antenatal care", <i>Lancet</i> , Nov, 357:9268, pp. 1551-1564.	Y	Y	No effect on maternal/fetal outcomes	Cost reduction, not included
		Actual reduction in number of visits, clearly sustainable.				
Home-based neonatal care	H	Bang, A.T. et al. 1999. "Effect of home-based neonatal care and management of sepsis on neonatal mortality: field trial in rural India", <i>Lancet</i> , December, 354:9194, pp. 1955-1961.	Y	N	Avert 1 death per 18 neonates cared for	No
		Control villages were not randomly chosen: Villages were used as controls because suitable women could not be found, or population was less than 300.				
Nutritional Supplements		Summary: Behrman, Jere R. and James C. Knowles. 2003. "Assessing the Economic Returns to Investing in Youths in Developing Countries", mimeo, University of Pennsylvania, and Nemer L., H. Gelband, P. Jha. 2001. "The Evidence Base for Interventions to Reduce Malnutrition in Children Under Five and School-age Children in Low and Middle-Income Countries", Commission on Macroeconomics and Health, Working Group 5, Working Paper 11				
De-worming drugs	E H	Kremer, Michael and Edward Miguel. 2004. "Worms: Identifying Impacts on Education and Health in the Presence of Treatment Externalities", <i>Econometrica</i> . January, 72:1 pp. 159-217.	Y	Y	Improved health and school participation, even in neighborhood schools	Yes
		Self-sustainable because "medical treatment was delivered to schools by Kenya Ministry of Health public health nurses and ICS public health officers" and this was done within the community setting (we see some children not participating).				
Iron supplementation	E H	Bobonis, Gustavo; Edward Miguel and Charu Sharma, 2003 "Iron Deficiency Anemia and School Participation." Pratham Research Conference in the USA, January 30-31, 2004. Texas, USA.	Y	Y	Increased participation, reduced absenteeism	Yes
		Existing pre-school network used.				
Zinc	H	Brown KH et al. 2002. "Effect of supplemental zinc on the growth and serum zinc concentrations of prepubertal children: a meta-analysis of randomized controlled trials." <i>American Journal of Clinical Nutrition</i> . June, 75:6, pp. 1062-71.	N	Y	Positive responses in health and weight	No. Not program
		Not self-sustainable because of overtly heavy fieldworker involvement - "in most cases, confirmation that the supplements were successfully delivered to the study subjects".				
Oral Iodized Oil supplementation	H	Claudine Cobra et al. 1997. "Infant Survival Is Improved by Oral Iodine Supplementation", <i>Journal of Nutrition</i> . April. 127:4, pp. 574-578.	Y	Y	Improved infant survival	Yes
		Added to the ongoing EPI program.				
Vitamin A supplementation	H	Grotto, I., et al. 2003. "Vitamin A supplementation and childhood morbidity from diarrhea and respiratory infections: a meta-analysis." <i>Journal of Pediatrics</i> . March, 142:3, pp. 297-304.	N	Y	"No consistent overall protective effect on the increase of diarrhea"	No
		Many clinical trials carried out over the years. The above meta-analysis shows that the net impact is not consistently positive.				

Supplements for lactating women	H	Anne Tinker, Kathleen Finn, and Joanne Epp. 2000. "Improving Women's Health: Issues and Interventions", Washington, DC: The World Bank.	Y	N	Reduced low weight babies	No. Not random
		Supplements were provided daily on a <i>volunteer</i> basis at a centralized location - there was no compulsion to consume it.				
Folate supplementation (Literature review)	H	Mahomed K. 1997. "Folate supplementation in pregnancy (Cochrane Review)". In: <i>The Cochrane Library</i> , August. Oxford: Update Software.	-	-	Not enough evidence to evaluate effectiveness on clinical outcomes	No
Anti-platelet supplementation (Literature review)	H	Knight M et al. 2000 "Antiplatelet agents for preventing and treating pre-eclampsia (Cochrane Review). In: <i>The Cochrane Library</i> , February, 2000:2. Oxford: Update software, 2000.	-	-	Timing and dosage knowledge insufficient	No
Supplemental feeding (Literature review)	H	Rush D. 2000. "Nutrition and maternal mortality in the developing world". <i>American Journal of Clinical Nutrition</i> . 72(suppl), pp. 212S-240S.	-	-	Insufficient knowledge to decide if nutritional supplements are good overall.	No
Tuberculosis						
		Summary: Borgdorff, M. W., K. Floyd and J. F. Broekmans. 2001. "Interventions to reduce tuberculosis mortality and transmission in low and middle-income countries: effectiveness, cost-effectiveness, and constraints to scaling up", Commission on Macroeconomics and Health, Working Group 5, Working Paper 8.				
Short-course chemotherapy	H	China Tuberculosis Control Collaboration. 1996. "Results of directly observed short-course chemotherapy in 112,842 Chinese patients with smear-positive tuberculosis." <i>Lancet</i> , February, 347(8998), pp. 358-362.	Y	N	Increased cure rates	No. Not random.
BCG	H	Tuberculosis Research Center (ICMR), Chennai. 1999. "Fifteen year follow up of trial of BCG vaccines in south India for tuberculosis prevention", <i>Indian Journal of Medical Research</i> . August, 110, pp. 56-69.	Y	Y	No effect of BCG on TB	No. No effect.
Preventive therapy	H	Mwinga A., et al. 1998. "Twice weekly tuberculosis preventive therapy in HIV infection in Zambia." <i>AIDS</i> , December, 12:18, pp. 2447-57.	Y	Y	Drugs reduced TB infection for HIV-infected people in Zambia	Yes (under HIV inter-ventions)
Vaccinations						
	H	Here we look for evidence for the new vaccines that the Global Alliance for Vaccines and Immunization is trying to introduce. The Vaccine Fund. 2002. <i>Strategic Plan 2002-2006</i> . There is a strong evidence base for the more established basic vaccines. Here we assume that all the vaccines can be implemented as programs, due to the success of the Expanded Program for Immunization.				
Japanese encephalitis vaccine	H	Hoke, C.H., et al. 1988. "Protection against Japanese encephalitis by inactivated vaccines", <i>New England Journal of Medicine</i> . September, 319:10, pp. 608-14.	Y	Y	Reduced encephalitis attack rate	Yes
Pneumococcal vaccine	H	O'Brien, Katherine L., et al. 2003. "Efficacy and safety of seven-valent conjugate pneumococcal vaccine in American Indian children: group randomized trial." <i>Lancet</i> . August, 362:9381, pp. 355-361.	Y	Y	Prevents vaccine serotype invasive pneumococcal disease	Yes
Quadrivalent Vaccine	H	Pérez-Schael, Irene, et al. 1997. "Efficacy of the Rhesus Rotavirus-Based Quadrivalent Vaccine in Infants and Young Children in Venezuela." <i>New England Journal of Medicine</i> 1997, October, 337:17, pp. 1181-1189.	Y	Y	Protection against severe diarrhea and dehydration, and reduced hospital admissions	Yes

Malaria		Summarized in Meek, S., J. Hill, J. Webster. 2001. "The Evidence Base for Interventions to Reduce Malaria in Low and Middle-Income Countries", Commission on Macroeconomics and Health, Working Group 5, Working Paper 6.				
Indoor Spraying	H	Rowland M., et al. 2000. "Indoor residual spraying with alphacypermethrin controls malaria in Pakistan: a community-randomized trial", <i>Tropical Medicine and International Health</i> . 5:7, pp. 472-481.	Y	Y	Reduction in anopheline porous rates	Yes
Insecticide-treated nets	H	Shulman CE, et al. 1998. "A community randomized controlled trial of insecticide-treated bednets for the prevention of malaria and anaemia among primigravid women on the Kenyan coast" <i>Tropical Medicine and Internal Health</i> , March, 3:3, pp. 197-204.	Y	Y	No significant impact	No. Not significant.
Outdoor Spraying	H	Cutler, David. 2003. "The economic impacts of public health improvements", Powerpoint slides, personal communication.	Y	N	Increase in arable land	No. Not random.
HIV		Summarized in P. Jha et al. 2001. "The evidence base for interventions to prevent HIV infection in low and middle-income countries", Commission on Macroeconomics and Health, Working Group 5, Working Paper 2, and Behrman, Jere R. and James C. Knowles. 2003. "Assessing the Economic Returns to Investing in Youths in Developing Countries", mimeo, University of Pennsylvania.				
Voluntary HIV-1 counseling and testing	H	Coates et al. 2000. "Efficacy of voluntary HIV-1 counseling and testing in individuals and couples in Kenya, Tanzania, and Trinidad: a randomized trial". <i>Lancet</i> . July, 356:9224, pp. 103-112.	Y	Y	Reduced intercourse with non-primary partners	Yes
Condom provision in motel rooms	H	Egger M et al. 2000. "Promotion of condom use in a high-risk setting in Nicaragua: a randomized controlled trial". <i>Lancet</i> . June, 355:9221, pp. 2101-2105.	Y	Y	Increased condom use	Yes
Package, including drug supply, health education and STD reference clinic	H	Grosskurth, H. et al. 1995. "Impact of improved treatment of sexually transmitted diseases on HIV infection in rural Tanzania: randomized controlled trial". <i>Lancet</i> , August, 346:8974, pp. 530-536.	Y	Y	Reduced HIV incidence	Yes
Home-based mass antibiotic treatment	H	Wawer, M.J. et al. 1999. "Control of sexually transmitted diseases for AIDS prevention in Uganda: a randomized community trial." Rakai Project Study Group. <i>Lancet</i> , February, 353:9152, pp. 525-535.	Y	Y	No effect	No
Confidential HIV testing and condom promotion	H	Allen, S. et al. 1992. "Confidential HIV testing and condom promotion in Africa. Impact on HIV and gonorrhea rates." <i>Journal of the American Medical Association</i> . December, 268:23, pp. 3338-3343.	Y	Y	Increased condom use, reduced rates of Gonorrhea and HIV in urban Rwandan women	Yes
Short-course zidovudine for babies that are not breastfed	H	Shaffer N. et al. 1999. "Short-course zidovudine for perinatal HIV-1 transmission in Bangkok, Thailand: a randomized controlled trial." <i>Lancet</i> , March, 353:9155, pp. 773-780.	Y	Y	Reduced risk of mother-to-child HIV transmission	Yes
Short-course zidovudine for breastfed babies	H	Wiktor S.Z. et al. 1999. "Short-course oral zidovudine for prevention of mother-to-child transmission of HIV-1 in Abidjan, Côte d'Ivoire." <i>Lancet</i> , March, 353:9155, pp. 781-785.	Y	Y	Reduced risk of mother-to-child HIV transmission	Yes

Monetary Transfers		Unlike the health-based interventions, all monetary transfers actually happened in a community-based context, hence they must be self-sustainable.				
Decentralized Targeting	P	Galasso, Emanuela and Martin Ravallion. 2001. "Decentralized Targeting of an Anti-Poverty Program", forthcoming, <i>Journal of Public Economics</i> .	Y	N	Results better in more favorable conditions	No. Not program evaluation per se.
Old Age Pension	H	Duflo, Esther. 2003. "Grandmothers and Granddaughters: Old Age Pension and Intra-household Allocation in South Africa", <i>World Bank Economic Review</i> , June, 17:1, pp. 1-25	Y	N	Increased weight of girls	No. Not random.
Poor-Area Development Programs	P	Jalan, Jyotsna and Martin Ravallion. 1998. "Are There Dynamic Gains from a Poor-Area Development Program?", <i>Journal of Public Economics</i> , January, 67:1, pp. 65-85	Y	N	Enough to prevent decline, but not enough for convergence	No. Not random.
Microfinance		Unlike the health-based interventions, all micro-finance interventions actually happened in a community-based context, hence they must be self-sustainable.				
Village Level Microfinance	P	Kaboski, Joseph and Robert Townsend. 2002. "Policies and Impact: An Analysis of Village-Level Microfinance Institutions", mimeo, University of Chicago.	Y	N	Mixed	No. Not random, not effective.
Village Level Microfinance	P	Morduch, Jonathan. 1998. "Does Microfinance Really Help the Poor? New Evidence from Flagship Programs in Bangladesh", Research Program in Development Studies, mimeo, Princeton University.	Y	N	No	No. Not effective.
School inputs		Unlike the health-based interventions, all school-based interventions actually happened in a community-based context, hence they must be self-sustainable.				
School Construction	E	Duflo, Esther. 2001. "Schooling and Labor Market Consequences of School Construction in Indonesia: Evidence from an Unusual Policy Experiment", <i>American Economic Review</i> . September, 91:4, pp. 795-813.	Y	N	Increase in education and earnings	No. Not random.
School Construction	E	Duflo, Esther. 2001. "The Medium Run Effects of Educational Expansion: Evidence from a Large School Construction Program in Indonesia", forthcoming in <i>Journal of Development Economics</i> .	Y	N	Increase labor force participation, but reduce wages of older cohorts	No. Not random.
Flip Charts	E	Glewwe, Paul et al. 2000. "Retrospective vs. Prospective Analyses of School Inputs: The Case of Flip Charts in Kenya", NBER Working Paper 8018.	Y	N / Y	No evidence with prospective, positive with retrospective	No. No evidence using random Evaluation.
Uniforms	E	Kremer, Michael; Sylvie Moulin and Robert Namunyu. 2003. "Decentralization: A cautionary tale", mimeo, Harvard University.	Y	Y	Reduce dropout rates, without reducing test scores	Yes
Teacher incentives	E	Glewwe, Paul, Nauman Ilias and Michael Kremer. 2003. "Teacher Incentives", NBER Working Paper 9671.	Y	Y	No evidence, teachers teach to test	No. Not successful.
Textbook	E	Glewwe, Paul, Michael Kremer and Sylvie Moulin. 2003. "Textbooks and Test scores: Evidence from a Prospective Evaluation in Kenya", mimeo, Harvard University.	Y	Y	Raised test scores of the highest quintile, more likely to go to secondary school	No. Very skewed results.
Remedial Education Program	E	Abhijit Banerjee et al. 2003. "Remedying Education: Evidence from Two Randomized Experiments in India", Poverty Action Lab Paper No. 4.	Y	Y	Increased learning	Yes
Teacher	E	Chin, Aimee. 2002. "The returns to school quality	Y	N	Increased	No. Not

provision		when school quality is very low: Evidence from Operation Blackboard in India”, mimeo, University of Houston.			female primary school completion and literacy	random.
School incentives						
School meals	E H	Vermeersch, Christel. 2002. “School Meals, Educational Achievement and School Competition: Evidence from a Randomized Experiment”, mimeo, Harvard University.	Y	Y	Conditional test score improvement, Higher participation, cut into instruction time	No. Questionable effectiveness.
Progresa	E	Schultz, Paul. 2001. “School Subsidies for the Poor: Evaluating the Mexican Progresa Poverty Program”, Economic Growth Center, Center Discussion Paper No. 834.	Y	Y	Higher enrollments	Yes
Progresa	E	Behrman, Jere, Pilali Segupta and Petra Todd. 2001. “Progressing Through Progresa: An Impact Assessment of a School Subsidy Experiment”, Penn Institute for Economic Research, Working Paper 01-033.	Y	Y	Increased educational attainment, % attending junior secondary school	
Progresa	H	Behrman, Jere and John Hoddinott. 2001. “Program Evaluation with Unobserved Heterogeneity and Selective Implementation: The Mexican Progresa Impact on Child Nutrition”, Penn Institute for Economic Research, Working Paper 02-006.	Y	Y	0/- effect with means, + stature with Fixed Effects	
Progresa	H	Gertler, Paul and Simone Boyce. 2003. “An Experiment in Incentive-Based Welfare: The Impact of PROGRESA on Health in Mexico”, Royal Economic Society Annual Conference 2003.	Y	Y	Health improvements	
School vouchers	E	Angrist, Joshua et al. 2002. “Vouchers for Private Schooling in Columbia: Evidence from a Randomized Natural Experiment.” <i>American Economic Review</i> . December, 92:5, pp. 1535-58.	Y	Y	Higher completion, less repeating of grades	Yes
Hygiene education	E H	Haggerty, P.A et al. 1994. “Community-based hygiene education to reduce diarrhoeal disease in rural Zaire: impact of the intervention on diarrhoeal morbidity.” <i>International Journal of Epidemiology</i> , October, 23:5, pp. 1050-1059.	Y	Y	Reduced diarrhea	Yes. (Folded into other education Estimates)
Fertilizer						
Fertilizer adoption through an NGO program	O	Duflo, Esther and Michael Kremer. 2003. “Understanding Technological Choices: Fertilizers in Western Kenya”, power-point slides, mimeo.	Y	Y	Increased adoption, high rate of return	Yes
Roads						
Rural Road Construction	O	van de Walle, Dominique. “Impact Evaluation of a Rural Road Rehabilitation Project in Viet Nam”, Cited 11 February 2004. Available online (http://econ.worldbank.org/view.php?type=20&id=11865)	Y	N	Research in progress	Not random, research in progress.

C.3 Program costs

PROGRAM (Recurring annual expenditure)	Source and Method of Calculation	R ?	Cost (US\$ ₂₀₀₀ Millions)
Education			
Remedial teaching based on the Balsakhi model developed by Pratham	REF: Abhijit Banerjee et. al. 2003. "Remedying Education: Evidence from Two Randomized Experiments in India", Poverty Action Lab Paper No. 4.	Y	644
	CALC: Unit (Cost per child-year), <i>BalkhashiCost</i> , from the Pratham 2001-02 Reports [cited on 28 June 2003] Available at World Wide Web (www.pratham.org/reports). We use the mean of costs reported for the Dehli, Mumbai, and Pune regions. Country cost = <i>BalkashiPop</i> * <i>BalkhashiCost</i> * <i>GDPCorrection</i>		
Universal education based on a 40 : 1 pupil-teacher ratio	REF: Angrist, Joshua and Victor Lavy. 1999. "Using Maimonides' Rule to Estimate the Effect of Class Size on Children's Academic Achievement." <i>Quarterly Journal of Economics</i> . May, 114:2, pp. 533-575. ²	N	1544
	CALC: Country cost = <i>NotInSchool</i> * <i>SchoolCost</i>		
School inputs (Uniforms and textbooks)	REF: Kremer, Michael, Sylvie Moulin and Robert Namunyu. 2003. "Decentralization: A cautionary tale", mimeo, Harvard University.	Y	2268
	CALC: Unit <i>InputsCost</i> from Kremer et al. (2003, pp. 44). Transportation costs ignored. Country Cost = <i>InputsCost</i> * <i>SchoolAge</i> * <i>PPPCorrection</i>		
Schooling vouchers	REF: Angrist, Joshua et. al. 2002. "Vouchers for Private Schooling in Columbia: Evidence from a Randomized Natural Experiment." <i>American Economic Review</i> . December, 92:5, pp. 1535-58.	Y	1478
	CALC: Assumes that everyone is sufficiently motivated to achieve satisfactory performance, hence qualifying for the vouchers, and ignoring general equilibrium effects due to the resultant increase in private school fees. Unit <i>VoucherCost</i> used in our calculations is the increase in public educational expenditure per lottery winner, given in Angrist et al. (2002 pp. 1535) This is multiplied by four because at any time there are four cohorts in high school Country Cost = (4 / 15) * <i>VoucherCost</i> * <i>ChildPop</i> * <i>GDPCorrection</i>		
Monetary rewards to parents for sending children to school	REF: Behrman, Jere, Pilali Segupta and Petra Todd. 2001. "Progressing Through Progres: An Impact Assessment of a School Subsidy Experiment", Penn Institute for Economic Research, Working Paper 01-033.	Y	23142
	CALC: Assumes that if the subsidy is large enough, everyone will want to send their children to school, and therefore everyone will get the subsidy. Unit <i>SubsidyCost</i> is calculated from data given in Behrman et al. (2001, pp. 1). This is multiplied by seven because at any time there are four cohorts getting the subsidy.		

¹ Randomized? (Y)es/(N)o.

² Angrist and Lavy's (1999) paper does not deal with a randomized experiment.

		Country Cost = (7 / 15) * <i>SubsidyCost</i> * <i>ChildPop</i> * <i>GDPCorrection</i>	
Nutrition Supplementation			
Iron	<u>REF:</u> Bobonis, Gustavo, Edward Miguel and Charu Sharma, 2003 "Iron Deficiency Anemia and School Participation." Pratham Research Conference in the USA, January 30-31, 2004. Texas, USA.	Y	346
	<u>CALC:</u> Unit <i>IronCost</i> data from Miguel and Bobonis, private communication. The program covers 5 cohorts aged between 2 and 6. Country Cost = (5 / 15) * <i>IronCost</i> * <i>ChildPop</i> * <i>PPPCorrection</i>		
Albendazole (De-worming)	<u>REF:</u> Kremer, Michael and Edward Miguel. 2004. "Worms: Identifying Impacts on Education and Health in the Presence of Treatment Externalities", <i>Econometrica</i> . January, 72:1 pp. 159-217.	Y	36
	<u>CALC:</u> Unit <i>DewormCost</i> data from Miguel and Bobonis, private communication. The program covers 5 cohorts aged between 2 and 6. Country Cost = (5 / 15) * <i>DewormCost</i> * <i>ChildPop</i> * <i>PPPCorrection</i>		
The delivery of iron supplements and de- worming pills through the Pratham Delhi Health Program model	<u>REF:</u> See for eg. Gustavo et. al. (2003)	Y	270
	<u>CALC:</u> Unit <i>TransportCost</i> data from Miguel and Bobonis's, private communication. The program covers 5 cohorts aged between 2 and 6. Country Cost = (5 / 15) * <i>TransportCost</i> * <i>ChildPop</i> * <i>PPPCorrection</i>		
Iodine	<u>REF:</u> Claudine Cobra et al. 1997. "Infant Survival Is Improved by Oral Iodine Supplementation", <i>Journal of Nutrition</i> . April, 127:4, pp. 574-578	Y	12
	<u>CALC:</u> Unit <i>IodineCost</i> data from International Council for the Control of Iodine Deficiency Disorders. 1998. <i>IDD Newsletter</i> , August, 14: 3. The experiment involved a once-off supplement. Country Cost = (1 / 15) * <i>IodineCost</i> * <i>ChildPop</i> * <i>PPPCorrection</i>		
Additional Transportation Costs	<u>CALC:</u> No handle on the transportation cost for the Iodine intervention. Presumably some it can be loaded on to the EPI program.		
HIV			
Condom provision	<u>REF:</u> Allen, S. et al. 1992. "Confidential HIV testing and condom promotion in Africa. Impact on HIV and gonorrhea rates." <i>Journal of the American Medical Association</i> . December, 268:23, pp. 3338-3343	Y	135
Improving STD management	<u>REF:</u> Grosskurth, H. et al. 1995. "Impact of improved treatment of sexually transmitted diseases on HIV infection in rural Tanzania: randomized controlled trial". <i>Lancet</i> , August, 346:8974, pp. 530-536.	Y	428
Voluntary counseling and testing	<u>REF:</u> Coates et al. 2000. "Efficacy of voluntary HIV-1 counseling and testing in individuals and couples in Kenya, Tanzania, and Trinidad: a randomized trial". <i>Lancet</i> , July, 356:9224, pp. 103-112.	Y	116

Prophylaxis for opportunistic infections	REF: Mwinga A. et al. 1998. "Twice weekly tuberculosis preventive therapy in HIV infection in Zambia." <i>AIDS</i> . December, 12:18, pp. 2447-2457.	Y	40
Short course zidovudine regime	REF: Shaffer N., et al. 1999. "Short-course zidovudine for perinatal HIV-1 transmission in Bangkok, Thailand: a randomized controlled trial." <i>Lancet</i> , March, 353:9155, pp. 773-780.	Y	4
	CALC: For all the HIV interventions - Main data source: Kumaranayake L, Watts C. 2000. <i>Costs of Scaling HIV program Activities to a National Level for Sub-Saharan Africa: Methods and Estimates</i>. Prepared for African Development Forum 2000, United Nations Economic Commission for Africa. Her numbers are for Sub-Saharan Africa (SSA), a group different from the Low-Income Countries (LIC). The Commission of Macroeconomics and Health, Working Group 5 (2001, Paper 19) shows estimates of scaling-up for both SSA and LIC for their set of interventions. Total Cost = $\text{Kumaranayake SSA Cost} * \text{CMH LIC Cost} / \text{CMH SSA Cost}$		
Malaria: Spraying	REF: Rowland M, et al. 2000. "Indoor residual spraying with alphacypermethrin controls malaria in Pakistan: a community-randomized trial", <i>Tropical Medicine and International Health</i> . 5:7, pp. 472-481.	Y	1627
	CALC: Malaria assumed to be prevalent in all the LICs. Unit <u>SprayCost</u> data from Verlé, et al. 1999. "Control of Malaria Vectors: cost analysis in a province in northern Vietnam", <i>Tropical Medicine & International Health</i>. Feb, 4:2, pp. 139-45. Country Cost = $\text{Population} * \text{SprayCost} * \text{PPPCorrection}$		
Fertilizer	REF: Duflo, Esther and Michael Kremer. 2003. "Understanding Technological Choices: Fertilizers in Western Kenya", power-point slides, mimeo	Y	1848
	CALC: Agriculture statistics and Fertilizer usage data from the 2003 World Development Indicators. We find the lowest <u>Fertilizer Consumption (100 grams per hectare of arable land)</u> and <u>Cereal Yield (kg per hectare)</u> between the 4 benchmark regions: [European Monetary Union], [High Income (nonOECD)], [High Income (OECD)] and the [United States]. We assume that a country is fertilizer deficient if both (a) fertilizer consumption and (b) the cereal yields are lower than the minimum value than the corresponding values (1045.875 and 2161.885) for the two benchmarks. For such countries we set <i>Deficient?</i> = 1, 0 otherwise. Unit <u>FertilizerCost</u> data from IFDC. 2001. <i>Monthly Bulletin of the International Institute of Soil Fertility Management (IFDC-Africa)</i>. February, 14:2. It does not have country specific price estimates: hence we are unable to correct for PPP differences. The recommended fertilizer consumption figure is also in the ballpark of the average recommendations of the net fertilizer (sum of different chemical compounds) across different countries, regions and crops, as seen in Wichmann, W (eds.), <i>World Fertilizer Use Manual</i>, available online at (http://www.fertilizer.org/ifa/publicat/html/pubman/manual.htm),		

	International Fertilizer Industry Association.		
	Country Cost = <i>Deficient?</i> * (1045.875 - <i>Fertilizer Consumption</i>) * <i>Land Area</i> * <i>Land Area Arable (% of total)</i>		
Vaccination	REF: Hoke, C.H. et al. 1988. "Protection against Japanese encephalitis by inactivated vaccines", <i>New England Journal of Medicine</i>. September, 319:10, pp. 608-14. O'Brien, Katherine L., et al. 2003. "Efficacy and safety of seven-valent conjugate pneumococcal vaccine in American Indian children: group randomized trial." <i>Lancet</i>. August, 362:9381, pp. 355-361. Pérez-Schael, Irene et al. 1997. "Efficacy of the Rhesus Rotavirus-Based Quadrivalent Vaccine in Infants and Young Children in Venezuela." <i>New England Journal of Medicine</i> 1997, October, 337:17, pp. 1181-1189.	Y	461
	CALC: The Vaccine Fund. 2002. <i>Strategic Plan 2002-2006</i>. Washington D.C.: The Vaccine Fund. It states that 5069 million of additional spending is required between 2001 and 2011. We divide this number by 11.		
TOTAL (excl. urban water provision and PROGRESA-style subsidies for school attendance)			11257 million ≈ 11 billion
TOTAL (excl. urban water provision)			34399 ≈ 34 billion
Urban water provision	REF: Cutler, David and Grant Miller. 2003. "Clean water measures in American History", power-point slides, private communication.	N	73
	CALC: For more details see the Water Construction section.		
TOTAL (incl. Urban water provision)			34472 million ≈ 34 billion

One-off Infrastructure Expenditure		R	One time cost (US\$ ₂₀₀₀ Million)
Urban water construction	REF: Cutler, David and Grant Miller. 2003. "Clean water measures in American History", power-point slides, private communication.	N	4454
Rural water construction		N	10867
	CALC: For each continent, the World Health Organization and United Nation's Children Fund, 2000. <i>Global Water and Sanitation Assessment, 2000 Report</i>, New York: United Nation's Children Fund, provides the unit costs (<i>WaterCost</i>) of possible interventions for (a) rural water construction (one off), (b) urban water construction (one off), and (c) urban water provision (per annum). Because of South Asia's huge size,		

	we make a special effort to find cost information for this sub-continent (details on website). For our estimates, we choose to use the cheapest cost option. The Assessment also provides country level statistics on the Urban (<i>UrbanPop</i>) and Rural (<i>RuralPop</i>) population, and the percentage of urban (<i>UNoAccess</i>) and rural (<i>RNoAccess</i>) people without access to water. Urban Country Cost = $PopCorrect * UrbanPop * UNoAccess * UnitCost$ Rural Country Cost = $PopCorrect * RuralPop * RNoAccess * UnitCost$		
Total water construction			15321 million ≈ 15 billion

R: Randomized

Details of the assumptions behind the above numbers

Many bold and dubious assumptions have been gone into the construction of this estimate of the cost of interventions for Low Income Countries. The fact that there may be economies of scale in expanding existing programs is ignored, as is the fact that the places that currently have these programs may not be randomly selected. Simplistic GDP and PPP corrections are used to correct for inter-country cost differences, and general equilibrium effects are ignored. We further assume linear population growth rates between 2000 and 2015, and that population is distributed equally amongst all ages between birth and 15 years. While the time scale used here is the average annual cost between 2000 and 2015, future inflation is ignored. We try to err on the side of finding a low number.

The construction of variables: The original data is from the 2002 Human Development Indicators in United Nations Development Program. 2002. *2002 Human Development Report*. New York: Oxford University Press, (denoted in Underlined Italics, for example, *Total Population, 2000*), unless specified otherwise.

Country population estimate:

$$Population = 1,000,000 * (\textit{Total Population (millions), 2000} + \textit{Total Population (millions), 2015})/2$$

Population growth factor estimate:

$$PopCorrect = (\textit{Total Population (millions), 2000} + \textit{Total Population (millions), 2015}) / (2 * \textit{Total Population (millions), 2000})$$

Child population estimate:

$$ChildPop = (1/100) * \textit{Population under age 15 (as \% of total), 2000} * Population$$

Percentage that needs remedial education (Method 1): Assumes that completion rates are proportional to the number of students in school. In other words, with universal education, the remedial rate will drop.

$$Remedial1 = 1 - \textit{Primary Completion Rate} / \textit{Combined Enrollment Rate}$$

Percentage that needs remedial education (Method 2): Assumes that remedial rate will stay constant even with full enrollment – children do not attend school because they are unable to cope.

$$Remedial2 = (100 - \textit{Primary Completion Rate}) / 100$$

Population that needs the Balkashi program: (A primary school length of 6 years is assumed)

$$BalkashiPop = (6 / 15) * ChildPop * (Remedial1 + Remedial2) / 2$$

Number of School aged population not in school:

$$NotInSchool = (6 / 15) * ChildPop * (100 - \textit{Combined Enrollment Data}) / 100$$

Cost per child year of Education: (Wage assumed at 3.6 times GDP per capita, at a 40:1 teacher-pupil ratio, and allowing recurrent non-teacher costs to comprise 33% of the expenditure)

$$SchoolCost = 3.6 * 1.5 * \textit{GDP (US\$ Billions.) 2000} / (\textit{Total Population (millions.) 2000} * 40)$$

Correction for costs of different services: We expect service costs to differ across countries, and assume these costs are proportional to GDP per capita. Let the country that we use for the point estimate be country X. Let the country that we want to estimate costs for be country Y. Then

$$GDPperCap = \textit{GDP (US\$ Billions.) 2000} / \textit{Total Population (millions.) 2000}$$
$$GDPCorrection = GDPperCap_Y / GDPperCap_X$$

Correction for costs of different goods: We expect goods costs to differ across countries, and assume these costs are proportional to the country's PPP. Let the country that we use for the point estimate be country X. Let the country that we want to estimate costs for be country Y. Then

$$PriceLevelrelUS = \textit{GDP (US\$ Billions.) 2000} / \textit{GDP (PPP US\$ Billions.) 2000}$$
$$PPPCorrection = PriceLevelrelUS_Y / PriceLevelrelUS_X$$

Missing countries: We do not have detailed data for 7 LIC countries. Hence to extrapolate our cost estimates to include these "missing" countries, we make the following correction:

$$\text{World Wide LIC costs} = [\text{Sum of Country costs calculated (for non-missing LIC countries)}]$$
$$* [\text{Sum of LIC population (2002 estimate)}]$$
$$/ [\text{Sum of non-missing LIC population (2002 estimate)}]$$

This correction ends up increasing our estimates by 3%

Appendix D

Appendix to Chapter 4

D.1 California Reapportionment propositions

D.1.1 1926 Proposition 20 “A Proportion-based plan”

Reapportionment Commission. Initiative

Creates reapportionment commission composed of Secretary of State, Attorney General and Surveyor General. If Legislature fails, at first session after each census, to adjust senatorial and assembly districts and reapportion representation as provided by Constitution, requires said commission to make such adjustment and reapportionment, and file same with Secretary of State, within three months after adjournment of such legislative session. Declares said commission shall make and file such reapportionment on basis of 1920 census within three months after this amendment takes effect.

Did not pass.

D.1.2 1926 Proposition 28 “A Federal plan”

Legislative Reapportionment. Initiative

For choosing legislators requires Legislature, immediately following each Federal census, and next Legislature using 1920 census, to divide State into forty senatorial and eighty assembly districts, comprising contiguous territory, with assembly districts as equal in population as possible, no county or city and county containing more than one senatorial district, and no senatorial district comprising more than three counties of small population; creates Reapportionment Commission, comprising Lieutenant Governor, Attorney General, Surveyor General, Secretary of State and State

Superintendent of Public Instruction, to make apportionment if Legislature fails to act.

Passed.

D.1.3 1928 Proposition 1 “Enacting the Federal plan”

Reapportionment of Legislative Districts. Referendum

Reapportions the legislative districts of the state, defining and establishing such districts, under authority of Section 6 of Article IV of Constitution adopted November 2, 1926.

Passed.

D.1.4 1948 Proposition 13 “A Proportion-based plan”

Senate Reapportionment. Initiative

Provides that Counties shall be represented in State Senate in proportion to population, but that no County shall have more than ten Senators. Eliminates present provision that no County shall contain more than one Senate District. Requires 1949 Legislature to reapportion Senate Districts according to population shown by 1940 Federal census, subsequent adjustments to be made following each decennial Federal census. Provides for election of all Senators in 1950, one-half of Senators to be elected every two years thereafter.

Did not pass.

D.1.5 1960 Proposition 15 “A compromised Proportion-based plan”

Senate Reapportionment. Initiative

Establishes and apportions 40 senatorial districts. Provides for election of all Senators in 1962, one-half of Senators to be elected every two years thereafter. Requires Legislature in 1961 to fix boundaries of districts in counties having more than one district on basis of population, area, and economic affinity, which may be re-fixed following each decennial federal census. Permits Legislature following 1960 and each subsequent decennial federal census to reapportion senatorial districts on same basis; provided no county shall have more than 7 districts and 20 districts be apportioned to designated counties.

Did not pass.

D.1.6 1962 Proposition 23 “A further compromised Proportion-based plan”

Senate Reapportionment. Initiative

Establishes and apportions 50 (instead of existing 40) senatorial districts. Provides for election of all senators in 1964, one-half of senators to be elected every two years thereafter. Additional districts allocated to existing single county districts based on population. Requires 1963 Legislature fix boundaries in counties having more than one district. Requires Legislature following 1970 and each subsequent decennial federal census to reapportion senatorial districts based on population, geographic area and economic affinity; provided no county shall have more than 6 districts and no district contain more than 3 counties.

Did not pass.

D.2 Construction of the fiscal preference index

All bond measures, not with setting up of boards to look after bonds. Also include all tax exemptions stuff. Some others that clearly spend money. From previous 5 years.

- 1922

1 Veterans' Validating Act. *3* Veterans' Welfare bond Act of 1921. *4* Land Settlement Bond Act of 1921. *7* Exempting Veterans from Taxation.

- 1924

14 Bonds. *15* Tax Exemption.

- 1926

1 Veteran's Welfare Bond Act of 1925. *11* Exempting Secondary Schools from Taxation. *12* Tax Exemption for Veterans and Others. *22* Exempting Forest Trees from Taxation. *25* Taxation and Local Assessment Exemption.

- 1928

2 Approving California Olympiad Bond Act. *4* Approving California State Park Bonds Act. *17* Grade Separation Bonds.

- 1944

1 The Veterans Bond Act of 1943. *2* Taxation. Veterans Exemption. *4* Taxation Exemption of Religious, Hospital, and Charitable Organizations. *9* Funds for Elementary Schools. *11* Retirement Payments, Gross Income Tax.

- 1946
 - 1 The Veterans' Bond Act of 1946. 3 Public Schools.
- 1948
 - 1 Veteran's Tax Exemption. 4 Aged and Blind Aid.
- 1956
 - 1 Veterans Bond Act of 1956. 2 School Bonds. 3 State Construction Program Bonds.
- 1958
 - 1 The Veterans Bond Act of 1958. 2 School Bonds. 3 State Construction Program Bonds.
- 1960
 - 1 California Water Resources Development Bond Act. 3 Disabled Veterans' Tax Exemption.
 - 11 Veterans' Tax Exemption.
- 1962
 - 2 Veterans' Tax Exemption in Housing Project. 3 Veterans' Tax Exemption. 10 Welfare Exemption: Leased Property. 12 Aid to Widows of Veterans. 13 College Exemption: Extension of.

D.3 Survey descriptions for the individual level data

D.3.1 Measure of government expenditure, *Fiscal*

1960: As you may know, in November Californians will be asked to vote on a bond issue to finance the state water program. Here is a copy of the measure as it will appear on the ballot. (Card C) If you are voting today, do you think you would vote for this or against it.¹

1962: The first Proposition 13 - The one dealing with an extension of tax exemptions to non-profit colleges. As it appears on the ballot, the proposition will be stated this way (Hand Card B and allow respondent time to read it) Now, if you were voting today on Proposition 13, would you vote yes or no (if undecided) Just suppose that you had to make up your mind today, would you vote yes or no.²

¹Question 16 from California poll 6004 as recorded in the 1960-1961 codebooks, Volume 1. Same questions for polls 6004-6007

²Question 22 from California poll 6204 as recorded in Codebook 6201. Same questions for poll 6205

D.3.2 Reapportionment, *Reapport*

1960: There will also be a measure on the ballot to reapportion the California state senate. Here is the way this measure will appear on the ballot. (Card D) If you were voting today, do you think you would vote for this or against it.³

1962: What about Proposition 23 - The one dealing with state senate reapportionment - as it appears on the ballot, the proposition will be stated this way (Hand card C and allow respondent time to read it) Now, if you were voting today on Proposition 23, would you vote yes or no (if undecided) Just suppose that you had to make up your mind today, would you vote yes or no.⁴

Fiscal and *reapport* take on values of 1 and 0 for Yes and No replies respectively. For the Probit we drop don't knows and won't votes on these two issues.

D.3.3 *Urban*

For the 1960 data: *Urban* = 0 if respondent lives in "Farm" or "Rural non-farm" area (i.e. population less than 2,500), *Urban* = 1 otherwise. For the 1962 data: *Urban* = 0 if respondent lives in "Rural places" or "Other rural" area (i.e. population less than 2,500), *Urban* = 1 otherwise.

D.3.4 *Party Affiliation*

Republicans are assigned a value of -1, Democrats 1, while Others and Independents are assigned a value of 0.

D.3.5 *South*

South as defined in survey.

D.3.6 *Urban * South*

Interaction term of *Urban* and *South*.

D.4 Propositions used for robustness checks

Just like the measures of fiscal preferences, these indexes take the demeaned average of the relevant propositions in the 5 years preceding each reapportionment vote for the County and District level

³Question 17 from California poll 6004 as recorded in the 1960-1961 codebooks, Volume 1

⁴Question 23 from California poll 6204 as recorded in Codebook 6201. Same questions for poll 6205

regressions, and only in 1926 for the town level data.

- Communism

1926:13 Suffrage.

1956:13 Repealing Alien Land Law. *1962:24* Subversive Activities.

- Religious Conservatism

1926:17 Requiring Bible in Schools.

1956:6 Church Exemption: Parking Lots. *1958:16* Taxation of School Property of Religious and Other Nonprofit Organizations.

- Pure Liberalism

1924:11 Klamath River Fish and Game District. *1928:21* Prohibiting Certain Acts with Animals and Use of Certain Instruments to Control Them.

1944:12 Right of Employment. *1946:11* Fair Employment Practices Act. *1948:6* Regulation of Commercial Fishing. *1948:15* Fish Nets.

1956:4 Oil and Gas Conservation Act. *1958:18* Employer-Employee Relations.

- Liquor

1922:2 Prohibition Enforcement Act.

1948:2 Local Control and Enforcement of Intoxicating Liquors. *1948:12* Local Control of Intoxicating Liquors.

1956:5 Alcoholic Beverage Establishments. *1962:18* Alcoholic Beverage Control Appeals Board.

- Criminal

1956:16 Civil and Criminal Appeals.